City charges for competing delivery carriers: exact best responses under probability-box travel times and leader pricing without equilibrium search

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Abstract

A city that charges delivery carriers faces a bilevel problem with three stacked difficulties. Each carrier solves an NP-hard routing problem whose travel times are known only through probability boxes, with unknown dependence between road segments. The carriers compete in a Cournot-type game in which the price falls with total deliveries and travel times rise with vehicle density. The city's charges are usually found by solving one equilibrium per grid point, which costs about two thousand routing solves per city. We give three algorithms. Budget-resource labelling prices routes exactly under the dependence-robust criterion and, inside branch-and-price, makes each carrier's integer plan exact. Dual implementation pricing targets a social optimum and finds charges that make it an equilibrium by cutting planes whose separation step is one best response per carrier. Aggregate-response pricing sets charges from observed zone totals only and contracts at a rate that does not depend on the number of zones, carriers or routing data. On nine metropolitan areas built from OpenStreetMap and gridded population, three random draws each, dual implementation pricing used 4.9–9.4% of the grid's routing calls in the three cities with a full grid and reached 104–135% of its gain, partly because it uses zone-specific charges; aggregate-response pricing reached 90.8–126% there. Pure equilibria exist under exposure-based congestion, and deciding their existence is strongly NP-complete under vehicle-count congestion. Equilibria, charges and the price of anarchy vary strongly with the random draw of the city slice.

Keywords: bilevel optimization, vehicle routing, imprecise probabilities, congestion games, branch-and-price, road pricing

1. Introduction

Several delivery firms serve the same neighbourhoods, sell to overlapping customers and slow one another down on the same streets. City authorities respond with instruments that act on all of them: congestion and low-emission zones, kerbside rules, and charges or subsidies per delivery [1, 2, 3, 4]. Choosing such an instrument means predicting how the carriers react, and the reaction is itself the outcome of routing decisions that are hard to compute.

We treat the city as the leader of a bilevel problem. Three difficulties are stacked.

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(1) *Each follower solves an NP-hard routing problem under imprecise probabilities.* A carrier has a few weeks of travel-time observations. These support bounds on the distribution function of each arc time (a probability box) and say nothing about the dependence between arcs. The robust cost of a route is its worst-case expected loss over all joint laws compatible with the boxes, and the carrier's best response is a vehicle routing problem with this cost.

(2) *The followers play a congestion game.* The price a carrier obtains falls with the total deliveries in a zone (Cournot competition) and travel times rise with vehicle density. If congestion depends on vehicle-hours (exposure), the game has an exact potential and pure equilibria exist. If it depends on vehicle counts, pure equilibria may fail to exist and deciding whether one exists is strongly NP-complete (Section 5).

(3) *The leader's problem is a bilevel program whose lower level is the game above.* The usual approach evaluates a grid of charges and computes one equilibrium per grid point. With 63 grid points this took 1 921–2 419 routing-oracle calls per city and seed in our experiments, each call an NP-hard problem.

This paper gives algorithms for each layer and tests them on nine metropolitan areas. We state at once what is new and what is not.

Contributions, in order, with their novelty level.

(i) *The problem and the model* (Section 3): a bilevel model with a public leader above carriers who choose routing plans, interact through zone aggregates of deliveries and vehicle-hours, and plan with probability-box travel times. The ingredients are standard; the combination, and the two congestion regimes (exposure and vehicle count) with their different existence properties, are what we set up.

(ii) *Budget-resource labelling (BRL)* (Section 4.1): exact pricing of routes under probability boxes with unknown dependence, and, with branch-and-price, exact integer plans for the carriers. The reduction to a comonotone worst case and the stop-loss decomposition are classical [5], and a comonotone reduction for ambiguous demand is known in routing [6]. The new element is to use the per-arc budget of the decomposition as a *decision resource* of the pricing problem, one per kink of the loss function, together with a dominance rule restricted to a computable completion window and its exactness proof. Branch-and-price itself is standard machinery.

(iii) *Leader pricing without equilibrium search* (Sections 4.4 and 4.5). *Dual implementation pricing (DIP)* computes a socially good target profile and then finds charges under which it is an equilibrium, by cutting planes whose separation step is one best response per carrier. *Aggregate-response pricing (ARP)* lets the leader see only the zone aggregates of the market after each posted charge, and a marginal tariff variant needs no iteration. Both rest on classical ideas: the Pigou–Cournot tax, toll polyhedra, inverse optimisation, trial-and-error tolls and marginal-contribution tariffs [7, 8, 9, 10, 11, 12, 13]. What is new is limited to the following: the potential–welfare identity of the lumpy Cournot–congestion game with loss bounds in terms of lot asymmetry; the use of the carriers' own

NP-hard best responses as the separation step; a contraction rate $\varrho = \max_z \kappa_z/(b_z + \kappa_z)$ for the observation-only iteration that does not depend on the number of zones, carriers, lots, routing costs or probability boxes; and the shift of the implementing charges by a Γ-maximin surcharge.

(iv) *Theory* (Section 5): a credal absorption theorem, which gives a belief width bounded independently of the number of carriers above which a mixed equilibrium rounds to an exact pure imprecise equilibrium, and shows that the bound is attained (closely related to 14, who obtain an approximate equilibrium of the precise game); the split between exposure-based congestion (pure equilibria always exist) and vehicle-count congestion (strongly NP-complete existence), which is incremental to Apt et al. [15]; and Γ-maximin beliefs as a surcharge game. The discrete core of these results is checked in Lean 4 without unproved steps; Section 5.9 says what that does and does not cover.

(v) *Evidence* (Sections 6 and 7): nine metropolitan areas built from OpenStreetMap, gridded population (GHSL, epoch 2025) and web-sourced operating parameters with quality grades; three random draws (seeds) of each city slice; an exact integer oracle; and results on algorithm performance, equilibria, charges, the price of anarchy and imprecision, reported with their spread over seeds. The instance pipeline, with its parameters, is released as a benchmark generator for multi-carrier routing games.

What is not new, and what we do not claim. The comonotone upper prevision of a route, the stop-loss decomposition, label-setting for elementary shortest paths, branch-and-price and ng-routes are classical. The approximate pure equilibrium of nonconvex aggregative games with an error that does not grow with the number of players is due to Liu et al. [14]. The hardness of pure-equilibrium existence by a kernel reduction is of the kind in Apt et al. [15]. The tangent charge, the marginal-contribution tariff and trial-and-error pricing are classical; the closest neighbour of ARP in the adaptive-pricing literature is Maheshwari et al. [13, 16], which treats atomic Cournot games with continuous strategies. We do not claim a forecast for any city: the instances are a stylised slice of each city with proxy demand and several weak parameters, and the price-of-anarchy values are empirical, computed from the equilibria that our methods found. We do not claim optimal charges (the leader's optimum is defined with optimistic equilibrium selection), a guarantee about which equilibrium the carriers select, or any guarantee for the pure game under ARP unless its fixed point is reached. We do not claim results for vehicle-count congestion at city scale. Zone-market clearing (ZMC), an earlier method of ours for equilibrium computation, did not outperform plain best response on the city games; Remark 5 and Appendix C report this.

Organisation. Section 2 places the work in the literature, Section 3 defines the model, Section 4 gives BRL, the equilibrium engine, DIP and ARP, Section 5 the equilibrium theory, Section 6 the data, and Section 7 the computational study. Sections 8 and 9 give implications and limits. Proofs are in the appendices.

2. Related work

Competing carriers and leaders above carriers. Carriers competing in sequential auctions for requests were studied by simulation by Figliozzi et al. [17, 18]. Stylised games treat shared

collection points and parcel lockers [19, 20], and ride-hailing and crowdsourcing models give equilibrium evidence on competition for workers and customers, with queues instead of routes [21, 22, 23, 24, 25]. Cerulli et al. [26] place a platform above carriers who solve profitable tour problems, but the carriers do not interact. Catalano and Migliore [27] cast terminal design as a Stackelberg game. Curb management supplies pricing leaders [4, 28], and policy work documents the restrictions cities use [1, 2, 3]. Collaborative routing [29, 30, 31] gives savings that bound what cooperation could recover, not a price of anarchy.

Routing under uncertainty. Robust and distributionally robust routing keeps the combinatorial structure when the ambiguity set is suitable [32, 33, 34, 35, 36]. Closest to BRL, Ghosal et al. [6] show that a capacitated routing problem with ambiguous demand marginals and unknown dependence reduces, through a comonotone worst case, to a scenario problem solved by branch-and-cut. The comonotone worst case and the stop-loss decomposition of comonotone sums are due to Meilijson and Nádas [5] [see also 37, 38], and worst-case marginals of a probability box under monotone losses are standard [39]. Branch-price-and-cut under mean-covariance travel times [40] and path dominance by second-order stochastic dominance for independent arc times [41] address other dependence models. None of these adds competition among carriers.

Games, aggregation and ambiguity. Potential and congestion games guarantee pure equilibria [42, 43], and selfish-routing theory bounds their inefficiency [44, 45, 46]. A carrier chooses tours, so its strategy set is finite and nonconvex. Liu et al. [14] prove, with a Shapley–Folkman argument in the tradition of Starr [47], that nonconvex aggregative games have approximate pure equilibria whose error does not grow with the number of players. Approximate pure equilibria in large games go back to Kalai [48] and Azrieli and Shmaya [49], and finding them can be PPAD-complete [50]. Equilibrium under ambiguous beliefs has been studied as equilibrium in beliefs [51, 52], as pure equilibrium in beliefs with ambiguous payoffs [53], and as robust-optimisation equilibrium [54, 55]; Crespi et al. [55] show that robust equilibria are approximate equilibria of the nominal game, the converse direction of our absorption result. Feik et al. [56] study convex aggregative games with an optimal-transport ambiguity ball. Deciding pure-equilibrium existence is NP-complete in polymatrix and integer programming games [15, 57].

Tolls, implementation and adaptive pricing. Corrective taxes for market power and externalities go back to Buchanan [7] and Barnett [58]. Toll polyhedra that implement a target flow are classical for nonatomic routing [8, 59], cost-balancing tolls exist for atomic games [60], and Bhaskar et al. [61] compute implementing tolls from an equilibrium oracle. The cutting-plane loop of DIP is an inverse-optimisation cutting plane [9, 62, 63], and best-response cuts are used for equilibria of integer programming games [64, 65]. Marginal-contribution tariffs make welfare the potential of the game [11, 66]. Trial-and-error implementation of marginal-cost pricing from observed flows is due to Yang et al. [10], evolutionary implementation of pricing to Sandholm [12], and adaptive incentive design for atomic Cournot games with continuous strategies to Maheshwari et al. [13, 16]. The closest leader-side method for combinatorial games updates tolls by zeroth-order steps around an equilibrium solver, including freight carriers with hard tour problems [67]; like a grid, it needs an equilibrium solve per outer step. Our novelty relative to these is stated in Sections 4.4 and 4.5. Our searches were web and

index searches, with snippet-level reading for several neighbours; a Scopus and Web of Science search has not been done.

3. Model

Notation: $[n] = \{1, \dots, n\}$; for $v \in \mathbb{R}^d$ and $w \in \mathbb{R}^d_{>0}$, $\|v\|_{w,\infty} = \max_k |v_k|/w_k$ and $\|v\|_M^2 = \langle v, Mv\rangle$.

3.1. Carriers, plans and congestion

The city is divided into zones $z \in [Z]$. There are N carriers. Carrier i has a finite plan set $\mathcal{P}_i$; a plan is a set of depot-to-depot routes over the carrier's customer zones. A plan p determines deliveries $q_{iz}(p)$, vehicles $n_{iz}(p)$ and vehicle-hours $\tau_{iz}(p)$ in each zone, and a routing cost $C_i(p)$ (Section 3.2). Write $Q_z = \sum_i q_{iz}$ and $T_z = \sum_i \tau_{iz}$. Inverse demand in zone z is $\alpha_z - b_z Q_z$ with $b_z > 0$, and a vehicle-hour costs $\gamma(\xi_z + c_z X_z)$, where $\gamma > 0$ is the hourly cost, ξ_z the free-flow travel-time factor with mean μ_z, $c_z > 0$ the congestion slope and X_z the congestion variable. The city charges s_z per delivery and θ_z per unit of the congestion variable. Carrier i's profit is

$$u_i = \sum_z \left[q_{iz}(\alpha_z - b_z Q_z) - \gamma \tau_{iz}(\xi_z + c_z X_z) - s_z q_{iz} - \theta_z x_{iz}\right] - C_i(p). \tag{1}$$

Two congestion regimes are distinguished.

- **D1 (exposure-based):** $x_{iz} = \tau_{iz}$ and $X_z = \sum_i \tau_{iz}$.
- **D2 (vehicle-count):** $x_{iz} = n_{iz}$ and $X_z = \sum_i n_{iz}$.

Under D1 the carriers' interaction depends on the same quantity that determines the carrier's own congestion cost. Under D2 the externality depends on a different quantity (counts) than the carrier's own exposure (hours), which breaks the potential structure (Section 5).

For the theory it is convenient to write the interaction abstractly. Let $d = 2Z$ with coordinates (z, Q) and (z, X).

Assumption A1 (Finite plans)**.** Plan $p \in \mathcal{P}_i$ has a contribution $a_i(p) \in \mathbb{R}^d_{\geq 0}$ to the aggregate, an exposure $\varphi_i(p) \in \mathbb{R}^d_{\geq 0}$ to the aggregate, a traffic exposure $t_i(p) \in \mathbb{R}^Z_{\geq 0}$ and a private profit $\pi_i(p)$. At a pure profile σ, $A(\sigma) = \sum_i a_i(\sigma_i)$ and $A_{-i} = A - a_i(\sigma_i)$.

Assumption A2 (Affine coupling)**.** There are a diagonal $M \geq 0$ and $\gamma > 0$ such that, for a rival aggregate $m \in \mathbb{R}^d_{\geq 0}$ and traffic factor $\xi \in \mathbb{R}^Z_{\geq 0}$, carrier i obtains $u_i(p; m, \xi) = \pi_i(p) - \langle \varphi_i(p), Mm\rangle - \gamma\langle t_i(p), \xi\rangle$.

Equation (1) has this form with $a_{i,(z,Q)} = \varphi_{i,(z,Q)} = q_{iz}$, $M_{(z,Q)} = b_z$, $a_{i,(z,X)} = x_{iz}$, $\varphi_{i,(z,X)} = \tau_{iz}$, $M_{(z,X)} = \gamma c_z$ and $t_{iz} = \tau_{iz}$. The game is *of type D1* if $\varphi_i = a_i$ for all carriers and plans, and *of type D2* otherwise. Social welfare, with charges treated as transfers, is

$$W(\sigma) = \sum_z \left[\alpha_z Q_z - \tfrac{b_z}{2} Q_z^2 - \gamma \mu_z T_z - \gamma c_z T_z X_z\right] - \sum_i C_i(\sigma_i), \tag{2}$$

that is, consumer surplus minus the congestion cost that carriers impose on each other, minus routing costs; in the experiments the traffic term is part of the routing cost and the social cost is $\mathrm{SC} = -W$. The affine coupling excludes power-law congestion such as BPR curves, which we linearise (Section 6).

3.2. The routing layer

For every arc a the carrier knows a probability box: distribution functions $\underline{F}_a \le \overline{F}_a$ on a bounded support $[0, B_a]$, for example a Dvoretzky–Kiefer–Wolfowitz band around the empirical distribution of n observed days, $\underline{F}_a(x) = \max\{F_n(x) - \varepsilon, 0\}$ with $\varepsilon = \sqrt{\ln(2/\alpha)/(2n)}$ [68, 69]. The dependence between arcs is unknown, so the joint credal set is the Fréchet class $\mathcal{M}$ of all laws whose a-th marginal lies between $\underline{F}_a$ and $\overline{F}_a$. A route of duration $T_r = \sum_{a \in r} X_a$ incurs a convex piecewise-linear loss

$$\phi(T) = c_0 + \gamma T + \sum_{j=1}^{H} \rho_j \, (T - \Theta_j)^+ , \qquad \gamma \ge 0, \ \rho_j > 0, \ 0 \le \Theta_1 < \cdots < \Theta_H. \tag{3}$$

In the experiments $H = 1$: a fixed cost, an hourly cost and an overtime penalty after the shift length. The robust route cost is the upper prevision $\overline{\mathbb{E}}[\phi(T_r)] = \sup_{P \in \mathcal{M}} \mathbb{E}_P \phi(T_r)$, and $C_i(p) = \sum_{r \in p} \overline{\mathbb{E}}[\phi(T_r)]$. A carrier's best response to prices on its deliveries and hours is therefore a prize-collecting routing problem in which every route is priced by $\overline{\mathbb{E}}[\phi]$. We call this problem the *carrier oracle*:

Assumption A3 (Oracle)**.** For each i there is an oracle that, for price vectors $\lambda^a, \lambda^\varphi \in \mathbb{R}^d$, $\nu \in \mathbb{R}^Z$ and a diagonal $Q \succeq 0$, returns a plan in $\arg\max_{p \in \mathcal{P}_i} \{\pi_i(p) + \frac{1}{2}\langle a_i(p), Q a_i(p)\rangle - \langle \lambda^a, a_i(p)\rangle - \langle \lambda^\varphi, \varphi_i(p)\rangle - \langle \nu, t_i(p)\rangle\}$.

A best response at the belief (m, ξ) is the call $(0, Mm, \gamma\xi, 0)$. Section 4.1 gives an exact implementation of the oracle. We count *oracle calls* as the cost unit of every method below.

3.3. The leader's problem

The city chooses charges u from an admissible set $\mathcal{U}$. For a charge u let $\mathrm{NE}(u)$ be the set of pure Nash equilibria of the carriers' game. With optimistic selection the leader's problem is

$$\min_{u \in \mathcal{U}, \ \sigma} \ \mathrm{SC}(\sigma) \qquad \text{s.t.} \qquad \sigma \in \mathrm{NE}(u). \tag{4}$$

In the D1 instance used for computation (Section 4.4), each carrier serves its customer zones in all-or-nothing lots, and the charges reach the game through one number per zone. The three methods compared in Section 7 differ in what the leader may use: the *grid* and DIP use the carriers' routing oracle, ARP uses only the zone aggregates the market produces after each posted charge.

4. Algorithms

The algorithms form a stack. BRL (Section 4.1) and branch-and-price (Section 4.2) implement the carrier oracle exactly. An equilibrium engine (Section 4.3) calls the oracle to compute a pure equilibrium for a given charge. DIP (Section 4.4) and ARP (Section 4.5) choose the charges; the grid baseline calls the engine once per grid point.

4.1. Budget-resource labelling (BRL)

Let $\overline{Q}_a(u) = \inf\{x : \underline{F}_a(x) \geq u\}$ be the upper quantile of arc a, $\bar{x}_a = \int_0^1 \overline{Q}_a(u)\,du$ and $\overline{\Pi}_a(d) = \int_0^1 \big(\overline{Q}_a(u) - d\big)^+ du$, the stop-loss transform of the lower distribution function. $\overline{\Pi}_a$ is convex and nonincreasing with slopes in $[-1, 0]$.

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Theorem 1 (Upper prevision and budget decomposition)**.** *For every route r,*

$$\overline{\mathbb{E}}[\phi(T_r)] = \int_0^1 \phi\Big(\sum_{a\in r} \overline{Q}_a(u)\Big)du = c_0 + \sum_{a\in r} \gamma\bar{x}_a + \sum_{j=1}^{H} \rho_j \min\Big\{\sum_{a\in r} \overline{\Pi}_a(d_a^j) :\ \sum_{a\in r} d_a^j \leq \Theta_j\Big\}.$$

The supremum is attained by the comonotone coupling of the lower CDFs; each minimum is attained, at an integer vector when quantiles and thresholds are integers.

The first equality and the stop-loss identity for comonotone sums with fixed marginals are due to Meilijson and Nádas [5] (see also 37, 38), and the choice of the lower CDF as worst marginal is standard [39]. We re-prove the statement in Appendix B because the algorithm relies on the inequality-budget form and on integrality. With a common quantile grid of L levels (weights ω_l), the route profile $S_{r,l} = \sum_{a\in r} q_{a,l}$ gives $\overline{\mathbb{E}}[\phi(T_r)] = c_0 + f_r(T_{\max})$ for $H = 1$, where $f_r(D) = \gamma \sum_l \omega_l S_{r,l} + \rho \sum_l \omega_l \,(S_{r,l} - D)^+$.

Theorem 2 (Budget-resource reformulation)**.** *Let y be dual prices and define the arc costs*

$$c_a(d^1, \ldots, d^H) = \gamma\bar{x}_a + \sum_j \rho_j \overline{\Pi}_a(d^j).$$

The dependence-robust pricing problem

$$\min_{r\in\mathcal{R}} \Big\{\overline{\mathbb{E}}[\phi(T_r)] - \sum_{i\in r} y_i\Big\}$$

equals an elementary shortest path problem in which every arc chooses a consumption d_a^j of each of H budget resources, paid through the separable convex cost c_a and subject to the budget constraints $\sum_{a\in r} d_a^j \leq \Theta_j$, $j \leq H$. The number of resources is H; it does not depend on L, the sample size or the number of arcs.

A scenario view of the same problem, which is what the comonotone reduction of Ghosal et al. [6] gives, needs the L level sums $(S_{p,l})_l$ as resources and compares labels pointwise. BRL instead compares labels through their cost functions, but only where a completion can still reach. For a partial path (label) p ending at v with level profile S_p, collected duals $y(p)$, visited set $\mathcal{V}(p)$ and load $\ell(p)$, concatenation with a completion q is an inf-convolution, $f_{p\oplus q}(T_{\max}) = \min_D\{f_p(D) + f_q(T_{\max} - D)\}$ (Lemma 46). Let $C_{\min}(v)$ be a lower bound on the duration of any completion from v and define the completion window $\mathcal{W}(p) = [\underline{D}(p), \overline{D}(p)]$ with $\overline{D}(p) = \min\{S_{p,L}, T_{\max} - C_{\min}(v)\}$ and $\underline{D}(p) = \min\{S_{p,1}, \overline{D}(p)\}$.

Algorithm 1 BRL pricing for the dependence-robust routing problem ($H = 1$)

Require: graph, demands, capacity; level weights ω; arc level profiles $q_{a,\cdot}$; $\gamma, \rho, T_{\max}, c_0$; duals y

$C_{\min}(v) \leftarrow$ shortest path from v to the depot with arc lengths $q_{a,1}$
buckets $\mathcal{B}_v \leftarrow \emptyset$; queue $\leftarrow$ {depot label}
while queue nonempty **do**
    pop p; record the closed route $p \oplus (v, 0)$ with reduced cost $c_0 + f_{p\oplus(v,0)}(T_{\max}) - y(p)$
    for each customer $j \notin \mathcal{V}(p)$ that fits the capacity **do**
        $p' \leftarrow$ extension of p to j ($S_{p'} = S_p + q_{(v,j),\cdot}$, $y(p') = y(p) + y_j$)
        if some $o \in \mathcal{B}_j$ functionally dominates p' **then** discard p'
        else delete the labels of $\mathcal{B}_j$ dominated by p'; insert and push p'
return recorded routes sorted by reduced cost

Definition 3 (Functional dominance)**.** Label p_1 functionally dominates p_2, both ending at v, if $\mathcal{V}(p_1) \subseteq \mathcal{V}(p_2)$, $\ell(p_1) \le \ell(p_2)$ and $(f_{p_1}(D) - y(p_1)) - (f_{p_2}(D) - y(p_2)) \le 0$ for all $D \in \mathcal{W}(p_2)$.

The difference is piecewise linear with breakpoints at the level sums, so the test needs at most $2L + 2$ evaluations.

Theorem 4 (Validity and exactness)**.** *Functional dominance implies that every completion of p_2 is a completion of p_1 with no larger reduced cost. The labelling algorithm that discards dominated labels (Algorithm 1) returns the minimum reduced cost exactly, with a route attaining it.*

The window is what makes the rule exact and strong at the same time. The slope of f_p lies in $[-\rho, 0]$ and that of $D \mapsto f_q(T_{\max} - D)$ in $[0, \rho]$, so an optimal split never lies above $\overline{D}(p)$, and below $S_{p,1}$ the difference of two label functions is nondecreasing (Lemma 47). Quantile-vector dominance and sector dominance are special cases of functional dominance, and neither converse holds (Proposition 48). Under independent arc times concatenation is a convolution of laws rather than an inf-convolution and the method does not apply. The natural shortcut of fixing a single risk level for all routes gives a lower bound but is not exact (Proposition 49).

On the city instances the pricing runs over ng-routes [70]: the visited set is replaced by the ng-memory, and Theorem 4 holds verbatim, with the weaker ng-relaxation bound. Column generation [71, 72] with BRL pricing returns the exact LP bound of the dependence-robust routing problem when no negative reduced cost remains; an integer program over the generated columns (price-and-branch) gives a feasible plan but is not an exact integer method. Section 4.2 makes the integer step exact.

Prior art and the claim. The upper-prevision formula and the stop-loss decomposition are classical (Meilijson–Nádas; Theorem 1 is re-proved in Appendix B because the algorithm uses its inequality-budget form and its integrality). Label-setting with resources is standard, and the comonotone reduction for ambiguous marginals is in Ghosal et al. [6]. We did not find in the literature (i) the per-arc budget used as a *decision resource* of the pricing problem (Theorem 2) or (ii) the completion-window functional dominance with its exactness proof

(Theorem 4). A referee may reasonably describe BRL as Meilijson–Nádas plus resource-constrained labelling. Our answer is empirical and structural: the dominance rule is what keeps the label sets small (Section 7.1), and with independent arc times the method does not apply.

4.2. Branch-and-price with BRL pricing

Price-and-branch, an integer program over the columns of the root relaxation, can miss the integer optimum. We therefore embed the BRL pricer in branch-and-price. The master is a set-partitioning model for the routing benchmark, or a prize-collecting set-packing model for the carrier oracle, with one column pool shared by all nodes. Each node solves its LP by column generation with the exact BRL pricer, restricted to the node's allowed arcs with $C_{\min}$ recomputed on them; functional dominance remains valid because the restrictions are local.

- *Branching* is on the aggregated arc flow f_{ij}: $f_{ij} = 0$ deletes the arc; $f_{ij} = 1$ deletes the other arcs leaving i and entering j. Integral arc flows imply an integral plan.
- *Fleet size.* When every customer must be served, a row for the minimum number of routes is added at the root and a fractional route count is branched on first.
- *ng-routes* [70]. The visited set is replaced by the ng-memory, and Theorem 4 holds verbatim with the weaker relaxation. If a node solution contains a non-elementary column, the ng-sets along its cycle are enlarged; reported plans are elementary.
- *Bounds.* The node LP value and the Lagrangian bound during column generation are valid lower bounds, marked invalid if a pricing call hits the label limit. Upper bounds come from integral nodes and from an integer program over the generated elementary columns. Nodes are chosen by best bound; the method stops at a relative gap of 10^{-6}.

Since ng-routes relax elementary routes, a zero gap proves optimality for the elementary problem. This is standard branch-and-price machinery; its purpose here is to remove the inexact integer step, not to add a branching rule. Column generation [71, 72] with exact pricing returns the exact LP bound, which also gives every integer plan a valid optimality gap.

4.3. The equilibrium engine

For a given charge, a pure equilibrium of the D1 game is computed in two phases. In the first phase, zone-market clearing (ZMC) queries every carrier's oracle at a common zone-price vector for eight rounds, using the sign of the clearing residual to update the prices (Appendix C). A restricted master over the collected plans and derandomised rounding give a start profile. In the second phase, sequential best responses (one oracle call each) run until no carrier improves; since D1 has an exact potential (Proposition 26) this terminates at a pure Nash equilibrium. Each carrier's column pool is kept across calls, because route costs do not depend on prices. In the city experiments a carrier chooses node-disjoint routes from its depot over 22 customer zones; each served zone adds its whole lot (all-or-nothing) together with prize, own Cournot and congestion terms, charges and prices as node costs, and each route costs its BRL upper expectation.

Remark 5 (ZMC and its negative result). ZMC has a clearing identity that bounds the potential gap and each carrier's regret by the clearing error, and a price-bracketing rate that does not depend on d (Appendix C). On synthetic games with non-separable plan sets, it gave the smallest potential gap in 17 of 20 instances at an equal call budget (Table F.17). On the nine city games it did not beat plain best response: it used 36–65 calls, against 8–17 for best response, and was strictly better than all other methods in 0 cities (Section 7.2). We use it as the starting phase of the engine and make no claim for it.

4.4. Dual implementation pricing (DIP)

A direct approach searches a grid of charges and solves one equilibrium per grid point; with 63 points this costs between 1 921 and 2 419 oracle calls per city and seed in the three cities where the exact grid is complete (Section 7.4.1). DIP replaces the search by an implementation step. It computes the profile the city wants and then computes charges under which that profile is an equilibrium, using the carriers' own oracle to separate candidate charges. No equilibrium is solved for the leader's decision.

Setting. We use the D1 instance of Section 3 in cost form. Carrier i serves a subset of its customer zones with all-or-nothing lots $d_{iz} > 0$, so $q_{iz} \in \{0, d_{iz}\}$ and $q_{iz}^2 = d_{iz} q_{iz}$; vehicle-hours are $\tau_{iz} = r\, q_{iz}$ for one ratio $r > 0$ (assumption (S3)); $R_i(p) \geq 0$ is the robust routing cost of plan p; $g_z = \gamma c_z$. The city posts s_z per delivery and t_z per vehicle-hour. Carrier i's cost and the social cost (charges are transfers) are

$$J_i(p; \sigma_{-i}) = R_i(p) + \sum_z \big[q_{iz}(-\alpha_z + b_z(q_{iz} + Q_{-i,z}) + s_z) + \tau_{iz}(g_z(\tau_{iz} + X_{-i,z}) + t_z) \big],$$

$$\mathrm{SC}(\sigma) = \sum_i R_i(\sigma_i) + \sum_z \big[-\alpha_z Q_z + \tfrac{b_z}{2} Q_z^2 + g_z X_z^2 \big].$$

Write $\kappa_z = g_z r^2$ (congestion slope in delivery units), $m_z = b_z + \kappa_z$ (the slope the carriers face) and $\bar{d}_z$ for the mean lot in zone z.

Proposition 6 (One instrument per zone)**.** *(a) For every plan, $\sum_z (s_z q_{iz} + t_z \tau_{iz}) = \langle u, q_i \rangle$ with $u_z = s_z + r t_z$, so the game depends on the charges only through $u \in \mathbb{R}^Z$. (b) Every u is realised by a uniform per-delivery charge $s = \min_z u_z$ over zones with lots and non-negative zone congestion charges $t_z = (u_z - s)/r$. (c) The grid of Section 7.4 is the two-parameter family $u = s\mathbf{1} + \vartheta\, r\, g \odot X^{\mathrm{NE}}$.*

Under u the game has the exact potential $\Phi_u(\sigma) = \sum_i R_i(\sigma_i) + \sum_z [-\alpha_z Q_z + \frac{m_z}{2}(Q_z^2 + \sum_i q_{iz}^2) + u_z Q_z]$ (Proposition 26).

Lemma 7 (Potential–welfare identity)**.** *Fix $Q^\circ \in \mathbb{R}^Z$, let $u_z^{\tan} = \kappa_z Q_z^\circ - \frac{m_z}{2} \bar{d}_z$ and $v = u - u^{\tan}$. For every pure profile σ,*

$$\Phi_u(\sigma) = \mathrm{SC}(\sigma) - \sum_z \tfrac{\kappa_z}{2}(Q_z - Q_z^\circ)^2 + L(\sigma) + \langle v, Q \rangle + \sum_z \tfrac{\kappa_z}{2}(Q_z^\circ)^2, \qquad L(\sigma) = \sum_{i,z} \tfrac{m_z}{2}(d_{iz} - \bar{d}_z) q_{iz}.$$

Algorithm 2 DIP: dual implementation pricing

Target. Column generation on the convexified social cost (the carriers' oracle at social prices), rounding and coordinate polish $\Rightarrow$ pure profile σ^* with aggregate Q^*
Reference. $u^{\mathrm{ref}} \leftarrow \kappa \odot Q^* - \frac{m}{2} \odot \bar{d}$ (tangent charge); cuts $\mathcal{C} \leftarrow \emptyset$; $u \leftarrow u^{\mathrm{ref}}$
repeat
    for each i: $b_i \leftarrow$ best response to σ^*_{-i} under $u \triangleright N$ routing calls, no equilibrium solve
    for each i with $J_i^u(b_i) < J_i^u(\sigma_i^*)$: add $\langle q_i(b_i) - q_i(\sigma_i^*), u\rangle \geq J_i^0(\sigma_i^*) - J_i^0(b_i) + \varepsilon$ to $\mathcal{C}$
    if no cut was added: **return** u $\quad\triangleright\ \sigma^*$ is an exact pure equilibrium under u
    $u \leftarrow \arg\min\{\|u - u^{\mathrm{ref}}\|_1 : u \in \mathcal{U} \text{ satisfies } \mathcal{C}\}$ (LP); **if** infeasible: **break**
until
Fallback (only if $P(\sigma^*) = \emptyset$): best-response dynamics from σ^* under the least-violating charge and under u^{ref}; keep the better equilibrium
Instruments. $s = \min_z u_z$, $t_z = (u_z - s)/r$ (Proposition 6)

The *tangent charge* $u^{\tan}$ is the classical corrective tax of a Cournot market with an externality: the marginal external damage at the target aggregate minus a market-power subsidy [7, 58]. Two details are specific to all-or-nothing lots: the subsidy is half a lot, $\frac{m_z}{2}\bar{d}_z$ per delivery, and it includes the carriers' own congestion. With carrier-specific charges $U_{iz} = \kappa_z Q^*_{-i,z} - \frac{b_z}{2} d_{iz}$ every carrier's cost differences equal the social-cost differences, so every social optimum is an equilibrium; this is the classical marginal-contribution charge [11, 66] (Proposition 56 in Appendix D). A city cannot charge per carrier, and the next results measure what flat zone charges lose. In the convexified (mixed) game let x^* minimise the convexified social cost $\overline{\mathrm{SC}}$, with aggregate $\bar{Q}^*$.

Theorem 8 (Loss under asymmetric lots)**.** *Let $u = \kappa \odot \bar{Q}^* - \frac{m}{2} \odot \bar{d}$. Every convexified equilibrium x^u satisfies $0 \leq \overline{\mathrm{SC}}(x^u) - \overline{\mathrm{SC}}(x^*) \leq (1+\varrho)E$, where $\varrho = \max_z \kappa_z/(b_z + \kappa_z) < 1$ and $E = \sum_{i,z} \frac{m_z}{2} |d_{iz} - \bar{d}_z|\, d_{iz}$ measures how unequal the lots in a zone are.*

Theorem 9 (Exact implementation with symmetric lots)**.** *If all lots in a zone are equal, the tangent charge makes x^* a convexified equilibrium, and every convexified equilibrium under it has aggregate $\bar{Q}^*$ and social cost $\overline{\mathrm{SC}}(x^*)$.*

Sketch. The identity holds for the convexified potential and social cost with the same algebra. Because $\overline{\mathrm{SC}}$ is linear plus a quadratic in $\bar{Q}$, its expansion around x^* is exact; comparing it with the identity at a minimiser of the potential bounds the aggregate deviation by the lot-asymmetry term L, and $|L| \leq E$. With equal lots $L \equiv 0$.

The pure game. For a pure target σ^* and an admissible box $\mathcal{U}$ of charges let

$$P(\sigma^*) = \big\{u \in \mathcal{U} :\ \langle q_i(p) - q_i(\sigma_i^*), u\rangle \geq J_i^0(\sigma_i^*; \sigma^*_{-i}) - J_i^0(p; \sigma^*_{-i})\ \ \forall i,\ \forall p \in \mathcal{P}_i\big\},$$

with J_i^0 the cost at zero charges. Since $J_i^u = J_i^0 + \langle u, q_i\rangle$, $P(\sigma^*)$ is exactly the set of charges under which σ^* is a pure Nash equilibrium: a polyhedron, possibly empty. Toll polyhedra of this kind are classical for nonatomic routing [8, 59] and inverse optimisation [9].

Theorem 10 (Separation and finite termination)**.** *(a) $u \in P(\sigma^*)$ if and only if no carrier has a strictly improving best response to σ^*_{-i} under u; an improving best response gives that carrier's*

most violated constraint. (b) With $\varepsilon = 0$, Algorithm 2 stops after at most $1 + \sum_i (|\mathcal{P}_i| - 1)$ rounds, either with $u \in P(\sigma^*)$ or with an infeasible LP, which happens only if $P(\sigma^*) = \emptyset$. Each round costs N oracle calls and no equilibrium solve.

Proposition 11 (Single-level form)**.** *With optimistic selection, the leader's value is $V^\star = \min\{\mathrm{SC}(\sigma) : \sigma \text{ pure},\ P(\sigma) \neq \emptyset\} \geq \min_\sigma \mathrm{SC}(\sigma)$. If Algorithm 2 ends in its main loop, the leader's loss is at most the suboptimality of the target, $\mathrm{SC}(\sigma^*) - \min_\sigma \mathrm{SC}(\sigma)$; in particular DIP is at least as good as any grid over any subfamily of charges when its target is a social optimum.*

Theorem 12 (Loss of the fallback)**.** *For any charge u with $v = u - u^{\tan}$ and $Q^\circ = Q^*$, the equilibrium σ^u reached from σ^* by improving moves satisfies $\mathrm{SC}(\sigma^u) - \mathrm{SC}(\sigma^*) \leq \sum_z \frac{\kappa_z}{2}(Q^u_z - Q^*_z)^2 + L(\sigma^*) - L(\sigma^u) + \langle v, Q^* - Q^u \rangle$.*

Corollary 13 (Oracle calls)**.** *DIP uses the calls of the social target, N calls per cutting-plane round, and, only if $P(\sigma^*)$ is empty, the calls of the fallback dynamics. A grid of G points with an equilibrium solve of C_{eq} calls per point uses GC_{eq}.*

Theorem 14 (Imprecision shifts the charge)**.** *Under Γ-maximin credal boxes of width δ (Lemma 25), the charges that implement σ^* are $P(\sigma^*) - \hat{s}(\delta)$ with $\hat{s}_z(\delta) = \delta(b_z w_{(z,Q)} + r\, g_z w_{(z,X)})$. One run of DIP therefore gives $u(\delta) = u(0) - \hat{s}(\delta)$ for every width at no further oracle cost, and the leader attains the same social cost at every width as long as $u(\delta)$ is admissible. For E-admissibility, maximality and interval dominance, $u(0)$ works at every width.*

Proposition 15 (When one instrument suffices)**.** *A uniform per-delivery charge implements the convexified optimum under equal lots if the tangent charge is the same in every active zone. With the calibration of Section 6 ($\alpha_z = 1.5\,$fee, $b_z = \alpha_z/(2Q_z^{\mathrm{full}})$) and n_z carriers with equal lots in zone z, the congestion-free tangent charge is $u_z = -0.375\,\mathrm{fee}/n_z$: a subsidy of 37.5% of the fee in single-carrier zones and 19% in two-carrier zones. The congestion charge contributes only spatial differentiation.*

Proofs are in Appendix D. Theorem 10(a) is what makes the method cheap: the separation oracle is one best response per carrier at the fixed target, not an equilibrium solve. Selection remains optimistic: the target is an exact equilibrium under the returned charges, but other equilibria may coexist.

Prior art and the claim. The tangent charge is the Pigou–Cournot corrective tax [7, 58]; carrier-specific charges are marginal-contribution utilities [11, 66]; toll polyhedra and implementing a target flow are classical [8, 59, 60, 61]; the loop is an inverse-optimisation cutting plane [9, 62, 63] with best-response cuts as used for integer programming games [64]. The closest leader-side method [67] needs an equilibrium solve per outer step. What remains new is modest: the exact potential–welfare identity of the lumpy Cournot–congestion game and the loss bounds it gives with the lot-asymmetry number E; the use of the carriers' own NP-hard best responses at the fixed target as the separation step; the translation of the implementing charges by the Γ-maximin surcharge; and the competitor-count rule.

Algorithm 3 ARP flat rule

Require: slopes b, g, r; damping θ

1: $u \leftarrow 0$; observe (Q, n) of the unpriced equilibrium
2: **repeat**
3: update the lot estimates $\bar{d}_z$ from (Q, n) and the previous observations
4: $u_z \leftarrow (1-\theta)u_z + \theta[\kappa_z Q_z - \frac{m_z}{2}\bar{d}_z]$ for all z
5: post u; the carriers best-respond from their current plans until nobody moves; observe (Q, n)
6: **until** $u = \kappa Q - \frac{m}{2}\bar{d}$ (fixed point) or the posted charge and the market state repeat (cycle)

4.5. Aggregate-response pricing (ARP)

DIP and the grid need the carriers' routing oracle, which a real city does not have. ARP asks how much of the leader's gain is reachable from what the city can observe. The city knows its own demand and congestion slopes (b, g, r) and hence $\kappa_z = g_z r^2$ and $m_z = b_z + \kappa_z$. It posts a per-delivery-equivalent charge $u \in \mathbb{R}^Z$ (realised, as in Proposition 6, by a uniform per-delivery charge plus non-negative zone congestion charges), the carriers settle in a pure equilibrium by their own best responses, and the city observes the deliveries Q_z and the number n_z of active carriers in each zone, nothing else.

The flat rule. With an estimate $\bar{d}_z$ of the mean lot, the city updates

$$u_z^{t+1} = (1-\theta_z)u_z^t + \theta_z\big[\kappa_z Q_z^t - \tfrac{m_z}{2}\bar{d}_z\big], \qquad \theta_z \in (0, 1]. \tag{5}$$

With $\theta = 1$ this is DIP's tangent charge evaluated at the observed aggregate instead of a computed target. The lot estimate is Q_z/n_z; when n_z changes by one between observations, the jump of Q_z is the lot of the carrier that entered or left (Lemma 59); for a zone never served nothing is learned and the city-wide mean observed lot is used.

The marginal tariff (ARP-M). The flat rule charges each delivery the externality at the observed aggregate. The exact externality of carrier i's lot on the others is $\kappa_z d_{iz} Q_{-i,z}$, and its market-power wedge is $\frac{b_z}{2}d_{iz}^2$; both are functions of the carrier's own deliveries and of the zone total, so an anonymous nonlinear tariff can bill them exactly, in the limit of re-tangenting at every transaction.

Theorem 16 (Marginal tariff; the mechanism is classical)**.** *Let every carrier pay, in every zone, $T_z(q_{iz}, Q_z) = \kappa_z q_{iz}(Q_z - q_{iz}) - \frac{b_z}{2}q_{iz}^2$. Then (a) $J_i^T(p; \sigma_{-i}) = \mathrm{SC}(p, \sigma_{-i}) - \mathrm{SC}(\emptyset_i, \sigma_{-i})$ for all i, p and σ_{-i}, so* SC *is an exact potential of the tariff game; (b) the pure equilibria are exactly the profiles that no single carrier can improve socially, every social optimum is a pure equilibrium, and best-response dynamics from any profile never increase* SC *and stop at such an equilibrium; (c) given the rivals, the tariff is linear in the carrier's own service indicators, so a best response is one oracle call with the zone prices shifted by $\kappa_z Q_{-i,z} - \frac{b_z}{2}d_{iz}$.*

The mechanism is the marginal-contribution tariff of Wolpert and Tumer [11], Sandholm [12] and Marden and Wierman [66]; what is stated here is its lumpy form in the city game and the consequence that DIP's implementation step is unnecessary under it. The tariff needs billing on own deliveries and the zone total. It leaves equilibrium selection open: the carriers'

dynamics stop at a unilateral social optimum, which may be a local optimum of SC. In the experiments we run it with the carriers' own best responses from the status quo (tariff+BR) and with the potential-minimising solver of the engine on the social-cost potential (tariff+SO).

Theory of the flat rule. Convexify the plan sets: let $\mathcal{X} = \prod_i \Delta(\mathcal{P}_i)$, $\bar{q}_i(x_i) = \mathbb{E}_{x_i} q_i$, $\bar{Q}(x) = \sum_i \bar{q}_i$ and $\mathcal{Q} = \bar{Q}(\mathcal{X})$. With a reference lot vector $\bar{d}$ define the carriers' aggregate value function $G(Q) = \min\{\sum_i \mathbb{E}_{x_i}[R_i + \sum_z \frac{m_z}{2}(d_{iz} - \bar{d}_z)q_{iz}] : \bar{Q}(x) = Q\}$ and $F(Q) = G(Q) + \sum_z[-\alpha_z Q_z + \frac{m_z}{2}Q_z^2 + \frac{m_z}{2}\bar{d}_z Q_z] + \iota_{\mathcal{Q}}(Q)$, where R_i is the routing cost. Let $L_{\bar{d}}(\sigma) = \sum_{i,z} \frac{m_z}{2}(d_{iz} - \bar{d}_z)q_{iz}$ be the lot-asymmetry term of Section 4.4 with reference lots $\bar{d}$, $E_{\bar{d}} = \sum_{i,z} \frac{m_z}{2}|d_{iz} - \bar{d}_z|d_{iz}$, and $g(Q) = \sum_z[\frac{\kappa_z}{2}Q_z^2 - \frac{m_z}{2}\bar{d}_z Q_z]$. All statements below are proved in Appendix E; $\|v\|_w^2 = \sum_z w_z v_z^2$.

Theorem 17 (The response is a proximal map)**.** *F is 1-strongly convex in $\|\cdot\|_m$. For every charge u, all convexified equilibria share one aggregate $Q(u) = \arg\min_Q\{F(Q) + \langle u, Q\rangle\} = \nabla F^*(-u)$, and $\|Q(u_1) - Q(u_2)\|_m \le \|u_1 - u_2\|_{m^{-1}}$. The leader's problem $\min \overline{\mathrm{SC}} + L_{\bar{d}}$ is the Fenchel dual $\min_u D(u)$, $D(u) = F^*(-u) + \sum_z (u_z + \frac{m_z}{2}\bar{d}_z)^2/(2\kappa_z)$, whose gradient $\kappa^{-1} \odot (u + \frac{m}{2}\bar{d}) - Q(u)$ is observable from the aggregate alone. Its unique minimiser u^* is the unique fixed point of $u \mapsto \kappa \odot Q(u) - \frac{m}{2} \odot \bar{d}$. With symmetric lots within zones the fixed point implements a convexified social optimum; in general it is within $E_{\bar{d}}$ of one.*

Theorem 18 (Instance-free contraction)**.** *Let $\varrho = \max_z \kappa_z/m_z < 1$ over zones with a lot. In the convexified game, (a) the plain rule ($\theta = 1$) satisfies $\|u^{t+1} - u^*\|_{\kappa^{-1}} \le \varrho\|u^t - u^*\|_{\kappa^{-1}}$; (b) the damped rule with $\theta = 2/(2+\varrho)$ contracts at ratio $\varrho/(2+\varrho) < 1/3$. The ratios depend on the instance only through ϱ, not on Z, N, the number of customers, the lots, the routing costs or the probability boxes.*

The rate comes from the Cournot slope b_z: without the demand side ($b = 0$), $\varrho = 1$ and there is no contraction, which is why the nonatomic literature uses averaging [10]. The flat rule is gradient descent on D in the κ^{-1} geometry; this identification, and the resulting rate in a lumpy NP-hard routing game, is the new part.

Theorem 19 (Pure game and identification)**.** *(a) If σ is a pure equilibrium under $u = \kappa \odot Q(\sigma) - \frac{m}{2} \odot \bar{d} + v$, then for every carrier i and plan p, with $\sigma' = (p, \sigma_{-i})$ and $\Delta Q = Q(\sigma') - Q(\sigma)$, $\mathrm{SC}(\sigma') \ge \mathrm{SC}(\sigma) + [L_{\bar{d}}(\sigma) - L_{\bar{d}}(\sigma')] + \sum_z \frac{\kappa_z}{2}\Delta Q_z^2 - \langle v, \Delta Q\rangle$: at an exact fixed point with symmetric lots, no carrier can lower the social cost unilaterally. (b) If the observed aggregate Q^t differs from the convexified response by e^t, then $\|u^{t+1} - u^*\|_{\kappa^{-1}} \le \varrho\|u^t - u^*\|_{\kappa^{-1}} + \|e^t\|_\kappa$, hence $\limsup \|u^t - u^*\|_{\kappa^{-1}} \le \sup \|e^t\|_\kappa/(1-\varrho)$; if the carriers settle in potential minimisers, $\|e^t\|_\kappa \le \sqrt{2\varrho\eta_t}$ with η_t the rounding gap of the potential. (c) The convexified response map determines F up to a constant, so the convexified optimum is all that an observation-only scheme can target; the pure social optimum is not identified (Example 58).*

Proposition 20 (Imprecise carriers)**.** *Under Γ-maximin beliefs of width δ the response to u is the precise response to $u + \hat{s}(\delta)$ (Lemma 25), so the flat rule needs no knowledge of δ and contracts at the same rates; it only moves the fixed point, and the loss in $\overline{\mathrm{SC}} + L_{\bar{d}}$ against the precise fixed point is at most $\frac{1}{2}\|\hat{s}(\delta)\|_{m^{-1}}^2$. Under E-admissibility of width δ, a pure target is implemented exactly once the posted charge is within the surcharge band $|u_z - u_z^*| \le \hat{s}_z(\delta)$ of an implementing charge (with a side condition on rival volumes, Appendix E), after at most $\lceil \log(\cdot)/\log(1/\varrho) \rceil$ rounds.*

Prior art and the claim. The central step, charge the externality at the observed aggregate and iterate, is classical: trial-and-error marginal-cost pricing [10], evolutionary implementation of pricing [12], and, for atomic Cournot games with continuous strategies, adaptive incentive design [13, 16], whose fixed points are socially optimal. Marginal-contribution tariffs are classical. What is new, as far as we found, is the proximal-map identification and the instance-free rates in the lumpy routing game (Theorems 17 and 18), the fixed-point and identification results for the pure game (Theorem 19), and the absorption of the leader's error by the carriers' credal width (Proposition 20). We do not claim a guarantee on which pure equilibrium the carriers select; Theorem 19(b) is conditional on potential minimisation.

5. Equilibrium theory

This section states the results on pure and imprecise equilibria; complete proofs are in Appendix A. Results labelled *classical* are known and cited.

5.1. Imprecise beliefs

Assumption A4 (Traffic box)**.** $\Xi = \prod_z [\underline{\mu}_z, \overline{\mu}_z] \subset \mathbb{R}^Z_{\geq 0}$ with widths $\varpi_z = \overline{\mu}_z - \underline{\mu}_z$; a nominal mean $\hat{\mu} \in \Xi$ is fixed and the true mean μ lies in Ξ.

Assumption A5 (Credal sets, clipped and consistent)**.** Fix weights $w \in \mathbb{R}^d_{>0}$ and a width $\delta \geq 0$. For $c \in \mathbb{R}^d_{\geq 0}$ let $K^\delta(c) = \{m \in \mathbb{R}^d_{\geq 0} : |m_k - c_k| \leq \delta w_k \ \forall k\}$. At a pure profile σ, carrier i's credal set $\mathcal{M}_i^\delta(\sigma)$ is the set of Borel probability laws P on $\mathbb{R}^d_{\geq 0} \times \mathbb{R}^Z_{\geq 0}$ with finite first moments such that $\mathbb{E}_P[m] \in K^\delta(A_{-i}(\sigma))$ and $\mathbb{E}_P[\xi] \in \Xi$.

The box is centred at the realised rival aggregate, which is the consistency requirement of equilibrium in beliefs [51], and it is clipped at zero, so every belief is a feasible aggregate. The width δ measures how much a carrier is unsure about its rivals' activity, in units w_k.

Definition 21 (Decision criteria; 73)**.** For a finite option set $\mathcal{P}$ and a set $\mathcal{M}$ of laws, with $\underline{\mathbb{E}} = \inf_{P \in \mathcal{M}} \mathbb{E}_P$ and $\overline{\mathbb{E}} = \sup_{P \in \mathcal{M}} \mathbb{E}_P$, an option p is *E-admissible* if it maximises $\mathbb{E}_P[u(\cdot)]$ for some $P \in \mathcal{M}$; *maximal* if no p' has $\underline{\mathbb{E}}[u(p') - u(p)] > 0$; *interval-undominated* (ID) if $\overline{\mathbb{E}}[u(p)] \geq \max_{p'} \underline{\mathbb{E}}[u(p')]$; and Γ-*maximin* if $\underline{\mathbb{E}}[u(p)] = \max_{p'} \underline{\mathbb{E}}[u(p')]$.

Definition 22 (Imprecise credal Nash equilibrium)**.** For $C \in \{E, M, ID, \Gamma\}$ and $\delta \geq 0$, $\mathrm{ICNE}_C(\delta)$ is the set of pure profiles σ in which every σ_i is C-admissible for $(\mathcal{P}_i, \mathcal{M}_i^\delta(\sigma))$ with payoff u_i. The existence threshold is $\delta_0^C = \inf\{\delta \geq 0 : \mathrm{ICNE}_C(\delta) \neq \emptyset\}$.

5.2. Reduction to mean boxes and the criteria

Write $\bar{u}_i(p; m, \xi)$ for the payoff of Assumption A2 evaluated at the means (m, ξ) and $K_i^\delta(\sigma) = K^\delta(A_{-i}(\sigma)) \times \Xi$.

Lemma 23 (Reduction of the credal set to its mean box)**.** *Under Assumptions A1–A5: (a) for every $P \in \mathcal{M}_i^\delta(\sigma)$ and every p, $\mathbb{E}_P[u_i(p)] = \bar{u}_i(p; \mathbb{E}_P m, \mathbb{E}_P \xi)$; (b) $P \mapsto (\mathbb{E}_P m, \mathbb{E}_P \xi)$ maps $\mathcal{M}_i^\delta(\sigma)$ onto $K_i^\delta(\sigma)$; (c) hence each criterion of Definition 21 coincides with the same criterion evaluated over the compact box $K_i^\delta(\sigma)$, and infima and suprema are attained.*

Lemma 24 (Nesting and monotonicity)**.** *For every $\delta \geq 0$,*

$$\mathrm{ICNE}_E(\delta) \subseteq \mathrm{ICNE}_M(\delta) \subseteq \mathrm{ICNE}_{ID}(\delta) \quad \textit{and} \quad \mathrm{ICNE}_\Gamma(\delta) \subseteq \mathrm{ICNE}_M(\delta).$$

For $C \in \{E, M, ID\}$ and $\delta \leq \delta'$, $\mathrm{ICNE}_C(\delta) \subseteq \mathrm{ICNE}_C(\delta')$. For $C = \Gamma$ the monotonicity fails (Example 40 in Appendix A).

The inclusions between criteria are due to Troffaes [73]. Monotonicity in δ says that more imprecision can only enlarge the set of E-admissible, maximal or interval-undominated equilibria; Γ-maximin behaves differently because the worst case moves with δ.

Lemma 25 (Γ-maximin surcharge)**.** *If $\varphi_i(p), t_i(p) \geq 0$, the lower expectation over $K^\delta(c) \times \Xi$ is attained, for every plan simultaneously, at the upper corner $(c + \delta w, \overline{\mu})$ and is linear in mixtures. Consequently $\mathrm{ICNE}_\Gamma(\delta)$ is the set of pure Nash equilibria of the precise game G^Γ_δ with private profits*

$$\pi_i^\Gamma(p) = \pi_i(p) - \delta\langle \varphi_i(p), Mw\rangle - \gamma\langle t_i(p), \overline{\mu}\rangle,$$

the same coupling, and traffic set to zero.

The surcharge $\delta\langle \varphi_i(p), Mw\rangle$ is what a cautious carrier adds to the cost of activity in zone k because it fears up to δw_k more rival activity than it observes. Section 7 uses it to compute Γ-maximin equilibria of the city games.

5.3. Existence under exposure-based congestion

Proposition 26 (Existence under D1 at every width)**.** *Under D1, for every $\hat{\xi} \in \Xi$,*

$$\Psi_{\hat{\xi}}(\sigma) = \textstyle\sum_i \left[\pi_i(\sigma_i) + \frac{1}{2}\|a_i(\sigma_i)\|_M^2 - \gamma\langle t_i(\sigma_i), \hat{\xi}\rangle\right] - \frac{1}{2}\|A(\sigma)\|_M^2$$

is an exact potential of the precise game at traffic $\hat{\xi}$. Every maximiser lies in $\mathrm{ICNE}_E(\delta)$ for all $\delta \geq 0$, so $\delta_0^E = \delta_0^M = \delta_0^{ID} = 0$, and $\mathrm{ICNE}_\Gamma(\delta) \neq \emptyset$ for all $\delta \geq 0$.

Sketch. For a unilateral change of carrier i from contribution a to a' with rival aggregate R, the quadratic terms give $\frac{1}{2}\|a'\|_M^2 - \frac{1}{2}\|a\|_M^2 - \frac{1}{2}\|R + a'\|_M^2 + \frac{1}{2}\|R + a\|_M^2 = -\langle a' - a, MR\rangle$, which equals the change of the coupling term when $\varphi_i = a_i$. A maximiser of an exact potential is a pure equilibrium [43], and the centre of every credal box is the realised aggregate. For Γ, G^Γ_δ is again of type D1.

5.4. Credal absorption

Let x^* be a mixed equilibrium of the precise game at traffic $\hat{\xi} \in \Xi$ (it exists by 74; under affine coupling, mixed equilibria are exactly the profiles in which every plan in the support is a best response to the mean rival aggregate, Proposition 41). Write B_j for carrier j's best-response set at x^*, $\bar{a}_j = \sum_p x_j^*(p) a_j(p)$, $\bar{A}^*_{-i} = \sum_{j \neq i} \bar{a}_j$, and define the spreads $s_{jk} = \max_{p,p' \in B_j} |a_{jk}(p) - a_{jk}(p')|$ and $r_{jk} = \max_{p \in B_j} |a_{jk}(p) - \bar{a}_{jk}|$, with $\bar{\sigma} = \max_{j,k} s_{jk}/w_k$, $\bar{\rho} = \max_{j,k} r_{jk}/w_k$, $d_L = \dim \operatorname{span} \bigcup_j (a_j(B_j) - a_j(B_j))$ and $k^\sharp = \min\{N, d_L\}$. Let $\sigma_{\max} = \max_{i,k} \operatorname{range}_{p \in \mathcal{P}_i} a_{ik}(p)/w_k$, so that $\bar{\rho} \leq \bar{\sigma} \leq \sigma_{\max}$. The *anchored radius* of a pure profile $p \in \prod_j B_j$ is $\hat{\delta}(p) = \max_i \|A_{-i}(p) - \bar{A}^*_{-i}\|_{w,\infty}$.

Theorem 27 (Credal absorption)**.** *Assume A1–A5 and let x^* be a mixed equilibrium at $\hat{\xi} \in \Xi$.*

(a) Absorption. *For every* $p \in \prod_j B_j$ *and every* $\delta \geq \hat{\delta}(p)$, $p \in \mathrm{ICNE}_E(\delta) \subseteq \mathrm{ICNE}_M(\delta) \subseteq \mathrm{ICNE}_{ID}(\delta)$.

(b) Radius. $\min_{p \in \prod_j B_j} \hat{\delta}(p) \leq \min\big\{ \max_i \max_k \sum_{j \neq i} r_{jk}/w_k,\ k^\sharp \bar{\sigma} + \bar{\rho},\ \bar{\sigma}\sqrt{k^\sharp \ln(2d)/2} + \bar{\rho} \big\}$.

(c) Uniform existence. $\delta_0^{ID} \leq \delta_0^M \leq \delta_0^E \leq \sigma_{\max} \min\{N-1,\ d+1,\ 1+\sqrt{d\ln(2d)/2}\}$.

(d) Construction. *A profile achieving the second bound is obtained from* x^* *by one linear program (a vertex of the polytope of Shapley–Folkman decompositions) followed by an arbitrary choice for at most* $k^\sharp$ *carriers; one achieving the third bound by independent sampling of these carriers, with success probability at least* $1 - 2d\exp(-2t^2/(k^\sharp \bar{\sigma}^2))$ *per draw for target* $t + \bar{\rho}$*; one achieving the first bound is any* $p \in \prod_j B_j$.

Sketch. Every carrier in $p \in \prod_j B_j$ plays a best response to the *anchor* $\bar{A}^*_{-i}$, which is nonnegative. If $\delta \geq \hat{\delta}(p)$, the anchor lies in the clipped box around the realised aggregate $A_{-i}(p)$, so the Dirac law at the anchor belongs to the credal set and witnesses E-admissibility. The radius bounds follow from the decomposition $A_{-i}(p) - \bar{A}^*_{-i} = \sum_{j \in J}(a_j(p_j) - y_j) - (a_i(p_i) - \bar{a}_i)$, where J is the set of at most $k^\sharp$ carriers left mixed at a vertex of the Shapley–Folkman polytope (Theorem 39); the sum is bounded deterministically by $k^\sharp \bar{\sigma}$ or, after independent sampling, by Hoeffding's inequality [75] and a union bound. The own term $a_i(p_i) - \bar{a}_i$ cannot be removed (it is the "+1" in the bound of 14).

Corollary 28 (ε-Nash in the precise game; cf. 14)**.** *A profile* p *as in Theorem 27(a) is an* ε*-Nash equilibrium of the precise game at any* $\mu \in \Xi$ *with* $\varepsilon_i \leq \hat{\delta}(p) L_i(p) + \gamma \sum_z \varpi_z T_{iz}(p)$*, where* $L_i(p) = \max_{p'} \sum_k w_k M_k |\varphi_{ik}(p') - \varphi_{ik}(p_i)|$ *and* $T_{iz}(p) = \max_{p'} |t_{iz}(p') - t_{iz}(p_i)|$.

Remark 29 (Novelty and regime). Corollary 28 restates, in our notation, the N-independent approximate equilibrium of Liu et al. [14], who also use a Shapley–Folkman argument on the generators of a convexified equilibrium; we claim no novelty for it. What Theorem 27 adds is that under E-admissibility the rounding error is *absorbed* by the belief width: the rounded profile is an *exact* imprecise equilibrium, not an approximate one. The proof of part (a) is short; the content of the result lies in the bound, its attainment (Proposition 31), its failure for Γ-maximin (Remark 30), and the contrast with the complexity result below. In the city instances $N \leq 6$ and $d = 2Z \geq 84$, so $k^\sharp = N$, the Shapley–Folkman step selects nothing, and the operative bound is the support-selection bound $(N-1)\bar{\rho}$. Shapley–Folkman matters only when the number of players exceeds the dimension, for example with per-vehicle or crowdsourced players.

Remark 30 (Γ-maximin). By Lemma 25, $\mathrm{ICNE}_\Gamma(\delta)$ is the pure-equilibrium set of a *precise* game, so absorption does not apply to it, and under D2 that set can be empty (Proposition 31 at $\delta = 0$).

5.5. Tightness

Proposition 31 (The support-selection bound is attained)**.** *Let* $N = 2$*, one zone, D2,* $M = \gamma c = 1$ *and weight* $w > 0$*. Carrier 1 has plans* $A_1 : (n, \tau, \pi) = (1, 4, 6)$ *and* $B_1 : (3, 1, 0)$*; carrier 2 has* $X : (1, 1, 0)$ *and* $Y : (3, 4, 6)$*; payoffs are* $\pi - \tau \cdot n_{\mathrm{rival}}$*. Then the game has no pure Nash equilibrium,* $\delta_0^E = \delta_0^M = 1/w$*, and the bound* $(N-1)\bar{\rho}$ *of Theorem 27(b) equals* $1/w$.

In each of the four pure profiles exactly one carrier has a strict incentive to deviate at the realised count, and it becomes indifferent only at a belief that differs from the realised count by one vehicle.

Proposition 32 (Linear dependence on the number of zones is necessary)**.** *For every $Z \geq 1$ and $N \geq 2$ there is a D2 game with Z zones in which every pure profile has $\max_i \varepsilon_i \geq 3\lceil Z/2 \rceil$, while every profile of Theorem 27(a) satisfies $\hat{\delta}(p) L_i(p) \leq 3Z$ $(w = 1)$.*

5.6. Complexity under vehicle-count congestion

Theorem 33 (NP-completeness under D2)**.** *Consider the problem of deciding $\mathrm{ICNE}_C(\delta) \neq \emptyset$ for a game given by explicit plan lists with rational data, M, w, a degenerate traffic box and a rational $\delta \geq 0$. For each $C \in \{E, M, ID, \Gamma\}$ the problem is in NP. For every digraph D with maximum out-degree $\Delta^+ \geq 1$ one constructs in polynomial time a D2 instance with two plans per carrier and one zone per arc such that, for all $\delta \in [0, 1/(2\Delta^+ + 1))$, $\mathrm{ICNE}_E(\delta) = \mathrm{ICNE}_M(\delta) = \mathrm{ICNE}_{ID}(\delta) = \mathrm{ICNE}_\Gamma(\delta)$ is the set of profiles that play "in" exactly on a kernel of D. Hence the problem is strongly NP-complete for each criterion, already at $\delta = 0$, and for every fixed $\delta \in [0, 1/5)$.*

Sketch. Deciding whether a digraph has a kernel (an independent set that absorbs every other vertex) is NP-complete [76], also for planar digraphs with in- and out-degree at most two [77]. Each vertex becomes a carrier with an "out" plan of zero data and an "in" plan that puts one vehicle on each out-arc zone and two vehicles on each in-arc zone, with exposure one on out-arcs and a small exposure on in-arcs. The gain of "in" is positive if no out-neighbour plays "in" and at most -1 otherwise, and the sign is constant on every credal box of width below $1/(2\Delta^+ + 1)$, so all four criteria select the realised strict best response.

Corollary 34 (Inapproximability of the existence threshold)**.** *Unless* P = NP*, no polynomial algorithm approximates δ_0^E of a D2 instance within any fixed factor, or within an additive error below $1/5$.*

Pure-equilibrium existence is known to be NP-complete in polymatrix games and integer programming games [15, 57]; Theorem 33 is a direct kernel reduction specialised to the rank-one D2 structure and extended to four credal criteria and a range of widths. Together with Proposition 26 it gives a complexity split between the two congestion regimes: charging or modelling congestion by exposure (vehicle-hours in a zone) yields a potential game with guaranteed pure imprecise equilibria, while counting vehicles does not. Under D2 the mixed equilibrium that Theorem 27 rounds is itself hard to compute: every bimatrix game embeds as a D2 instance with $N = 2$, so the step is PPAD-hard [78, 79] (Proposition 42).

5.7. Efficiency

Lemma 35 (Imprecise equilibria are approximate equilibria)**.** *If $\sigma \in \mathrm{ICNE}_C(\delta)$ and $\mu \in \Xi$, then σ is an ε-Nash equilibrium of the precise game with $\varepsilon_i \leq \delta L_i(\sigma) + \gamma \sum_z \varpi_z T_{iz}(\sigma)$ for $C \in \{E, M\}$ and $\varepsilon_i \leq 2\delta H_i + 2\gamma \Theta_i$ for $C = ID$, with $H_i = \max_p \sum_k w_k M_k \varphi_{ik}(p)$ and $\Theta_i = \max_p \sum_z \varpi_z t_{iz}(p)$.*

Proposition 36 (Monotone efficiency curves)**.** *For $C \in \{E, M, ID\}$ and $W > 0$,* $\mathrm{PoA}_C(\delta) = W^*/\min_{\mathrm{ICNE}_C(\delta)} W$ *is nondecreasing and* $\mathrm{PoS}_C(\delta) = W^*/\max_{\mathrm{ICNE}_C(\delta)} W$ *nonincreasing in δ, and both are step functions with at most $\prod_i |\mathcal{P}_i|$ steps. For $C = \Gamma$ neither monotonicity holds in general.*

Proposition 37 (Price of stability via the potential, D1)**.** *Under D1 with zero charges and $\mu = \hat{\mu}$, let σ^* maximise W. Every profile σ with $\Psi_{\hat{\mu}}(\sigma) \geq \max \Psi_{\hat{\mu}} - \eta$ satisfies $W(\sigma) \geq W^* - \sum_z \left[\frac{b_z}{2} \sum_i q_{iz}^{*2} + \frac{\gamma c_z}{2}(1 - \frac{1}{N}) T_z(\sigma)^2\right] - \eta$. In particular a potential maximiser, which lies in every* $\mathrm{ICNE}_E(\delta)$*, bounds* $\mathrm{PoS}_C(\delta)$ *for all δ and $C \in \{E, M, ID\}$.*

A smoothness-based price-of-anarchy bound also transfers to imprecise equilibria through Lemma 35 (Proposition 43), but it is conditional: we do not have smoothness constants for the Cournot–congestion game with arbitrary private routing costs. The price-of-anarchy values in Section 7 are therefore empirical.

5.8. Exact thresholds and the rounding algorithm

For any pure profile p, the exact E-threshold of carrier i,

$$A_i^E(p) = \min\{\delta : p_i \text{ is E-admissible at width } \delta\},$$

is the value of a linear program over the box, so $p \in \mathrm{ICNE}_E(\delta)$ if and only if $\delta \geq \max_i A_i^E(p)$ (Theorem 44). Theorem 27 and this threshold computation define an algorithm with four phases: relax (a mixed equilibrium by Frank–Wolfe on the concave relaxed potential under D1, or by a double oracle under D2; 80), sparsify (a vertex of the Shapley–Folkman polytope), round, and compute the exact width. Under D2 its output satisfies the bounds of Theorem 27(b) (Theorem 45), but its relaxation phase is PPAD-hard in general. Under D1 the relaxation-and-rounding technique is classical [81, 82, 83]. The full rounding algorithm was not run on the city games, which use the D1 engine of Section 4; the exact thresholds of Theorem 44 were used in the tests of Section 7.6.

Remark 38 (Nonaffine congestion). With a power congestion term the expected payoff under mixed rival strategies is no longer a function of the mean aggregate; Lemma 23 and the exact credal statements fail, although relaxed equilibria still exist. We do not treat this case.

5.9. What is machine-checked

A Lean 4 file with Mathlib (`lean/CCU/LeanTargets.lean` in the repository) formalises the finite aggregative game with clipped credal boxes and E-admissibility. It compiles with no unproved steps (no `sorry`), and every theorem in it depends only on the standard axioms `propext`, `Classical.choice` and `Quot.sound`. The following discrete core statements are machine-checked: E-admissibility implies maximality, and $\mathrm{ICNE}_E(\delta)$ is monotone in δ (Lemma 24, case E); the core of Theorem 27(a) (a profile whose plans are best responses to nonnegative anchors within the box is in $\mathrm{ICNE}_E(\delta)$); the Shapley–Folkman lemma for finite sets in a finite-dimensional real space, with the dimension of the space in place of d_L (Theorem 39; the formal proof lifts the sets with indicator coordinates and applies Carathéodory's theorem); the exact potential identity under D1 (Proposition 26); the combinatorial core of Theorem 33 (with integer-scaled gains, the equilibrium conditions hold exactly on kernels); the lower-bound core of Proposition 31; and the counting core of Proposition 32. The file covers the credal

and aggregative-game results only; BRL, DIP, ARP and the computational results are not formalised. Lean checks these core statements, not the full theorems: the radius bounds of Theorem 27(b)–(c), the Hoeffding step, the NP-completeness statement itself (Mathlib has no framework for polynomial-time reductions) and the algorithmic results are not formalised.

6. Data and calibration

Road networks and zones. For each of nine cities (New Delhi, Cairo, Moscow, Buenos Aires, New York, Mumbai, London, Paris, Berlin) we use an OpenStreetMap extract [84] distributed by BBBike (replication timestamp 16 October 2026). The drivable major-road network (motorway to tertiary, with links) is kept as an undirected graph with class-based free-flow speeds and lane counts (OSM `lanes` tag where present, class defaults otherwise). Population comes from the GHS-POP R2023A grid, epoch 2025, at 30 arc-seconds [85]. Zones are blocks of 4×4 raster cells (about 3.7 km) inside the extract's bounding box with at least one inhabitant; zone-to-zone free-flow times are shortest paths between population-weighted centroids plus a fixed access time. Points of interest from the same extracts give a depot score per zone used to place carrier depots. Table F.12 (Appendix F) summarises the instances; Z ranges from 42 to 593.

Operating parameters. Per-city parameters were collected from public web sources on 23 October 2026 and graded A (official or primary statistic), B (industry report or country-level proxy) or C (rough estimate or proxy from elsewhere). Table F.14 in Appendix F lists the values used; the repository file `data/parameters/SOURCES.md` gives source, URL, grade and note for each number. Congestion levels are those of the TomTom Traffic Index 2025 [86]; volume-delay curves use the BPR form with $\alpha = 0.15$, $\beta = 4$ [87], linearised at the observed congestion level.

Weak numbers. Parcels per capita are country-level figures. Driver wages are minimum-wage floors or job-site averages, not courier payroll. Drops per vehicle and van capacities outside the United States are proxies. Delivery fees in Cairo, Moscow and Buenos Aires are consumer or reference rates, not carrier revenue per parcel, and the European fees are regulator revenue per parcel. Moscow is absent from the 2025 TomTom index and uses a 2020 value. Market shares are available only for Paris and Berlin; elsewhere carriers have equal shares. The sensitivity of the equilibria to the five weakest parameters is in Table F.15.

Calibration rules and simplifications. Each carrier serves 22 customer zones sampled around its depot with weight proportional to population times a gravity decay in travel time, so carriers overlap in some zones. A lot is 10–50% of a vehicle's daily drops, proportional to the zone's parcel demand. Inverse demand has intercept $\alpha_z = 1.5\times$ fee and slope $b_z = \alpha_z/(2Q_z^{\text{full}})$. The congestion slope c_z is the derivative of the BPR curve at the TomTom level, with capacity from lane-kilometres, scaled by a slice multiplicity m_z (the ratio of the city's real delivery vehicle-hours to the modelled ones) so that the slice faces the congestion of the whole city. Travel-time multipliers of nine sectors follow a lognormal law matching the TomTom all-day and peak congestion; each carrier observes 30 days, from which DKW boxes at confidence $1 - \alpha$ are built. The route loss (3) uses the vehicle-hour cost (wage plus fuel) per hour, twice that as fixed cost, three times that per overtime hour, and the shift length as deadline. The simplifications are:

(S1) all-or-nothing lots: a carrier serves a customer zone fully or not at all, which makes the Cournot game lumpy;

(S2) the integer routing step: the main game experiments use branch-and-price with a time limit per call; some earlier experiments use price-and-branch (Section 7);

(S3) vehicle-hours are proportional to the lot, so within a zone a congestion charge on hours and a delivery charge are collinear;

(S4) linear congestion (D1);

(S5) oracle exactness at city scale is checked against an independent oracle, not proved.

Further proxies are the sampled customer zones instead of whole-city demand, equal market shares in seven cities, the slice multiplicity, and a *broad* social cost that adds the delay imposed on background traffic (population times vehicle ownership times one hour per day, valued at the wage, times the added slowdown, divided by m_z). The broad objective is reported but not used for conclusions.

Instances as a benchmark. The pipeline (extract download, zones, travel times, depots, parameters with grades, p-box construction, game instances with seeds) is in the repository and regenerates all instances from the raw extracts. The instances are small (22 customer zones per carrier, four to six carriers) so that exact oracles are possible; they are intended as a common test bed for algorithms on multi-carrier routing games, not as validated city models.

7. Computational study

Research questions.

RQ1 Is BRL pricing exact, and does its completion-window dominance reduce the work compared with other labelling rules? Does branch-and-price give exact integer plans on the city instances? (Section 7.1)

RQ2 How do equilibria computed with an exact oracle differ from those computed with price-and-branch, and how do the equilibrium engine's components compare? (Sections 7.2 and 7.3)

RQ3 How do DIP and ARP compare with the grid in oracle calls and in the leader's gain, and what do they lose? (Section 7.4)

RQ4 What do the equilibria, charges, price of anarchy and imprecision effects look like on the nine cities, and how stable are the findings over seeds? (Sections 7.3–7.5)

RQ5 Do the theoretical bounds hold on the city equilibria, and does vehicle-count congestion break pure equilibria in practice? (Section 7.6)

Protocol. All numbers are generated from the result files of the repository by `make_tables.py`. Experiments use three seeds (0, 1, 2). A seed redraws the depots, the customer zones of each carrier, the travel-time sectors and the 30 observed days, hence the probability boxes; calibration inputs are the same in every seed. With three seeds we report the mean with the minimum and maximum, or the values per seed, as a rough indication of spread, not a confidence interval. Seeds do not cover the uncertainty in the calibration inputs, which is examined separately (Table F.15).

Oracle and coverage. The base equilibria, the grid and DIP comparisons, the price of anarchy and the imprecision sweeps use the exact branch-and-price oracle with a limit of 20 seconds per call and a 120-second retry. Of 41,469 oracle calls, 41,125 (99.17%) were proven optimal; 344 remained unproven, with a remaining gap of at most 14 USD on plans worth hundreds to tens of thousands of USD. Coverage is partial in places (Table F.13, Appendix F):

- base equilibria: all 27 city-seeds;
- grid: 27 city-seeds, with the full 63-point exact grid only for New Delhi, Buenos Aires, London. For Cairo, Moscow, New York, Mumbai, Paris, Berlin only 7 to 17 grid points per seed were solved exactly, chosen by screening on the price-and-branch grid, so the grid's gains there are lower bounds (marked †);
- price of anarchy: 27 city-seeds; in New York, Moscow and Berlin the social-optimum benchmark of the broad objective used price-and-branch;
- imprecision: 27 city-seeds; all 12 widths for New Delhi, Buenos Aires, Mumbai, London, and 6 widths for Cairo, Moscow, New York, Paris, Berlin;
- DIP: all 27 city-seeds; for Berlin seed 2, Moscow seed 2, New York seed 2 the comparison grid is the price-and-branch grid;
- ARP: four cities (New Delhi, Buenos Aires, Mumbai, London), three seeds each (12 runs); Mumbai's grid is a reduced one.

The algorithm comparison of the engine (Table F.11), the ZMC checks, the theory tests of Section 7.6 and the tiny-game verification still use the price-and-branch oracle; the sensitivity scenarios use the exact oracle but are compared with a price-and-branch base case. Every reported equilibrium is a pure Nash equilibrium of the game with the exact oracle up to unproven calls (Section 7.7).

7.1. BRL and branch-and-price

Exactness. The upper-prevision formula agrees with a linear program over all joint laws on 40 random instances to a relative error of $4.5 \cdot 10^{-14}$. For scale, the comonotone worst case exceeds the value under independent worst marginals by 4.5% on average and up to 14.1%. BRL pricing equals complete enumeration on every test instance, including 18 of 18 sub-instances built from the nine cities. Fixing one risk level for all routes is suboptimal in 9 of 20 small exact comparisons, by up to 6.42% (Proposition 49).

Labelling rules. Table 1 compares labelling rules on random instances with $L = 33$ levels and nine sectors. BRL takes between 0.004 and 0.065 seconds in the range shown, while sector dominance times out from the smallest size and quantile-vector and scenario labelling from $n = 15$. With dual prices scaled by 2.5, BRL prices $n = 25$ in 1.8 s while the risk-neutral labelling timed out; this reflects deadline pressure in those instances, where windows collapse to points, not a general complexity property.

Branch-and-price on the metro instances. On the nine metro routing instances (22 customer zones, $L = 51$, ng-routes with eight neighbours, seed 0), branch-and-price proved optimality (gap 0) in 9 of nine cities (Table 2), in at most 18 minutes on one core (Moscow, 7487 nodes); pricing takes about 64% of the time. Against complete enumeration on 800 random

Table 1: BRL pricing on random instances (nine sectors, $L = 33$ levels, capacity 6, two instances per size). Time in seconds (labels kept); TO: 60 s limit. The risk-neutral labelling solves a different (expected-value) problem and is shown for reference; SAA scenarios optimise the sample-average objective.

n	risk-neutral	BRL	sector (K)	quantile (L)	SAA scenarios
10	0.0050 (19)	**0.0040 (16)**	TO	2.2 (4946)	5.4 (7904)
10	0.017 (36)	**0.0080 (16)**	TO	2.1 (4391)	4.7 (7715)
15	0.17 (264)	**0.019 (48)**	TO	TO	TO
15	0.029 (69)	**0.010 (24)**	TO	TO	TO
20	0.39 (714)	**0.049 (97)**	TO	TO	TO
20	1.4 (1730)	**0.047 (89)**	TO	TO	TO
25	0.46 (650)	**0.059 (94)**	TO	TO	TO
25	0.49 (693)	**0.065 (109)**	TO	TO	TO

formulations (up to ten customers), branch-and-price returned the optimum in 800 of 800, while price-and-branch was strictly suboptimal in 39 of them, by up to 7.3%. An earlier benchmark with price-and-branch found the BRL plan worse than a heuristic optimised on the observed days in London, Paris and Berlin; that deficit came from the inexact integer step. With branch-and-price the London optimum has the same robust value as the heuristic plan; the optimum is strictly better than the best of three heuristics in Moscow (-1.1%), Buenos Aires (-2.4%), New York (-10%), Mumbai (-0.32%), Paris (-0.10%), Berlin (-0.92%), and in New Delhi, Cairo, London the best heuristic was already optimal. The instances are small (three to six customers per route), Moscow's congestion is imputed, and larger instances would likely need cutting planes.

The robust optimum is not the best plan under the calibrated law that generated the data. Evaluated under that law (20,000 draws), the branch-and-price plan is worse than the sample-average plan in Buenos Aires (+7.0%), New York (+4.8%), Berlin (+6.8%). The worst-case cost of the robust optimum lies between 10% and 45% above its expected cost under the calibrated law. This conservatism is a property of the criterion (sixty observed days, support cap, unknown dependence), not of the algorithm; a user who trusts an estimated law should not adopt the dependence-robust criterion.

7.2. Equilibrium engine and exactness of the oracle

Effect of the inexact integer step. The game experiments were first computed with price-and-branch. Tested against a fresh exact oracle, those base equilibria were exact in New Delhi, Buenos Aires, Mumbai, London in every seed but not in 10 of the 27 city-seeds: Cairo (3 of 3 seeds, up to 0.82 USD); Moscow (2 of 3 seeds, up to 0.83 USD); New York (1 of 3 seeds, up to 4.0 USD); Paris (2 of 3 seeds, up to 2.0 USD); Berlin (2 of 3 seeds, up to 3.5 USD). In those cases a carrier could gain at most 4.0 USD per day with a plan the pooled integer program had not generated. City-mean welfare of the base equilibria changes by at most 0.14% outside Buenos Aires, and by -1.4% in Buenos Aires, where seed 0 moves to a different equilibrium (-5.4%); single seeds elsewhere change by up to 0.53% with both signs. The exact oracle thus changes which equilibrium is reached rather than improving welfare systematically. It takes three to twelve times longer per equilibrium.

Table 2: Branch-and-price with BRL pricing on the nine metro routing instances (seed 0, 22 customer zones, robust objective in driver-hours). Root LP: column-generation bound; P&B plan: price-and-branch plan; B&P optimum and remaining gap; nodes and time on one core; difference of the optimum from the best of three heuristics under the robust objective, and from the sample-average heuristic plan under the calibrated law. *Moscow congestion imputed.

City	root LP	P&B plan	B&P optimum	gap %	nodes	time s	vs best heur. %	vs SAA, true law %
New Delhi	63.03	63.18	63.18	0	65	1.63	0	0
Cairo	40.39	40.42	40.42	0	215	5.30	0	0
Moscow*	42.01	43.65	43.17	0	7487	1 063	-1.1	-0.57
Buenos Aires	47.29	48.37	48.13	0	1659	166	-2.4	7.0
New York	46.10	46.10	46.10	0	1	8.44	-10	4.8
Mumbai	53.82	54.08	53.91	0	177	3.59	-0.32	-0.31
London	51.04	52.03	51.38	0	59	21.2	0	0
Paris	37.73	38.32	38.25	0	4305	715	-0.10	-0.026
Berlin	51.85	52.67	52.06	0	3	5.63	-0.92	6.8

Engine components. On the city games (Table F.11 in Appendix F, price-and-branch oracle) ZMC uses between 36 and 65 calls, about ten per carrier, and its potential gap coincides with those of Frank–Wolfe and column generation with eight rounds to three digits in most cities. Plain best response needs only 8 to 17 calls and is as good as ZMC (within 10^{-3}) in Cairo, Moscow, New York, Paris, Berlin; it is worse in New Delhi, Buenos Aires, Mumbai, London. ZMC therefore does not beat best response on these games. Reasons we can see: only 32 to 89 coordinates are active, so the dimension that favours ZMC in the synthetic instances is absent; the games are mildly nonconvex; and carriers share few zones. ZMC's dimension independence is not tested at city scale.

7.3. Unpriced equilibria

Table 3 reports the unpriced equilibria. The cities fall into two groups. In Cairo, Moscow, New York, Paris, Berlin the carriers serve at least 85% of customer demand. In New Delhi, Buenos Aires, Mumbai, London they serve between 18% and 45%, because the delivery fee is low relative to the vehicle-hour cost or the shift is short. The added travel-time slowdown caused by delivery vehicles ranges from 0.77% to 17.6% (Moscow). Figure F.5 (Appendix F) shows equilibrium zone prices of seed 0. The bound η_{rel} on the relative potential gap is below 1% in 3 cities and reaches 44% in the low-service cities, where the potential is close to zero and the dual bound is loose.

7.4. Leader methods: grid, DIP and ARP

The grid. The baseline leader chooses a congestion charge per vehicle-hour, $t_z = \vartheta \gamma c_z X_z^{\text{NE}}$ with $\vartheta \in \{0, 0.5, 1, 2, 4, 8, 16\}$, and a per-delivery charge s in $\{-0.6, \ldots, 0.6\}$ times the fee (negative: subsidy): a two-parameter family of 63 points, one warm-started equilibrium each. Table 4 gives the reduction of social cost relative to no charge for the best exact grid point using only the congestion charge, only the delivery charge, or both (Figure F.4 in Appendix F shows the same gains).

Congestion charge alone. It gains little in all but one of the 27 city-seed cases: in 26 of them it gains less than 1.7% of social cost, the largest being 1.34% (London, seed 0). The exception is Buenos Aires, seed 0, with 6.51%. With price-and-branch this case had given

Table 3: Unpriced equilibria with the exact oracle (mean over three seeds; seed standard deviation in parentheses). Served: share of customer demand delivered; slowdown: added mean travel-time slowdown; price: mean zone price (USD per parcel); veh.-h: vehicle-hours per day; welfare W in USD per day of the slice; calls: oracle calls; $\text{regret}^{\text{LP}}$: guaranteed upper bound on any carrier's gain from deviating (USD), nonzero only where an oracle call was not proven optimal; η_{rel}: upper bound on the relative potential gap.

City	N	Z	served	slowdown %	price	veh.-h	welfare	calls	$\text{regret}^{\text{LP}}$	η_{rel}
New Delhi	5	120	0.26 (0.01)	4.2	0.803	49.1	82.7 (7.0)	48	$7.1 \cdot 10^{-15}$	0.12
Cairo	4	42.0	1.0 (0)	3.6	0.885	125	1 066 (20)	39	0.0020	$-2.4 \cdot 10^{-4}$
Moscow	5	160	0.99 (0.004)	18	2.76	170	6 756 (154)	47	1.2	0.0036
Buenos Aires	4	208	0.36 (0.03)	0.77	0.941	43.6	272 (19)	39	$2.1 \cdot 10^{-14}$	0.24
New York	4	234	0.99 (0.002)	7.8	6.82	112	20 003 (1 441)	39	4.8	0.0050
Mumbai	5	143	0.45 (0.02)	4.6	0.751	83.1	153 (10)	47	$1.8 \cdot 10^{-14}$	0.046
London	5	348	0.18 (0.03)	6.7	4.67	24.8	570 (101)	48	$1.1 \cdot 10^{-13}$	0.44
Paris	5	198	0.85 (0.04)	5.0	4.66	105	5 332 (437)	50	0.76	0.039
Berlin	6	593	0.89 (0.01)	3.7	3.36	127	8 619 (256)	58	0.90	0.029

Table 4: Grid leader, main objective, exact oracle (3 seeds). Gain of the best charge over no charge (% of |social cost|), mean [min, max] over seeds, with the congestion charge only, the delivery charge only, or both. Delivery share: gain with the delivery charge alone divided by the gain with both, per seed, shown where the combined gain is at least 0.5%. Subsidy: seeds in which the best point with both instruments has $s < 0$. Served share and city revenue (USD/day; negative is a net subsidy) at that point. †: only a subset of the grid solved exactly, so gains are lower bounds and shares are ratios of lower bounds.

City	cong. only	deliv. only	both	deliv. share by seed	subsidy	served	revenue
New Delhi	0	10.3 [5.0, 17]	14.5 [12, 17]	0.41 / 0.97 / 0.65	3/3	0.39 [0.36, 0.42]	-54.7 [-64, -48]
Cairo†	0	0	0	–	0/3	1.0	0
Moscow†	0	0.215 [0.13, 0.39]	0.215 [0.13, 0.39]	–	3/3	1.00 [1.00, 1.0]	-2 411 [-3 375, -747]
Buenos Aires	2.49 [0.43, 6.5]	35.4 [27, 40]	35.4 [27, 40]	1.0 / 1.0 / 1.0	3/3	0.74 [0.65, 0.79]	-413 [-528, -226]
New York†	0.00448 [0, 0.013]	0.352 [0.27, 0.46]	0.352 [0.27, 0.46]	–	3/3	1.0	-4 024 [-8 085, -1 891]
Mumbai†	0	2.09 [0.37, 3.3]	2.16 [0.81, 3.3]	0.46 / 1.4 / 0.79	3/3	0.55 [0.49, 0.58]	-64.4 [-91, -20]
London	0.446 [0, 1.3]	49.3 [31, 67]	59.7 [34, 90]	0.92 / 0.89 / 0.74	3/3	0.49 [0.39, 0.54]	-1 359 [-1 803, -808]
Paris†	0.0412 [0, 0.12]	3.47 [2.8, 4.7]	3.57 [2.8, 4.7]	1.0 / 0.91 / 1.0	3/3	0.95 [0.89, 0.99]	-2 712 [-4 160, -849]
Berlin†	0	1.85 [1.6, 2.2]	1.98 [1.6, 2.4]	0.93 / 1.0 / 0.90	3/3	0.94 [0.93, 0.97]	-3 334 [-5 258, -1 921]

1.67%, so the reading "below about 1.7% everywhere" did not survive. The difference comes from the baseline: the unpriced equilibrium selected with the exact oracle in Buenos Aires seed 0 is a worse one (welfare 252.2 against 266.7 USD per day), while the best congestion-charge equilibrium is about as good as before. Across the three seeds the gain is 6.5 / 0.43 / 0.53%; the value of a congestion charge alone can depend on which unpriced equilibrium the market is in.

Per-delivery instrument. In 18 city-seed cases the combined gain is at least 0.5% of social cost. In all of them the best point has a subsidy (18 of 18). The delivery charge achieves at least half of the combined gain in 16 of these cases; the exceptions are Mumbai seed 0 (0.46), New Delhi seed 0 (0.41). Shares are New Delhi 0.41–0.97, Buenos Aires 1.0 in every seed, London 0.74–0.92 in the full-grid cities; in cities with a reduced grid they are ratios of lower bounds and not reliable to two digits (the share exceeds one in Mumbai seed 1).

Where the gain is large. The gain is material only where the unpriced equilibrium under-delivers. In Cairo, Moscow, New York, where service is nearly complete, it is at most 0.46% in every seed. In the low-service cities it varies widely between seeds: in London the combined gain ranges from 34.5% to 90.1% and in New Delhi from 12.2% to 17.4%. London, Buenos

Table 5: DIP against the grid, per city and seed (0 / 1 / 2). Grid pts: exact grid points available (pb: the 63-point price-and-branch grid). Gain share: DIP gain divided by the best grid gain (n/a: no charge improves on the unpriced equilibrium). Calls: DIP's oracle calls as a percentage of the calls of the grid run, and of a 63-point grid at the same calls per point. Rounds: cutting-plane rounds (1: the tangent charge implemented the target without any cut). Exact: runs in which the target is an exact pure equilibrium under the returned charges. PoA of the grid's best point and of DIP's priced equilibrium.

City	grid pts	gain share	calls % (grid run)	calls % (63 pts)	rounds	exact	PoA grid	PoA DIP
New Delhi	63 / 63 / 63	1.23 / 1.35 / 1.14	5.6 / 4.9 / 5.5	5.6 / 4.9 / 5.5	2 / 1 / 2	3/3	1.02 / 1.05 / 1.02	1.00 / 1.00 / 1.00
Cairo	13 / 17 / 13	n/a / n/a / n/a	11 / 8.5 / 12	2.3 / 2.3 / 2.4	1 / 1 / 1	3/3	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00
Moscow	12 / 10 / 63^{pb}	1.10 / 1.00 / 1.00	13 / 13 / 2.9	2.6 / 2.1 / 2.9	1 / 1 / 1	3/3	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00
Buenos Aires	63 / 63 / 63	1.09 / 1.05 / 1.08	6.3 / 6.3 / 5.4	6.3 / 6.3 / 5.4	1 / 1 / 1	3/3	1.03 / 1.01 / 1.02	1.00 / 1.00 / 1.00
New York	7 / 7 / 63^{pb}	0.999 / 0.999 / 1.04	24 / 23 / 2.9	2.6 / 2.6 / 2.9	1 / 1 / 1	3/3	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.000
Mumbai	15 / 14 / 14	3.25 / 1.20 / 1.28	24 / 31 / 26	5.6 / 6.9 / 5.8	1 / 2 / 2	3/3	1.02 / 1.00 / 1.00	0.997 / 0.997 / 0.995
London	63 / 63 / 63	1.04 / 1.31 / 1.06	7.2 / 9.4 / 8.7	7.2 / 9.4 / 8.7	3 / 3 / 3	3/3	1.01 / 1.08 / 1.02	1.00 / 1.00 / 0.995
Paris	14 / 13 / 13	1.02 / 1.06 / 1.05	22 / 23 / 21	4.9 / 4.8 / 4.4	3 / 1 / 2	3/3	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00
Berlin	13 / 13 / 63^{pb}	1.03 / 1.03 / 1.13	17 / 15 / 4.1	3.5 / 3.2 / 4.1	1 / 1 / 1	3/3	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00

Aires and New Delhi have gains above 10% in every seed, but the magnitudes are not pinned down. The mechanism visible in the model is under-provision caused by fixed route costs and market power, which a subsidy corrects, while the congestion externality among carriers is small; within a zone the two instruments are collinear (S3), so the congestion charge adds only spatial differentiation. Subsidies cost the city money; since charges are transfers in the welfare definition, this fiscal cost is not part of the social cost.

With the broad objective, which adds the background-delay proxy, the best response of the leader in New Delhi, Mumbai, London is to stop deliveries altogether in every seed (Table F.16, Appendix F). This is an artifact of the proxy and draws no conclusion from us.

7.4.1. DIP against the grid

DIP was run with the exact oracle on all nine cities and three seeds (27 runs) and compared with the grid of the same city and seed (Table 5, Figure 1). The gain share is DIP's social-cost gain over the unpriced equilibrium divided by the gain of the best grid point; calls count every routing-oracle query of each method with the same counter. DIP searches a richer family than the grid (one charge per zone against two parameters), so a gain share above one is not a like-for-like win; restricted to the grid's family, DIP reaches on average 1.0 of the grid gain, with the worst case 0.50 (Mumbai, seed 0, a reduced grid).

Gain and cost. In the three cities with a full exact grid (New Delhi, Buenos Aires, London, 9 runs), DIP reaches 104% to 135% of the grid-optimal gain (mean 115%) with 4.9% to 9.4% of the grid's oracle calls (mean 6.6%), that is 105 to 221 calls against 1 921 to 2 419. Over all runs with a positive gain, DIP reaches at least 0.999 of the best evaluated grid point and uses at most 9.4% of the calls of a 63-point grid. In Cairo no charge improves on the unpriced equilibrium and DIP leaves it in place. The priced equilibrium attains the social cost of the target; its empirical price of anarchy is between 0.995 and 1.000 (below one where DIP's multi-start target is slightly better than the benchmark), against 1.01 to 1.08 for the grid's best point in the full-grid cities.

Exactness. In all 27 runs the cutting-plane loop ended with charges under which the target is an exact pure equilibrium (regret against a fresh exact oracle at most $9.1 \cdot 10^{-13}$ USD); the fallback was never needed. The tangent charge alone implemented the target in 18 runs; the

others needed at most 3 rounds and 4 cuts. The uniform part of the charges is a subsidy of 0.37 to 0.50 of the fee, and the zone charges follow the competitor-count rule of Proposition 15.

Qualifiers.

- Against the reduced exact grids of six cities (7–17 screened points), DIP uses more than 10% of their calls in most runs (Cairo 8.5–12%, Moscow 13%, New York 23%, Mumbai 24–31%, Paris 21–23%, Berlin 15–17%), although at most 9.4% of a 63-point grid.
- Wall-clock time is not always lower: the median ratio to the grid's time is 0.22, but in London seed 0 DIP took 247 s against 157 s, because the social-pricing calls of column generation are harder branch-and-price instances than warm-started equilibrium calls.
- Selection is optimistic. The equilibrium reached by the grid's protocol under DIP's charges is the target in 21 of 27 runs and in all runs has at least 0.999 of the grid's gain. In 210 enumerated tiny games the equilibrium under DIP's charges was unique in only 35% of the games.
- The result is limited by the social target. In the tiny games DIP reached the exact leader optimum over all zone charges in 93% of the games, and in 100% when given the exact social optimum as target, so each miss there was a miss of the social solver; it used 23 to 64 calls per game against 1 323 to 1 794 for the grid.
- With Γ-maximin carriers, the shifted charges of Theorem 14 made the target an exact equilibrium in every run at $\delta = 0.2$ and 0.8 without further oracle calls (largest regret $9.1 \cdot 10^{-13}$ USD). At $\delta = 0.8$ the required uniform subsidy is 0.98 to 3.6 times the fee, which is not a realistic policy.

7.4.2. ARP: charges from observed aggregates

ARP was run with the exact oracle on London, Buenos Aires, New Delhi and Mumbai, three seeds each (Table 6, Figure 2). The carriers best-respond day to day from their current plans after each posted charge, and all their oracle calls during the run are counted. The status-quo equilibrium (40–60 calls) is common to all methods and not counted, as for DIP. The fixed-point and iteration statements are tested on the convexified tiny games (Table 7 and below).

City runs. From (Q_z, n_z) alone, ARP reaches 90.8% to 293% of the grid's gain (mean 125%) after 3–7 posted charges and 22–67 calls, against 104%–325% (mean 134%) for DIP with 105–221 calls; in the three full-grid cities the figures are 90.8%–126% (ARP) and 104%–135% (DIP), and ARP falls below the grid's gain in 1 of 9 runs. ARP matches DIP within two percentage points in 5 of 12 runs; in the other 7 runs it is below DIP by up to 32 points of the grid's gain (the largest gaps are in runs where the iteration cycles or the grid is a reduced one). The clear miss is New Delhi seed 0 (90.8% against DIP's 123%), where the iteration cycles between two equilibria and $\varrho = 0.40$, the largest of the sample. The flat rule reached an exact fixed point in 6 runs (three rounds in 5 of them) and cycled in 6; the cycles are two-equilibrium cycles of the lumpy response, and the successive-averages variant does not remove them (90.8%–127% on the 6 cycling runs). The relative change of the charge vector falls to between 0.37 and 0.63 of its initial value after the first round and to between 0.044

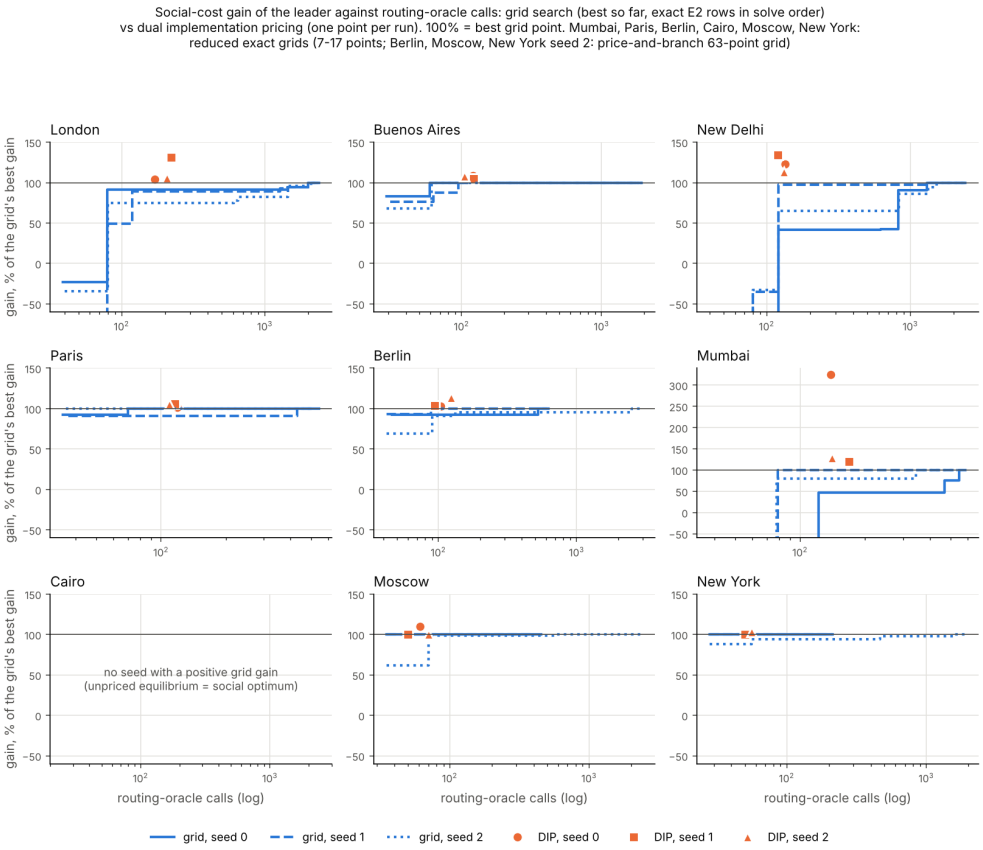


Figure 1: Leader's social-cost gain against routing-oracle calls: best grid point found so far (lines, exact grid rows in solve order) and DIP (one marker per run); 100% is the best grid point. Six cities have reduced exact grids; for Berlin seed 2, Moscow seed 2, New York seed 2 the grid is the price-and-branch grid.

Table 6: Leader methods on four cities (exact oracle). ϱ: contraction constant of Theorem 18. ARP: flat rule, status after how many rounds (a cycle is a repeated charge and market state). Gains are percentages of the best exact grid point's gain; tariff+BR: marginal tariff with the carriers' own best responses; tariff+SO: tariff with the potential solver; DIP for reference. Calls: oracle calls of the carriers during the method. Mumbai's grid is a reduced exact grid (gains are relative to a lower bound).

City	seed	ϱ	ARP status / rounds	ARP %	calls	tariff+BR %	calls	tariff+SO %	calls	DIP %	calls	grid calls
New Delhi	0	0.40	cycle / 5	90.8	53	98.7	11	92.5	51	123	134	2404
New Delhi	1	0.19	fixed point / 3	126	27	135	10	127	51	135	119	2418
New Delhi	2	0.40	cycle / 6	105	60	103	9	114	48	114	132	2419
Buenos Aires	0	0.029	fixed point / 3	108	24	108	9	108	40	109	121	1921
Buenos Aires	1	0.016	fixed point / 3	105	22	105	10	105	40	105	122	1950
Buenos Aires	2	0.11	fixed point / 3	108	23	108	8	108	39	108	105	1943
Mumbai	0	0.20	fixed point / 7	293	65	293	9	263	51	325	139	586
Mumbai	1	0.094	fixed point / 3	111	30	113	10	108	50	120	169	544
Mumbai	2	0.20	cycle / 5	128	47	111	10	118	49	128	141	537
London	0	0.10	cycle / 5	102	51	104	14	100	49	104	170	2376
London	1	0.10	cycle / 5	120	47	122	9	131	50	131	221	2356
London	2	0.10	cycle / 7	103	67	105	11	101	50	106	206	2378

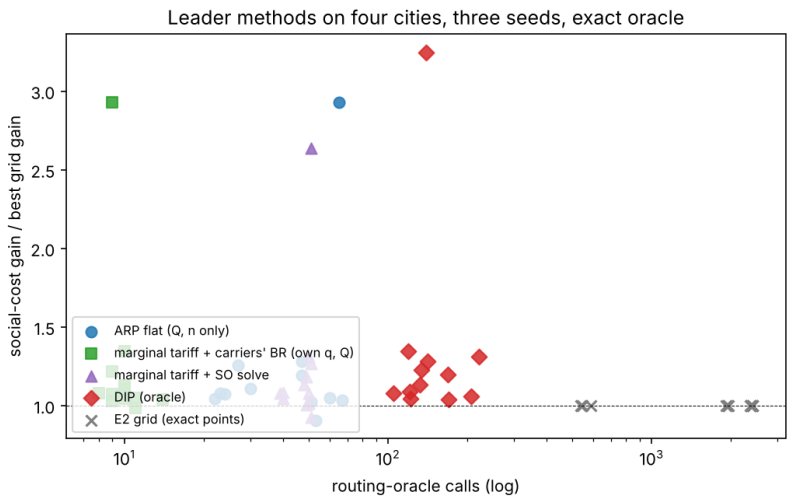


Figure 2: Leader methods on four cities and three seeds (exact oracle): gain relative to the best exact grid point against routing-oracle calls (log scale). The grid points are the exact grids of each city.

and 0.24 after the second, in every run; these ratios are measured on the pure game and are not directly comparable with ϱ.

The marginal tariff with the carriers' own best responses reaches 98.7% to 293% (mean 125%) with 8–14 calls and no city-side computation; it equals DIP within two points in 6 runs and never raised the social cost along the way in 12 of 12 runs, as Theorem 16 says. With the potential solver under the tariff, the result is the benchmark social optimum in some runs and slightly worse in others (the run from the standard start selects another local optimum of SC): 92.5%–263% with 39–51 calls. Every returned profile is an exact pure equilibrium of the posted charges (fresh-oracle regret at most $1.1 \cdot 10^{-13}$ USD for ARP and $1.4 \cdot 10^{-14}$ USD for the tariff; 0 unproven calls). The price of anarchy against the social optimum benchmark is 0.966–1.00 for ARP. Wall-clock time is 1.9–41 s, with median ratio 0.041 to DIP's time. ARP's uniform subsidy is 1.3–1.6 times the fee (DIP: 0.38–0.50), because the city-mean lot used for never-served zones spreads the zone charges more than DIP's customer-based mean lot.

Tiny games. On 210 exhaustively enumerated games (136 with a positive leader gain; congestion multipliers 1, 20, 60, median ϱ 0.013 / 0.22 / 0.53) Table 7 gives the share of the exact optimal gain. The flat rule reached a fixed point in 166 games (mean 2.3 rounds, at most 8) and cycled in 44, almost all at strong congestion. By congestion level its mean share is 0.72 / 0.64 / 0.23, the tariff's 0.93 / 0.79 / 0.58 and DIP's 1.00 / 0.96 / 0.93. Failures of the flat rule are of two kinds: the subsidy for never-served zones (the city-mean lot) attracts entry into zones that the optimum leaves unserved, and the carriers then settle in a worse local optimum; and two-charge cycles at strong congestion. Knowing the lots does not help (share 0.63): the limit is the information the aggregate carries about which profile is socially good (Theorem 19(c)). Under the tariff the set of pure equilibria is exactly the set of unilateral social optima (the best tariff equilibrium is the social optimum in 210 of 210 games), and the social-cost descent from the status quo never increased the social cost (210 of 210). No method has a worst-equilibrium guarantee.

Table 7: Tiny games with exhaustive enumeration, games with a positive leader gain. Mean share of the exact optimal gain, share of games in which the optimum is reached, games worse than the unpriced equilibrium, and mean oracle calls. For tariff rows, "worse than unpriced" compares with the potential-minimising unpriced equilibrium of the enumeration.

Method	city's information	mean share	optimum hit %	worse than unpriced	calls
ARP flat (plain)	(Q_z, n_z)	0.56	54	9	13
ARP flat (successive averages)	(Q_z, n_z)	0.64	57	8	51
ARP flat, lots known	(Q_z, n_z) + lots	0.63	54	15	12
tariff + carriers' best responses	own q, Q_z	0.80	71	3	5.4
tariff + potential solver	oracle	0.94	89	1	24
DIP	oracle	0.97	92	0	39
grid	oracle	0.83	67	0	1 492

Contraction constants. On 48 convexified tiny games with ϱ from 0.0062 to 0.81, with convexified responses computed exactly over all enumerated plans, the Lipschitz bound of Theorem 17 (observed maximum 0.99), the plain-step bound of Theorem 18(a) (observed maximum 1.00 of the bound) and the damped-step bound of part (b) (observed maximum 1.00 of the bound, that is tight) held in all games.

Reading. ARP is not better than DIP in solution quality. It is better in the information it needs (none about the carriers) and in cost (a fraction of DIP's calls and a few per cent of its wall-clock time), and it equals DIP in about half of the city runs. Damping helps the convexified rate but not the lumpy cycles. The tariff removes the implementation problem but not the selection problem.

7.5. Price of anarchy and imprecision

Empirical price of anarchy and stability. The social optimum is approximated by column generation on the convexified welfare problem followed by derandomised rounding and coordinate polish; this gives a feasible pure profile W_{SO} and, from the relaxation, an upper bound. The equilibria considered are those found by the engine, by best response from the empty and the full-service profile and from three random price vectors: between 1 and 6 distinct equilibria per city and seed. The price of anarchy and stability in Table 8 are therefore *empirical*: ratios of W_{SO} to the equilibria found, not bounds over all equilibria.

In Cairo, Moscow, New York, Mumbai, Paris, Berlin the empirical price of anarchy of the engine's equilibrium is at most 1.05 in every seed. It is larger in New Delhi, Buenos Aires, London in every seed; the ordering London > Buenos Aires > New Delhi holds in each seed, and across seeds the values range from 1.45 to 1.94 in London, 1.29 to 1.43 in Buenos Aires and 1.15 to 1.23 in New Delhi. London's value is therefore not pinned down. Compared with the price-and-branch run, city means move by at most 0.022 and the ordering of the cities is unchanged; the largest single change is Buenos Aires seed 0 (1.349 to 1.429), where a different unpriced equilibrium is selected. In 5 of the 27 city-seed games the price of anarchy of the worst equilibrium found exceeds the price of stability by more than 0.05: Mumbai seed 0 (PoS 1.02, worst 1.11), Buenos Aires seed 0 (PoS 1.32, worst 1.43), Buenos Aires seed 1 (PoS 1.41, worst 1.54), London seed 0 (PoS 1.52, worst 1.61), London seed 1 (PoS 1.42, worst 1.67). Equilibrium selection thus matters in some games. With the grid's best charges the equilibrium comes closer to the rounded optimum, not uniformly: the shortfall is at most

Table 8: Empirical price of anarchy and stability, main objective, exact oracle (3 seeds). W_{SO}: rounded social optimum (USD/day), mean [min, max]; #NE: distinct equilibria found; PoA of the engine's equilibrium, of the worst equilibrium found, and PoS (best found), per seed; priced gap: shortfall of the equilibrium under the grid's best charges from W_{SO} (%), per seed.

City	W_{SO}	#NE	PoA (ZMC eq.)	PoA (worst NE)	PoS	priced gap %
New Delhi	97.7 [86, 106]	4–5	1.15 / 1.23 / 1.16	1.15 / 1.24 / 1.20	1.12 / 1.23 / 1.16	2.4 / 4.9 / 1.6
Cairo	1 066 [1 046, 1 086]	2–3	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00	0 / 0 / 0
Moscow	6 771 [6 639, 6 929]	2	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00	0.012 / 0 / 0
Buenos Aires	375 [361, 390]	3–6	1.43 / 1.42 / 1.29	1.43 / 1.54 / 1.31	1.32 / 1.41 / 1.27	2.5 / 1.3 / 1.7
New York	20 072 [18 604, 21 466]	1–3	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00	0 / 0 / 0
Mumbai	158 [148, 170]	4–6	1.02 / 1.04 / 1.04	1.11 / 1.08 / 1.04	1.02 / 1.04 / 1.04	1.5 / 1.2 / 0.39
London	927 [882, 991]	5–6	1.57 / 1.45 / 1.94	1.61 / 1.67 / 1.98	1.52 / 1.42 / 1.94	1.4 / 7.4 / 2.2
Paris	5 530 [5 023, 5 879]	3–5	1.03 / 1.03 / 1.05	1.03 / 1.03 / 1.05	1.03 / 1.02 / 1.05	0.047 / 0.19 / 0.23
Berlin	8 800 [8 570, 9 137]	2–4	1.02 / 1.02 / 1.03	1.02 / 1.02 / 1.03	1.02 / 1.02 / 1.02	0.054 / 0.054 / 0.46

0.4% in every seed in Cairo, Moscow, New York, Paris, and reaches 7.4% in London, seed 1 and 4.9% in New Delhi seed 1. The relaxation bound exceeds the rounded optimum by up to 25% (London), so the true price of anarchy could be larger by that margin, and all values refer to the lumpy lots (S1).

Imprecision. Two kinds of imprecision are varied (Table 9, Figure F.3). The first is the width of the travel-time probability boxes, through the DKW level, from $\varepsilon = 0$ to $\varepsilon = 0.41$; it acts through the BRL route costs. At the widest box, over all cities and seeds, deliveries fall by between 0% and 78% and welfare by between 2.7% and 82%. The grouping of cities is the same in every seed: Cairo loses no deliveries, Moscow and New York a few per cent, Paris and Berlin somewhat more, Mumbai more, and New Delhi, Buenos Aires and London about half or more; the magnitudes in the low-service cities vary by seed (London: -60.0 / -56.1 / -78.2% in seeds 0, 1, 2). The second is the credal width about rivals' aggregates under Γ-maximin, through the surcharge of Lemma 25 with w equal to the unpriced equilibrium aggregate, from $\delta = 0$ to $\delta = 0.80$; deliveries fall by between 31% and 100% and welfare by between 22% and 100%. Relative to the price-and-branch run, the city means of these widest-width effects change by at most 1.4 percentage points and single seeds by up to 6.0.

Deliveries are not always monotone in the width: of 54 (city, seed, sweep) series, 3 contain at least one rise, by up to 50 deliveries, all in Buenos Aires. Proposition 36 does not apply: for Γ-maximin no monotonicity holds in general, and for the probability-box sweep the reported equilibrium is a single one that can switch as the width changes. In every case a pure equilibrium was found, every run converged, and the largest regret against the pooled exact oracle was $9.1 \cdot 10^{-13}$. Imprecision acts like a tax on activity: it reduces deliveries, vehicle-hours and congestion together at the cost of welfare, and because vehicle-hours are proportional to deliveries (S3), the model cannot separate a congestion effect from a delivery effect.

7.6. Theory against data

The city games are of type D1, so the engine's profile is an exact pure equilibrium with E-threshold zero and Theorem 27 holds trivially for it. To test the theorems in a non-trivial way we computed the exact thresholds of Theorem 44 on mixed equilibria with belief weights w equal to the realised aggregate. These tests used the price-and-branch oracle; each threshold is

Table 9: Imprecision, exact oracle (3 seeds; mean [min, max]). Left block: widest probability box ($\varepsilon = 0.41$) against $\varepsilon = 0$, change in deliveries and welfare (%). Right block: widest Γ-maximin width ($\delta = 0.80$) against $\delta = 0$. Monotone series: (seed, sweep) series in which deliveries never rise with the width. †: three exact widths per sweep.

City	Δdeliv. %	ΔW %	Δdeliv. %	ΔW %	monotone series
New Delhi	-53.5 [-55, -51]	-60.4 [-61, -59]	-62.2 [-75, -49]	-51.2 [-65, -39]	6/6
Cairo†	0	-2.88 [-3.0, -2.7]	-31.4 [-33, -31]	-22.8 [-24, -22]	6/6
Moscow†	-1.35 [-2.1, -0.95]	-10.5 [-11, -9.9]	-63.7 [-68, -62]	-55.1 [-61, -52]	6/6
Buenos Aires	-52.3 [-57, -49]	-52.9 [-57, -47]	-64.9 [-65, -64]	-54.6 [-56, -52]	3/6
New York†	-2.62 [-3.0, -2.1]	-10.9 [-11, -11]	-63.2 [-69, -57]	-54.4 [-61, -47]	6/6
Mumbai	-29.4 [-34, -23]	-30.3 [-32, -27]	-59.0 [-62, -56]	-44.7 [-48, -41]	6/6
London	-64.8 [-78, -56]	-74.8 [-82, -70]	-89.5 [-100, -81]	-86.5 [-100, -76]	6/6
Paris†	-10.2 [-12, -6.9]	-17.3 [-18, -16]	-71.3 [-74, -67]	-60.4 [-64, -54]	6/6
Berlin†	-8.68 [-9.2, -8.1]	-17.8 [-18, -17]	-71.2 [-72, -70]	-59.6 [-61, -58]	6/6

Table 10: Theorems on the city games (three seeds). Mixed: seeds with a non-trivial mixed equilibrium. For those seeds (means): $\bar{\rho}$; the anchored three-term bound of Theorem 27(b); the N-uniform bound of Theorem 27(c); the largest empirical E-threshold over the tested profiles; and the ratio of the latter to the anchored bound. Γ-maximin (Lemma 25): drop in the share of demand served from the precise to the surcharge equilibrium at $\delta = 0.1$ and $\delta = 1$, and the maximality threshold of the surcharge equilibrium divided by δ at $\delta = 1$.

City	mixed	$\bar{\rho}$	anchored bd.	N-unif. bd.	max δ_E	ratio	drop δ=0.1	drop δ=1	M-thr./δ
New Delhi	3/3	1.0	1.61	10.7	0.64	0.41	0.0075	0.18	0.89
Cairo	0/3	–	–	–	–	–	0	0.47	0.99
Moscow	0/3	–	–	–	–	–	0.0021	0.73	0.97
Buenos Aires	1/3	0.96	1.76	8.80	0.83	0.47	0.082	0.31	0.93
New York	1/3	0.82	1.58	3.68	0.82	0.52	0.0090	0.71	0.98
Mumbai	2/3	0.74	1.23	10.6	0.53	0.43	0.049	0.34	0.92
London	3/3	0.85	1.56	13.6	0.82	0.53	0.033	0.17	0.76
Paris	3/3	0.88	1.39	8.43	0.69	0.49	0.036	0.74	0.96
Berlin	1/3	0.94	0.937	7.97	0.41	0.44	0.050	0.77	0.97

relaxed by the largest improvement observed at a verified equilibrium. Table 10 and Figure F.6 (Appendix F) summarise the results.

Credal absorption. Non-trivial mixed equilibria were constructed from pairs of distinct pure equilibria by solving the two-plan indifference conditions and checking with the oracle that every support plan is a best response over all plans. This succeeded in 14 of the 27 city-seed games; Cairo, Moscow have none, because almost all demand is served. On all 274 tested profiles from the products of the supports, no statement of Theorem 27 was violated: the empirical E-threshold never exceeded the anchored radius, the minimum anchored radius never exceeded the three-term bound, the existence threshold never exceeded the N-uniform bound, and the order of the thresholds held (0 violations). The bound is loose: the largest empirical threshold is 0.28 to 0.60 times the anchored bound (median 0.45) and 0.032 to 0.22 times the N-uniform bound. As Remark 29 predicts for $N \leq d$, the support-selection term is the smallest of the three in every case. The ratio shows no dependence on N in this

small sample and a mild increase with d; with 24 anchors this is exploratory. Without the best-response polish the rounded profile of the first engine phase is E-admissible at a finite width in only 10 of 27 games, because some carrier plays a strictly dominated plan. Limits: mixed equilibria were found only with two-plan supports, and thresholds are exact only up to the oracle gap.

Γ-*maximin.* In all 108 (city, seed, δ) cases the equilibrium of the surcharge game differs from the precise equilibrium, and in 106 of them the precise equilibrium is not Γ-admissible. The surcharge equilibrium is maximal at width δ, with a maximality threshold below δ in every case with a finite threshold (at $\delta = 1$ between 0.76 and 0.99 of δ in the city means; 2 Cairo cases are infinite because of oracle slack). Averaged over seeds, the share of demand served drops by 0 to 0.082 at $\delta = 0.1$ and by 0.17 to 0.77 at $\delta = 1$.

Vehicle-count congestion. The gadgets behave as proved: the example of Proposition 31 has no pure equilibrium and existence threshold exactly $1/w$ (3 of 3 weights), and the kernel construction of Theorem 33 reproduces the kernel sets for all four criteria on 223 of 223 digraphs. Among random bilinear D2 games with three players, 6 of 400 have no pure equilibrium, against 0 of 150 D1 controls. Small routing-integrated D2 games always had a pure equilibrium (no pure equilibrium in 0 of 540 game-scale cases). On the city games of Cairo, Mumbai and Paris (three seeds each) with a vehicle-count externality scaled by factors 1, 10, 100, 1 000, 10 000, all 180 best-response runs converged (0 cycles). Vehicle-count congestion did not prevent pure equilibria on the city instances; non-existence required adversarial plan menus. The D2 calibration is an assumption introduced for this test.

7.7. Verification

On 40 tiny games (two or three carriers, three or four customer zones) solved by exhaustive enumeration, the engine's profile is a pure equilibrium of the enumerated game in every case and reaches the global potential minimum in 40 of 40; best response reaches it in 35, Frank–Wolfe with rounding in 38 and column generation in 39 (Table F.18, Appendix F). These instances support correctness more than they separate the methods. At city scale, every base equilibrium, every verified grid best point (27 city-seeds) and every equilibrium of the price-of-anarchy experiment has regret at most $9.1 \cdot 10^{-13}$ USD against an independent fresh exact oracle (elementary routes, empty column pool). Because 344 oracle calls were not proven optimal, the guaranteed regret bound (incumbent of the verifying call against its proven lower bound) is nonzero for 10 profiles, at most 4.84 USD (a New York base case with welfare of about 20,000 USD per day). These checks rely on the same branch-and-price method as the oracle; the pricer itself was checked against complete enumeration (Section 7.1). The clearing identity of Theorem 50(a) held to $3.4 \cdot 10^{-12}$ in every round of every city in the price-and-branch run (Table F.19); this check was not repeated with the exact oracle.

8. Managerial implications

The model is stylised, several parameters are weak (Section 6) and the effects below are qualitative. They indicate what to test with local data, not what to do in any particular city.

Where pricing matters. In the model, the efficiency loss from competition is concentrated where service is incomplete. Where carriers already serve almost all demand (Cairo, Moscow, New York, Paris, Berlin), the empirical price of anarchy is close to one and charges change little; kerb space and background traffic are the more relevant levers there. Where service is partial, the instrument with the largest effect is a per-delivery subsidy, and in every material case the best flat charge was a subsidy. The size of the gain depends strongly on the sampled slice (London: 34.5% to 90.1% of social cost across three seeds), so the direction is better supported than the size. A congestion charge on delivery vehicle-hours alone recovered little in all but one case, because the congestion carriers impose on each other is small against the under-provision from fixed route costs and market power; the exception arose from a worse unpriced equilibrium. A subsidy has a fiscal cost that the welfare criterion with transfers does not count, access stays unequal in the low-service cities even at the best charges, and an objective that counts delay to other road users can reverse the recommendation, which argues for measuring that delay.

How a city could compute charges. A city does not need to search a grid. With the carriers' routing oracle, DIP gives zone charges at a few per cent of the grid's calls, with a readable structure (a subsidy of roughly 37.5% of the fee divided by the number of carriers in the zone, corrected for congestion). Without the oracle, ARP needs only the zone totals after each posted charge, and the marginal tariff needs billing on own deliveries and the zone total. The price of observation-only pricing is selection risk: the flat rule can cycle or settle in a worse local optimum, and the tariff stops at a unilateral social optimum. A pilot with a few posted charges is a cheap check of both. Zone-specific charges are harder to administer than a flat subsidy.

For carriers and for modelling. Wider travel-time boxes and wider beliefs about rivals reduce the volume a carrier finds worth serving. Under the dependence-robust criterion the planned cost is well above the expected cost under the simulated law, so a carrier that can estimate a law or knows the dependence between segments should use that information. For a modeller, the practical rule is that exposure-based congestion gives a potential game with guaranteed pure equilibria, whereas vehicle-count congestion can make existence NP-complete; on our instances that worst case did not occur.

9. Limitations

- *Model.* Lots are all-or-nothing (S1), so part of the measured inefficiency may be an artifact of indivisibility. Congestion is linear and, within a zone, proportional to deliveries (S3, S4). Demand is sampled, market shares are equal in seven cities, and the congestion coupling is scaled by a slice multiplicity. Most operating parameters are country-level or rough; delivery fee, drops per vehicle and the number of carriers move welfare by tens of per cent over their plausible ranges (Table F.15), and the sign of the best per-delivery instrument depends on the fee-to-cost ratio.
- *Exactness and coverage.* 344 of 41,469 oracle calls were not proven optimal; for 10 profiles the guaranteed regret bound is positive (at most 4.84 USD). The grid is exact in full only for New Delhi, Buenos Aires, London, so the grid's gains in the other six cities are

lower bounds; imprecision sweeps in Cairo, Moscow, New York, Paris, Berlin use three exact widths per sweep; the engine comparison, ZMC checks, theory tests and tiny-game verification use price-and-branch, whose equilibria were not exact in 10 of 27 city-seeds. Branch-and-price was tested only on instances with 22 customer zones.

- *Leader methods.* DIP is no better than its social target, its selection is optimistic, its closed forms need D1 and one hours-per-delivery ratio, and the required subsidies under strong ambiguity aversion are unrealistic. ARP was run on four cities, can cycle, can settle in a worse local optimum, and has no guarantee for the pure game unless its fixed point is reached; Theorem 19(b) is conditional on potential minimisation. Neither method has a worst-equilibrium guarantee. The comparison with the grid is on a two-parameter family against zone-specific charges.
- *Theory.* The credal results assume affine coupling. The absorption tests rely on two-plan mixed equilibria and thresholds exact up to the oracle gap; the full rounding algorithm was not run on real data; D2 was tested only with a calibration introduced for the test. The Lean formalisation covers the discrete core of the credal results only.
- *Evidence.* Three seeds give a rough indication of spread; they vary the sampled slice, not the calibration inputs. Several readings from an earlier single-seed, price-and-branch analysis did not survive the second and third seeds or the exact oracle (the role of the congestion charge in Mumbai, a priced equilibrium within 2.4% of the optimum everywhere, a price of stability always within 0.05 of the price of anarchy, a congestion-only gain below 1.7% everywhere). Price-of-anarchy values come from at most six equilibria per city and seed. The broad social cost is driven by a proxy and supports no conclusion. Novelty searches were web and index searches, partly at the level of abstracts; the closest neighbours (14, 53, 56, 67, 6) should be compared in full text.

Open questions include the complexity of finding a pure E-admissible equilibrium above the guaranteed width (we conjecture PPAD-hardness), smoothness constants for the Cournot–congestion game, an absorption result for nonaffine congestion, worst-equilibrium guarantees for the leader, and carrier-level data that would replace our proxies.

10. Conclusion

A city that prices competing delivery carriers faces NP-hard best responses under probability-box travel times, a congestion game among the carriers, and a bilevel search over charges. BRL prices routes exactly under the dependence-robust criterion, and with branch-and-price it proved optimal robust plans on all nine metro instances; the new element is the completion-window dominance, on top of a classical decomposition. DIP made its target an exact equilibrium in every run and, in the three cities with a full exact grid, reached 104–135% of the grid's gain with 4.9–9.4% of its calls, partly because its charges are zone-specific. ARP, which sees only zone totals, reached 90.8–126% there with 22–67 calls, contracts at a rate that does not depend on instance size in the convexified game, and in the pure game can cycle or settle in a worse local optimum. Pure equilibria exist under exposure-based congestion, deciding their existence is strongly NP-complete under vehicle-count congestion, and a belief width independent of the number of carriers turns a mixed equilibrium into an exact pure

imprecise one. On the nine cities, the empirical price of anarchy is near one where service is complete and larger, with a wide spread over seeds, where it is partial; a per-delivery subsidy captures most of the grid leader's gain in most material cases. These findings come from stylised slices with weak parameters and three seeds, and the novelty of each component is limited as stated in Section 1.

Data and code availability

Code, processed data, parameter sources, result files, figures and the scripts that generate the tables and numbers of this paper are available at `https://github.com/vinhqdang/Competing-Carriers-under-Uncertainty`. Raw OpenStreetMap extracts are distributed under the Open Database License; GHS-POP is distributed by the European Commission Joint Research Centre.

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Appendix A. Proofs for the equilibrium theory

The proofs in this appendix are taken from the theory notes in the repository (file `theory/main.tex`), with the notation of Section 3.

Appendix A.1. Classical tools

Theorem 39 (Shapley–Folkman, constructive vertex form; 47)**.** *Let $S_1, \dots, S_N \subset \mathbb{R}^d$ be finite and nonempty, $L = \operatorname{span} \bigcup_i (S_i - S_i)$ and $d_L = \dim L$. Index the elements of S_i by $\mathcal{P}_i$ through an onto map a_i. For $\bar{x} \in \sum_i \operatorname{conv} S_i$, the polytope $\Lambda(\bar{x}) = \{\lambda \in \prod_i \Delta(\mathcal{P}_i) : \sum_i \sum_p \lambda_i(p) a_i(p) = \bar{x}\}$ is nonempty, and at every vertex λ the set $J(\lambda) = \{i : |\operatorname{supp} \lambda_i| \geq 2\}$ has $|J(\lambda)| \leq \min\{N, d_L\}$.*

Proof. Nonemptiness: $\bar{x} = \sum_i y_i$ with $y_i \in \operatorname{conv} S_i$, and each y_i is a convex combination of points $a_i(p)$. $\Lambda(\bar{x})$ is a bounded polyhedron in standard form $\{\lambda \geq 0 : H\lambda = h\}$ with columns $c_{i,p} = (a_i(p), e_i) \in \mathbb{R}^d \times \mathbb{R}^N$ and $h = (\bar{x}, \mathbf{1})$. A feasible λ is a vertex if and only if the columns $\{c_{i,p} : \lambda_i(p) > 0\}$ are linearly independent. Fix a vertex and pick $p_i^0 \in \operatorname{supp} \lambda_i$ for each i. The family $\{c_{i,p_i^0}\}_i \cup \{c_{i,p} - c_{i,p_i^0} : p \in \operatorname{supp} \lambda_i \setminus \{p_i^0\}\}$ is obtained from the positive columns by an invertible (unitriangular) transformation, so it is linearly independent. Its second part consists of the vectors $(a_i(p) - a_i(p_i^0), 0) \in L \times \{0\}$, which number $\sum_i (|\operatorname{supp} \lambda_i| - 1) \geq |J(\lambda)|$. Linearly independent vectors in $L \times \{0\}$ number at most d_L. Hence $|J| \leq d_L$, and trivially $|J| \leq N$. □

Proof of Lemma 23. (a) $u_i(p; m, \xi)$ is affine in (m, ξ) with fixed coefficients, and P has finite first moments; linearity of the integral gives the identity. (b) Into: by definition. Onto: for $k = (m, \xi) \in K_i^\delta(\sigma)$, the Dirac law at (m, ξ) is supported in $\mathbb{R}^d_{\geq 0} \times \mathbb{R}^Z_{\geq 0}$ because $m \geq 0$ and $\xi \in \Xi \subset \mathbb{R}^Z_{\geq 0}$; its means are (m, ξ), so it belongs to $\mathcal{M}_i^\delta(\sigma)$. (c) Each criterion depends on P only through the numbers $\mathbb{E}_P[u(p)]$ and $\mathbb{E}_P[u(p') - u(p)]$, which are affine functions of the mean by (a). By (b), ranging over P is the same as ranging over $k \in K_i^\delta(\sigma)$, a product of compact intervals; continuous functions attain their extrema on it. □

Proof of Lemma 24. For a finite $\mathcal{P}$ and the compact convex box K with continuous affine payoffs: *$E \subseteq M$.* If p is optimal at k_0, then for every p', $\bar{u}(p'; k_0) - \bar{u}(p; k_0) \leq 0$, so $\min_k[\bar{u}(p') - \bar{u}(p)] \leq 0$. *$M \subseteq ID$.* If p is maximal, then for each p' there is k with $\bar{u}(p; k) \geq \bar{u}(p'; k)$; hence $\max \bar{u}(p; \cdot) \geq \bar{u}(p; k) \geq \bar{u}(p'; k) \geq \min \bar{u}(p'; \cdot)$. *$\Gamma \subseteq M$.* If $\min_k[\bar{u}(p'; k) - \bar{u}(p; k)] > 0$, then $\min \bar{u}(p') > \min \bar{u}(p)$, so p is not Γ-maximin. These inclusions hold carrier by carrier and give the inclusions between equilibrium sets. *Monotonicity.* $K^\delta(c) \subseteq K^{\delta'}(c)$ for every c, and the realised centre $A_{-i}(\sigma)$ does not depend on δ. E: a witness belief stays admissible. M: a dominance strict on the larger set is strict on the smaller one, so non-domination on the smaller set implies it on the larger one. ID: $\overline{\mathbb{E}}$ increases and every $\underline{\mathbb{E}}$ decreases as the set grows. □

Example 40 (Γ-equilibria are not nested)**.** Take one carrier with a fixed external aggregate $c = (1, 2)$, $d = 2$, $M = I$, $w = (2, 1)$, no traffic term, and two plans with $\pi = 0$, $\varphi(A) = (1, 0)$, $\varphi(B) = (0, 1)$. The lower values are $-(1 + 2\delta)$ for A and $-(2 + \delta)$ for B, so $\mathrm{ICNE}_\Gamma(0) = \{A\}$ and $\mathrm{ICNE}_\Gamma(2) = \{B\}$.

Proof of Lemma 25. A minimum of affine functions of the mixture x is concave, and

$$\min_k \textstyle\sum_p x(p)\bar{u}_i(p;k) \geq \sum_p x(p) \min_k \bar{u}_i(p;k)$$

because the left side picks one k for all p. With $\varphi, t \geq 0$ and $M \geq 0$, $\bar{u}_i(p;m,\xi)$ is coordinatewise nonincreasing in (m,ξ). The box $K^\delta(c)$ has the coordinatewise maximum $c+\delta w$, which lies in it (clipping only removes points below 0), and Ξ has the maximum $\overline{\mu}$. So $k^+ = (c+\delta w, \overline{\mu})$ minimises every $\bar{u}_i(p;\cdot)$ and the lower expectation is linear in the mixture. Substituting k^+ with $c = A_{-i}(\sigma)$ gives

$$\bar{u}_i(p;k^+) = \pi_i^\Gamma(p) - \langle \varphi_i(p), MA_{-i}(\sigma)\rangle,$$

the payoff of G_δ^Γ; Γ-maximin is then best response in G_δ^Γ. □

Proposition 41 (Relaxed equilibria are mixed equilibria)**.** *Under Assumption A2, x is a mixed Nash equilibrium of the finite game with payoffs $\bar{u}_i(\sigma_i; A_{-i}(\sigma), \hat{\xi})$ if and only if* $\operatorname{supp} x_i \subseteq \arg\max_p \bar{u}_i(p; \bar{A}_{-i}(x), \hat{\xi})$ *for all i. Such x exist.*

Proof. Under the product law x_{-i}, the expected payoff of a pure plan p is $\mathbb{E}[\bar{u}_i(p; A_{-i}, \hat{\xi})] = \bar{u}_i(p; \mathbb{E}A_{-i}, \hat{\xi})$ because $\bar{u}_i$ is affine in A_{-i}, and $\mathbb{E}A_{-i} = \bar{A}_{-i}(x)$ by linearity. A mixed profile is a Nash equilibrium if and only if each x_i is supported on pure best responses to x_{-i}. Existence follows from Nash [74]. □

Proof of Proposition 26. For a unilateral change $p \to p'$ of carrier i, put $a = a_i(p)$, $a' = a_i(p')$ and $R = A_{-i}$. Then $\Psi(p') - \Psi(p) = \pi_i(p') - \pi_i(p) + \frac{1}{2}\|a'\|_M^2 - \frac{1}{2}\|a\|_M^2 - \frac{1}{2}\|R+a'\|_M^2 + \frac{1}{2}\|R+a\|_M^2 - \gamma\langle t_i(p') - t_i(p), \hat{\xi}\rangle$, which equals $\pi_i(p') - \pi_i(p) - \langle a'-a, MR\rangle - \gamma\langle t_i(p') - t_i(p), \hat{\xi}\rangle = u_i(p') - u_i(p)$, using $\varphi_i = a_i$. A maximiser σ is a pure equilibrium [43]. The point $(A_{-i}(\sigma), \hat{\xi})$ lies in $K_i^\delta(\sigma)$ for every δ, so $\sigma \in \mathrm{ICNE}_E(\delta)$. For Γ, Lemma 25 shows that $\mathrm{ICNE}_\Gamma(\delta)$ is the equilibrium set of G_δ^Γ, which is again D1 with modified private profits and therefore has a potential and a pure equilibrium. □

Proof of Theorem 27. (a) Fix i and $\delta \geq \hat{\delta}(p)$. The anchor $\bar{A}^*_{-i}$ is a sum of convex combinations of nonnegative vectors, hence nonnegative. By definition of $\hat{\delta}(p)$, $|\bar{A}^*_{-i,k} - A_{-i,k}(p)| \leq \hat{\delta}(p) w_k \leq \delta w_k$ for all k, so $(\bar{A}^*_{-i}, \hat{\xi}) \in K_i^\delta(p)$. By Lemma 23(b) the Dirac law at this point belongs to $\mathcal{M}_i^\delta(p)$; under it the expected payoff is $\bar{u}_i(\cdot; \bar{A}^*_{-i}, \hat{\xi})$, which $p_i \in B_i$ maximises. Hence p_i is E-admissible for every i, and the inclusions follow from Lemma 24.

(b), first term. For any $p \in \prod_j B_j$, $A_{-i}(p) - \bar{A}^*_{-i} = \sum_{j\neq i}(a_j(p_j) - \bar{a}_j)$ and $|a_{jk}(p_j) - \bar{a}_{jk}| \leq r_{jk}$.

(b), second term. $\bar{A}^* \in \sum_j \operatorname{conv} a_j(B_j)$ because $\operatorname{supp} x_j^* \subseteq B_j$. Apply Theorem 39 with $\mathcal{P}_j$ replaced by B_j: a vertex λ of $\Lambda(\bar{A}^*)$ exists and can be found by linear programming. Let $J = J(\lambda)$, $|J| \leq k^\sharp$, and $y_j = \sum_p \lambda_j(p) a_j(p)$. For $j \notin J$, λ_j is a point mass at some $p_j \in B_j$ and $a_j(p_j) = y_j$; for $j \in J$ choose any $p_j \in \operatorname{supp}\lambda_j \subseteq B_j$. Since $\sum_j y_j = \bar{A}^* = \sum_j \bar{a}_j$,

$$A_{-i}(p) - \bar{A}^*_{-i} = \textstyle\sum_{j\in J}(a_j(p_j) - y_j) - (a_i(p_i) - \bar{a}_i). \tag{A.1}$$

For $j \in J$ both $a_{jk}(p_j)$ and y_{jk} lie in $[\min_{B_j} a_{jk}, \max_{B_j} a_{jk}]$, so the sum is at most $k^\sharp \bar{\sigma} w_k$ in coordinate k, and the last term is at most $\bar{\rho} w_k$.

(b), third term. If $J = \emptyset$ or $\bar\sigma = 0$, the sum vanishes. Otherwise draw $p_j \sim \lambda_j$ independently for $j \in J$. For each k the variables $(a_{jk}(p_j) - y_{jk})/w_k$, $j \in J$, are independent, have mean zero and range in an interval of length at most $\bar\sigma$. By Hoeffding's inequality and a union bound over k, $\Pr(\|E\|_{w,\infty} \geq t) \leq 2d\exp(-2t^2/(k^\sharp\bar\sigma^2))$, where E is the sum in (A.1). This is below one for $t > t_0 = \bar\sigma\sqrt{k^\sharp \ln(2d)/2}$; outcomes are finitely many, so some outcome has $\|E\|_{w,\infty} \leq t_0$, and $\hat\delta(p) \leq t_0 + \bar\rho$ for it.

(c) Relaxed equilibria exist (Proposition 41). Use $r_{jk} \leq s_{jk}$, $\bar\rho \leq \bar\sigma \leq \sigma_{\max}$, $k^\sharp \leq d$ and $\sum_{j\neq i} r_{jk}/w_k \leq (N-1)\sigma_{\max}$; the order of thresholds follows from Lemma 24. *(d)* This was shown in (b); linear programming finds a vertex in time polynomial in $\sum_j |B_j|$ and d. □

Proof of Corollary 28. Let $m^* = \bar A^*_{-i}$ and $m = A_{-i}(p)$. For any p', $\bar u_i(p'; m, \mu) - \bar u_i(p_i; m, \mu) = [\bar u_i(p'; m^*, \hat\xi) - \bar u_i(p_i; m^*, \hat\xi)] - \langle \varphi_i(p') - \varphi_i(p_i), M(m - m^*)\rangle - \gamma\langle t_i(p') - t_i(p_i), \mu - \hat\xi\rangle$. The bracket is nonpositive because $p_i \in B_i$; each term $|M_k(\varphi_{ik}(p') - \varphi_{ik}(p_i))(m_k - m^*_k)|$ is at most $w_k M_k |\Delta\varphi_{ik}|\hat\delta(p)$, and $|\mu_z - \hat\xi_z| \leq \varpi_z$. □

Proof of Proposition 31. For carrier 1, $\bar u(A_1; n) - \bar u(B_1; n) = 6 - 3n$, so A_1 is optimal iff $n \leq 2$; for carrier 2, $\bar u(Y; n) - \bar u(X; n) = 6 - 3n$, so Y is optimal iff $n \leq 2$. With two plans and a convex belief set, E-admissibility and maximality coincide: some n in the box makes the plan weakly better. In (A_1, X) carrier 2 sees $n = 1$ and needs a belief ≥ 2; in (A_1, Y) carrier 1 sees $n = 3$ and needs a belief ≤ 2; in (B_1, Y) carrier 2 sees $n = 3$ and needs ≤ 2; in (B_1, X) carrier 1 sees $n = 1$ and needs ≥ 2. Each requirement holds iff $\delta w \geq 1$, and the other carrier best-responds at the realised count. Hence $\mathrm{ICNE}_E(\delta) = \mathrm{ICNE}_M(\delta) = \emptyset$ for $\delta < 1/w$ and both equal the full profile set at $\delta = 1/w$; there is no pure Nash equilibrium. The unique mixed equilibrium makes each carrier indifferent, so the rival's mean count is 2, both mix $\frac{1}{2}$–$\frac{1}{2}$, $r_j = 1$, $\bar\rho = 1/w$ and $(N-1)\bar\rho = 1/w$. □

Proof of Proposition 32. Take two carriers with plan sets $\{A_1, B_1\}^Z$ and $\{X, Y\}^Z$, the one-zone data of Proposition 31 in each zone and no cross-zone interaction, and $N - 2$ carriers with a single null plan. Payoffs are sums over zones and plan sets are products, so a carrier's regret is the sum of its zone regrets. In each zone and each of the four zone profiles, exactly one of the two carriers has regret 3 and the other 0 (e.g. in (A_1, X) carrier 2 gains $2 - (-1) = 3$ by switching to Y). So $\varepsilon_1 + \varepsilon_2 = 3Z$ with both multiples of 3, and $\max \geq 3\lceil Z/2\rceil$. For the upper bound, the zone-wise mixed equilibrium has $\bar\rho = 1$, every $p \in \prod B_j$ has $\hat\delta(p) \leq 1$, and $L_i(p) \leq \sum_z 1 \cdot 1 \cdot |\Delta\tau_z| = 3Z$. □

Proof of Theorem 33. Membership. The witness is a profile σ. With K the clipped box and Ξ a point: E-admissibility of σ_i is feasibility of the linear system $\{m \in K : \bar u_i(\sigma_i; m) \geq \bar u_i(p'; m)\ \forall p'\}$; maximality requires $\max_{m\in K}[\bar u_i(\sigma_i; m) - \bar u_i(p'; m)] \geq 0$ for each p', a linear function maximised over a box coordinatewise; ID and Γ require maxima and minima of $\bar u_i(p; \cdot)$ over the box, also coordinatewise.

Construction. Let $V(D) = [n]$ and $\Delta^- = \max(1, \text{max in-degree})$. Create one zone per arc $e = (u, v)$, using only its vehicle coordinate, with $M_e = \gamma c_e = 1$, $w_e = 1$ and no deliveries. Set $\eta = 1/(2\Delta^- + 2)$. Carrier i has the plan *out* with all data 0, and the plan *in* with $\pi_i = 1$ and, for each out-arc $e = (i, j)$, vehicles $n_{ie} = 1$ and exposure $\tau_{ie} = 1$; for each in-arc $e = (h, i)$, vehicles $n_{ie} = 2$ and exposure $\tau_{ie} = \eta$; zero elsewhere. This is D2, since $\varphi = \tau \neq a = n$.

Payoff gap. The gain of *in* at belief m is $D_i(m) = 1 - \sum_{e=(i,j)} m_e - \eta \sum_{e=(h,i)} m_e$. At the realised aggregate of σ^S, $m_e = 2 \cdot \mathbf{1}[j \in S]$ for $e = (i,j)$ and $m_e = \mathbf{1}[h \in S]$ for $e = (h,i)$, so $D_i^{\text{real}} = 1 - 2o_i - \eta h_i$ with $o_i = |N^+(i) \cap S|$ and $h_i = |N^-(i) \cap S| \le \Delta^-$. If $o_i = 0$, $D_i^{\text{real}} \ge 1 - \Delta^-/(2\Delta^- + 2) > \frac{1}{2}$; if $o_i \ge 1$, $D_i^{\text{real}} \le -1$.

Robustness of the sign. Over K^δ, $|D_i(m) - D_i^{\text{real}}| \le \delta(\deg^+(i) + \eta \deg^-(i)) < \delta(\Delta^+ + \frac{1}{2})$, which is below $\frac{1}{2}$ for $\delta < 1/(2\Delta^+ + 1)$; clipping only shrinks the box. So D_i has the sign of D_i^{real} on the whole box.

All criteria select the realised strict best response. If $D_i > 0$ on the box, *in* is the unique optimum at every belief, hence the only E-admissible plan; *out* is strictly dominated everywhere, so *in* is the only maximal plan; and $\min_K \bar{u}(\text{in}) > 0 = \max_K \bar{u}(\text{out})$, so *out* is interval-dominated and *in* is the unique Γ-maximin plan. The case $D_i < 0$ is symmetric. So σ is admissible under every criterion iff each carrier plays *in* exactly when $o_i = 0$.

Equilibria are kernels. With $S = \{i : \sigma_i = \text{in}\}$, the condition reads $i \in S \Rightarrow N^+(i) \cap S = \emptyset$ (independence) and $i \notin S \Rightarrow N^+(i) \cap S \neq \emptyset$ (absorption). NP-hardness follows from Chvátal [76]; all numbers are polynomially bounded, so hardness is strong; with $\Delta^+ \le 2$ [77] the threshold is $1/5$. □

Proof of Corollary 34. On the instances of Theorem 33, $\delta_0^E = 0$ if D has a kernel; otherwise $\text{ICNE}_E(\delta) = \emptyset$ for all $\delta < 1/(2\Delta^+ + 1)$, so $\delta_0^E \ge 1/(2\Delta^+ + 1)$. A multiplicative approximation outputs 0 in the first case and a positive number in the second; an additive one separates the cases when its error is below $1/(2\Delta^+ + 1)$. □

Proposition 42 (The relaxation step is PPAD-hard under D2)**.** *Every bimatrix game $(R, C) \in \mathbb{Q}^{m\times n} \times \mathbb{Q}^{m\times n}$ is, in polynomial time, the precise game of a D2 instance with $N = 2$ and $d = m + n$. Hence computing a mixed equilibrium under D2 is PPAD-hard.*

Proof. Let $M = I_{m+n}$, no traffic term, $\bar{R} = \max R$ and $\bar{C} = \max C$. Carrier 1 has plans $p \in [m]$ with $a_1(p) = e_p$, $\varphi_1(p) = \sum_q (\bar{R} - R_{pq}) e_{m+q} \ge 0$ and $\pi_1(p) = \bar{R}$; carrier 2 has plans $q \in [n]$ with $a_2(q) = e_{m+q}$, $\varphi_2(q) = \sum_p (\bar{C} - C_{pq}) e_p \ge 0$ and $\pi_2(q) = \bar{C}$. Then $u_1(p; a_2(q)) = R_{pq}$ and $u_2 = C_{pq}$. By Proposition 41, relaxed equilibria are exactly the Nash equilibria of (R, C), and computing these is PPAD-complete [78, 79]. □

Proof of Lemma 35. Write $k^t = (A_{-i}(\sigma), \mu) \in K_i^\delta(\sigma)$ and $\Delta(p', k) = \bar{u}_i(p'; k) - \bar{u}_i(\sigma_i; k)$, affine in k. For any k in the (unclipped) box, $|\Delta(p', k) - \Delta(p', k^t)| \le \delta \sum_k w_k M_k |\Delta\varphi_{ik}| + \gamma \sum_z \varpi_z |\Delta t_{iz}|$. *E:* take the witness k, at which $\Delta(p', k) \le 0$. *M:* for each p', maximality and compactness give $k_{p'}$ with $\Delta(p', k_{p'}) \le 0$. *ID:* $\bar{u}_i(p'; k^t) - \bar{u}_i(\sigma_i; k^t) \le [\bar{u}_i(p'; k^t) - \min_K \bar{u}_i(p')] + [\max_K \bar{u}_i(\sigma_i) - \bar{u}_i(\sigma_i; k^t)]$, since $\max_K \bar{u}_i(\sigma_i) \ge \min_K \bar{u}_i(p')$, and each bracket is at most $\delta H_i + \gamma \Theta_i$. □

Proof of Proposition 36. The minimum over a growing finite set does not increase and the maximum does not decrease (Lemma 24). Each profile enters $\text{ICNE}_C(\delta)$ at a single threshold and never leaves, so the sets change at most at $\prod_i |\mathcal{P}_i|$ values. For Γ see Example 40. □

Proposition 43 (Price of anarchy under smoothness; conditional)**.** *Assume $W(\sigma) \ge \sum_i U_i(\sigma)$ for all σ, $W^* > 0$, and that the precise game is (λ, μ)-smooth: $\sum_i U_i(\sigma_i^*, \sigma_{-i}) \ge \lambda W(\sigma^*) - \mu W(\sigma)$ for all σ and an optimal σ^*. For $C \in \{E, M, ID\}$ let $\mathcal{E}(\delta) = \max_{\sigma \in \text{ICNE}_C(\delta)} \sum_i \varepsilon_i(\sigma)$ from Lemma 35. If $\lambda W^* > \mathcal{E}(\delta)$, then $\text{PoA}_C(\delta) \le (1 + \mu)/(\lambda - \mathcal{E}(\delta)/W^*)$.*

Proof. For $\sigma \in \mathrm{ICNE}_C(\delta)$: $W(\sigma) \geq \sum_i U_i(\sigma) \geq \sum_i [U_i(\sigma_i^*, \sigma_{-i}) - \varepsilon_i] \geq \lambda W^* - \mu W(\sigma) - \mathcal{E}(\delta)$. Rearrange. □

Proof of Proposition 37. With π_i as in Section 3 and $x = \tau$, direct expansion gives $W - \Psi_{\hat{\mu}} = \sum_z [\frac{b_z}{2} \sum_i q_{iz}^2 - \frac{\gamma c_z}{2}(T_z^2 - \sum_i \tau_{iz}^2)]$; the α, $\hat{\mu}$ and C_i terms cancel. By Cauchy–Schwarz, $\sum_i \tau_{iz}^2 \geq T_z^2/N$, and $\sum q^2 \geq 0$, so $W(\sigma) \geq \Psi_{\hat{\mu}}(\sigma) - \sum_z \frac{\gamma c_z}{2}(1 - \frac{1}{N}) T_z(\sigma)^2$. Next, $\Psi_{\hat{\mu}}(\sigma) \geq \Psi_{\hat{\mu}}(\sigma^*) - \eta$, and $\Psi_{\hat{\mu}}(\sigma^*) \geq W^* - \sum_z \frac{b_z}{2} \sum_i q_{iz}^{*2}$ because $T^2 \geq \sum \tau^2$ for $\tau \geq 0$. Membership of a potential maximiser in every $\mathrm{ICNE}_E(\delta)$ is Proposition 26, extended to M and ID by Lemma 24. □

Theorem 44 (Exact thresholds and regret brackets)**.** *Let p be a pure profile. (a) The exact E-threshold of carrier i is the optimal value of the linear program $A_i^E(p) = \min\{\delta \geq 0 : m \geq 0,\ \xi \in \Xi,\ |m_k - A_{-i,k}(p)| \leq \delta w_k\ \forall k,\ \bar{u}_i(p_i; m, \xi) \geq \bar{u}_i(p'; m, \xi)\ \forall p' \in \mathcal{P}_i\}$ ($+\infty$ if infeasible), and $p \in \mathrm{ICNE}_E(\delta)$ iff $\delta \geq \max_i A_i^E(p)$. With an exact oracle and implicit $\mathcal{P}_i$, Kelley's cutting-plane method on this program terminates finitely. (b) For anchors $m^i \geq 0$, $\xi^i \in \Xi$ with $p_i \in \arg\max \bar{u}_i(\cdot; m^i, \xi^i)$, $A_i^E(p) \leq \|A_{-i}(p) - m^i\|_{w,\infty}$. (c) For a heuristic plan h_i and a valid bound $\mathcal{U}_i \geq \max_{p'} \bar{u}_i(p'; m, \xi)$, the regret of p_i lies in $[\bar{u}_i(h_i) - \bar{u}_i(p_i),\ \mathcal{U}_i - \bar{u}_i(p_i)]$.*

Proof. (a) By Lemma 23, p_i is E-admissible at width δ iff the system with that δ is feasible; feasibility is upward closed in δ (Lemma 24) and the infimum of an LP is attained when finite. Kelley's method adds the oracle's best plan at the current (m, ξ) as a violated constraint; a plan already present cannot be violated, so cuts are distinct and finitely many. (b) The anchor is feasible with $\delta = \|A_{-i}(p) - m^i\|_{w,\infty}$. (c) Immediate from the definitions. □

Theorem 45 (Rounding guarantee under D2)**.** *Under D2, if the relaxation phase returns a mixed equilibrium x, the sparsify phase (a feasibility LP over $\Lambda(\bar{A}(x))$ with plan sets $\operatorname{supp} x_j$) and the round phase return $p \in \prod_j B_j(x)$ with $p \in \mathrm{ICNE}_E(\delta)$ for all $\delta \geq \hat{\delta}(p)$; $\hat{\delta}(p)$ satisfies the second bound of Theorem 27(b) for any choice in the round phase and the third for a successful sample. Sparsify and round run in time polynomial in $\sum_j |\operatorname{supp} x_j|$ and d.*

Proof. The sparsify LP's vertices have $|J| \leq k^\sharp$ by Theorem 39. The rest is Theorem 27(a), (b) and (d). □

The double-oracle relaxation [80] stops, with exact restricted solves and exact oracles, after at most $\sum_i |\mathcal{P}_i|$ rounds at a relaxed equilibrium: at termination every support plan attains the restricted maximum and no plan exceeds it, and each non-terminal round adds a new plan.

Appendix B. Proofs for BRL

Proof of Theorem 1. Write $S = \sum_{a \in r} X_a$ (arcs counted with multiplicity). Fix j and $\Theta = \Theta_j$.

Step 1 (pointwise inequality). For reals x_a, d_a with $\sum_a d_a \leq \Theta$, $(\sum_a x_a - \Theta)^+ \leq (\sum_a (x_a - d_a))^+ \leq \sum_a (x_a - d_a)^+$, because $t \mapsto t^+$ is nondecreasing and subadditive.

Step 2 (upper bound). Let $P \in \mathcal{M}$ with marginals F_a. Taking expectations, $\mathbb{E}_P (S - \Theta)^+ \leq \sum_a \mathbb{E}_{F_a} (X_a - d_a)^+$. Since $F_a \geq \underline{F}_a$ pointwise, F_a is first-order dominated by $\underline{F}_a$, and $x \mapsto (x - d)^+$ is nondecreasing, so $\mathbb{E}_{F_a} (X_a - d_a)^+ \leq \overline{\Pi}_a(d_a)$; likewise $\mathbb{E}_{F_a} X_a \leq \bar{x}_a$. Hence $\mathbb{E}_P \phi(S) \leq c_0 + \gamma \sum_a \bar{x}_a + \sum_j \rho_j \sum_a \overline{\Pi}_a(d_a^j)$ for every P and admissible d^j.

Step 3 (attainment). Let U be uniform on $(0,1)$, $\bar{X}_a = \overline{Q}_a(U)$ and $\bar{S} = \Sigma(U)$ with $\Sigma(u) = \sum_a \overline{Q}_a(u)$; then $\bar{X}_a \sim \underline{F}_a$ and the law of $(\bar{X}_a)$ belongs to $\mathcal{M}$. Let $u^* = \sup\{u \in (0,1] : \Sigma(u) \le \Theta\}$ ($\sup \emptyset = 0$). If $u^* = 1$, put $d_a = \overline{Q}_a(1)$; then $\sum_a d_a \le \Theta$ and both sides of Step 1 vanish almost surely. If $u^* < 1$, put $\ell_a = \overline{Q}_a(u^*)$ (or $\overline{Q}_a(0^+)$ if $u^* = 0$) and $h_a = \overline{Q}_a(u^{*+})$; then $\sum_a \ell_a \le \Theta$ when $u^* > 0$ and $\sum_a h_a \ge \Theta$. If $u^* = 0$ choose $d_a \le \ell_a$ with $\sum_a d_a = \Theta$; otherwise choose $d_a \in [\ell_a, h_a]$ with $\sum_a d_a = \Theta$. On $\{U \le u^*\}$ all $\bar{X}_a \le d_a$ and both sides of Step 1 are 0; on $\{U > u^*\}$ all $\bar{X}_a \ge d_a$ and $\sum_a \left(\bar{X}_a - d_a\right)^+ = \bar{S} - \Theta = \left(\bar{S} - \Theta\right)^+$. So Step 1 holds with equality almost surely, and the comonotone law attains the bound of Step 2; the minima are attained.

Step 4. $\mathbb{E}\phi(\bar{S}) = \int_0^1 \phi(\Sigma(u))du$ because $\bar{S} = \Sigma(U)$.

Step 5 (integrality). With integer quantiles and Θ, the numbers ℓ_a, h_a, $\overline{Q}_a(1)$ are integers and integer vectors with the required properties exist in each case of Step 3. □

Proof of Theorem 2. Substitute the budget form of Theorem 1 into the objective and merge the two minimisations. □

Lemma 46 (Concatenation is inf-convolution)**.** *For a label p and a completion q, $\overline{\mathbb{E}}[\phi(T_{p\oplus q})] - c_0 = f_{p\oplus q}(T_{\max}) = \min_D\{f_p(D) + f_q(T_{\max} - D)\}$.*

Proof. The stop-loss terms are separable over arcs. Split the budget vector into the arcs of p and of q, put $D = \sum_{a\in p} d_a$, minimise first over the arcs of p with budget D and then over those of q with budget $T_{\max} - D$, and apply Theorem 1 to each part. The mean terms add. □

Lemma 47 (Completion window)**.** *Let $C_{\min}(v) \le S_{q,1}$ for every completion q of labels ending at v. For every completion q, the minimum in Lemma 46 is attained at some $D^* \le \overline{D}(p)$; if $D^* < \underline{D}(p)$ then $D^* \le S_{p,1}$.*

Proof. Let $\Phi_q(D) = f_p(D) + f_q(T_{\max} - D)$. The map $D \mapsto f_q(T_{\max} - D)$ is convex and nondecreasing with slope in $[0, \rho]$, equal to ρ when $T_{\max} - D < S_{q,1}$; the slope of f_p lies in $[-\rho, 0]$ and is 0 for $D > S_{p,L}$. (i) On $D > T_{\max} - S_{q,1}$, Φ_q has slope ≥ 0, so a minimiser exists with $D \le T_{\max} - S_{q,1} \le T_{\max} - C_{\min}(v)$. (ii) On $D > S_{p,L}$, Φ_q has slope ≥ 0, so decreasing D to $S_{p,L}$ does not increase it. Applying (i) and then (ii) gives $D^* \le \overline{D}(p)$. The last claim is immediate from $\underline{D}(p) \le S_{p,1}$. □

Proof of Theorem 4. Validity. Let q be a completion of p_2; since $\mathcal{V}(p_1) \subseteq \mathcal{V}(p_2)$ and $\ell(p_1) \le \ell(p_2)$, it is a completion of p_1. By Lemma 47 there is a minimiser $D^* \le \overline{D}(p_2)$ for $p_2 \oplus q$. Let g be the difference in Definition 3. If $D^* \in \mathcal{W}(p_2)$, $g(D^*) \le 0$. Otherwise $D^* < \underline{D}(p_2) \le S_{p_2,1}$; on $(-\infty, S_{p_2,1}]$, f_{p_2} has slope $-\rho$ and f_{p_1} slope $\ge -\rho$, so g is nondecreasing there and $g(D^*) \le g(\underline{D}(p_2)) \le 0$. In both cases, by Lemma 46, $f_{p_1\oplus q}(T_{\max}) - y(p_1) \le f_{p_1}(D^*) + f_q(T_{\max} - D^*) - y(p_1) \le f_{p_2\oplus q}(T_{\max}) - y(p_2)$.

Exactness. Call a label processed if it is popped while not deleted. Let OPT be the optimum and Ω the set of pairs (o, q) with o a created label, q a completion of o and reduced cost of $o \oplus q$ equal to OPT. The pair (depot label, optimal route) is in Ω and the depot label is processed. Among pairs in Ω with o processed choose one with the fewest arcs $|q|$. If $|q| = 1$, closing o records a route of value OPT. If $|q| \ge 2$, let (v, j) be the first arc of q, q' the rest and $o_1 = o \oplus (v, j)$, which is created when o is processed; $(o_1, q') \in \Omega$. If o_1 is processed,

$|q'| < |q|$ contradicts minimality. Otherwise o_1 was discarded or deleted because a live label o_2 functionally dominates it; by validity $(o_2, q') \in \Omega$, and if o_2 is not processed it was deleted later by some o_3, with $(o_3, q') \in \Omega$ by transitivity. The moments at which these labels cease to be live strictly increase and labels are finitely many, so the chain ends at a processed label with $(o_k, q') \in \Omega$ and $|q'| < |q|$, a contradiction. □

Proposition 48 (Comparison with vector dominance)**.** *Assume $\mathcal{V}(p_1) \subseteq \mathcal{V}(p_2)$, $\ell(p_1) \le \ell(p_2)$ and $y(p_1) \ge y(p_2)$. (a) If $S_{p_1} \le S_{p_2}$ componentwise, p_1 functionally dominates p_2. (b) In the sector model $X_a = \sum_k \tau_{a,k}\xi_k$ with probability boxes on the multipliers, $\tau_{p_1} \le \tau_{p_2}$ componentwise implies $S_{p_1} \le S_{p_2}$. (c) Neither converse holds.*

Proof. (a) f is nondecreasing in each S_l. (b) The upper quantiles of the multipliers are nonnegative. (c) With $L = 2$, $\omega = (\frac{1}{2}, \frac{1}{2})$, $\gamma = 0$, $\rho = 1$ and equal duals: $S_{p_1} = (1, 1)$ and $S_{p_2} = (0, 2.2)$ give $f_{p_1} \le f_{p_2}$ on $\mathbb{R}$ (values $(1, 1.1)$, $(0, 0.6)$, $(0, 0)$ at $D = 0, 1, 2.2$, constant difference -0.1 below 0) although $S_{p_1,1} > S_{p_2,1}$. □

Proposition 49 (A single risk level is not exact)**.** *For every u, $\min_{\text{plan}} \sum_r \overline{\mathbb{E}}[\phi(T_r)] \ge \max_u \min_{\text{plan}} \sum_r [c_0 + \gamma \bar{x}_r + \rho(\mathcal{H}_r(u) - (1-u)T_{\max})]$, where $\mathcal{H}_r(u) = \int_u^1 \sum_{a \in r} \overline{Q}_a(v)\, dv$; the inequality can be strict and the best fixed-u plan need not be optimal.*

Proof. The maximum over u inside each route is at least its value at a common u, and $\min \max \ge \max \min$. Strictness is shown by exact computations by enumeration and set partitioning (Section 7.1); structurally, the optimal level of a route is its worst-case on-time level, which differs between routes of the same plan. □

Appendix C. Zone-market clearing (ZMC)

ZMC is the first phase of the equilibrium engine (Section 4.3). It is stated here with its guarantees and proofs; Remark 5 gives the negative empirical result.

ZMC targets the D1 game in cost form. Let $c_i(p) = -\pi_i(p) + \gamma \langle t_i(p), \hat{\xi} \rangle$ (charges included), $\tilde{c}_i = c_i - \frac{1}{2}\|a_i\|_M^2$, and assume $M_k > 0$ on the active coordinates. Then $\Phi(\sigma) = \sum_i \tilde{c}_i(\sigma_i) + \frac{1}{2}\|A(\sigma)\|_M^2 = -\Psi_{\hat{\xi}}(\sigma)$ is an exact potential (Proposition 26). For a common price $\lambda \in \mathbb{R}^d$, the pricing oracle returns $O_i(\lambda) = \arg\min_p \{\tilde{c}_i(p) + \langle \lambda, a_i(p) \rangle\}$, which is the oracle of Assumption A3 with $Q = M$. Let $V_i(\lambda)$ be its value, $\mathcal{D}(\lambda) = \sum_i V_i(\lambda) - \frac{1}{2}\|\lambda\|_{M^{-1}}^2$ the price dual of the convexified potential, λ^* its unique maximiser, and $\bar{\Phi}^* = \max \mathcal{D} \le \min \Phi$ (Lemma 53).

Theorem 50 (Clearing identity)**.** *Let $\sigma_i \in O_i(\lambda)$ for all i and let $e = M^{-1}\lambda - A(\sigma)$ be the clearing error (supply minus demand at price λ). Then (a) $\Phi(\sigma) - \mathcal{D}(\lambda) = \frac{1}{2}\|e\|_M^2$, so $\Phi(\sigma) - \min \Phi \le \frac{1}{2}\|e\|_M^2$; (b) carrier i's Nash regret is at most $\frac{1}{2}\sum_{k \in \mathcal{Z}_i} M_k e_k^2$, where $\mathcal{Z}_i$ is the set of coordinates in which its plans differ; (c) σ_i is an exact best response to the belief $m_i = M^{-1}\lambda - a_i(\sigma_i) = A_{-i}(\sigma) + e$, hence E-admissible at every width $\delta \ge \max_{k \in \mathcal{Z}_i} |e_k|/w_k$ (if $m_{ik} \ge 0$ on $\mathcal{Z}_i$); (d) $e = 0$ implies that σ minimises Φ and is a pure Nash equilibrium; (e) conversely, exact clearing is attainable if and only if the convexification of the potential is tight at a pure profile.*

Part (a) is Fenchel–Young equality and a standard duality-gap fact. The game-specific content is in (b) and (c): the own-contribution term of the absorption bound cancels, because

Algorithm 4 ZMC: zone-market clearing for pure equilibria of the D1 game

Require: pricing and best-response oracles; rounds T

$l \leftarrow 0$, $u \leftarrow MA^{\max}$, $\lambda \leftarrow (l+u)/2$; plan pools $\mathcal{Q}_i \leftarrow \emptyset$

for $t = 0, \dots, T-1$ **do**

  for all i in parallel: $\sigma_i^t \leftarrow O_i(\lambda)$; $\mathcal{Q}_i \leftarrow \mathcal{Q}_i \cup \{\sigma_i^t\}$ ▷ N routing solves

  $s \leftarrow A(\sigma^t) - M^{-1}\lambda$; update every bracket by the sign of s_k (Theorem 51)

  if t odd: $\lambda \leftarrow (l+u)/2$; **else** for each k: secant step on an isotonic fit of (λ_k, A_k), clipped to $[l_k, u_k]$

$\sigma^{\text{price}} \leftarrow$ the pricing profile with the smallest $\frac{1}{2}\|e^t\|_M^2$

$\sigma^{\text{round}} \leftarrow$ restricted master over the pools and derandomised rounding (no oracle calls)

$\sigma \leftarrow$ the better of the two; sequential best responses until no carrier improves

return σ

under D1 the own quadratic term turns a pricing-optimal plan into a best response to the anchor m_i. Finding a good pure equilibrium thus becomes a price search: find one λ at which the carriers' pricing answers nearly clear every zone. Because $\mathcal{D}$ is concave, the aggregate response is antitone in λ, and along each coordinate the sign of the residual $s_k = A_k(\lambda) - \lambda_k/M_k$ localises λ_k^* up to $M_k\chi_k$, where χ_k is the cross-zone variation: how far other coordinates' prices can move the response in coordinate k at a fixed own price.

Theorem 51 (Bracket invariant and dimension-free rate)**.** *Start from brackets $[0, M_k A_k^{\max}]$ and, after each round, move the lower (upper) end of bracket k to λ_k when $s_k > 0$ ($s_k < 0$). Then $\operatorname{dist}(\lambda_k^*, [l_k, u_k]) \le M_k\chi_k$ after every round. If the bracket midpoint is queried at least every other round, after T rounds (NT oracle calls, independent of d) every point of the bracket satisfies $|\lambda_k - \lambda_k^*| \le M_k A_k^{\max} 2^{-\lceil T/2 \rceil} + M_k\chi_k$ for all k. If plan sets are separable across coordinates, $\chi = 0$.*

The floor $M_k\chi_k$ is genuine when zones are complementary (Example 54). Algorithm 4 alternates midpoint steps (for the rate) with secant steps on isotonic fits of each coordinate's response (for speed), then rounds over the collected plans and finishes with best-response polish.

Theorem 52 (Guarantee of ZMC)**.** *With exact oracles, ZMC uses NT pricing calls before the polish; the polish stops after finitely many best-response calls at a pure Nash equilibrium σ with $\Phi(\sigma) \le \bar{\Phi}^* + \frac{1}{2}\min_t \|e^t\|_M^2$; and $\sigma \in \mathrm{ICNE}_C(\delta)$ for all $\delta \ge 0$ and $C \in \{E, M, ID\}$. Run on the surcharged game of Lemma 25, it returns an exact Γ-maximin equilibrium at width δ.*

Charges translate the supply line: under charges t the equilibrium gross price is 1-Lipschitz in t and convexified aggregates respond monotonically (Proposition 55), so brackets and plan pools warm-start the next evaluation when a city leader searches over charges.

Prior art and limits. Price decomposition with a pricing oracle is classical (Dantzig–Wolfe, column generation and bundle methods; 88); learning separable piecewise-linear concave approximations from sampled slopes is SPAR [89]; Walrasian prices with indivisible goods are computed from demand oracles by Paes Leme and Wong [90]; and Duguet et al. [65] relate the regret of price-allocation pairs to duality and integrality gaps in combinatorial markets, which

is the closest conceptual neighbour of Theorem 50(a)–(b). We claim novelty only for the own-term cancellation with credal anchors (Theorem 50(b),(c),(e)), the sign-bracket invariant with the cross-variation floor (Theorem 51) and the method they define. ZMC has no D2 version, does not bound the clearing error of pure answers a priori, and spends NT calls even on easy games.

Proofs

Lemma 53 (Price duality)**.** *Let* $\bar\Phi(x) = \sum_i \sum_p x_i(p)\tilde c_i(p) + \frac12\|\bar A(x)\|_M^2$ *on* $\prod_i \Delta(\mathcal{P}_i)$, $\bar\Phi^* = \min\bar\Phi$ *and* $\mathcal{B} = \prod_k [0, M_k A_k^{\max}]$ *with* $A_k^{\max} = \sum_i \max_p a_{ik}(p)$. *(a)* $\mathcal{D}$ *is strongly concave with a unique maximiser* λ^**, and* $\max\mathcal{D} = \bar\Phi^* \le \min_\sigma \Phi(\sigma)$. *(b) All minimisers of* $\bar\Phi$ *share the aggregate* $\bar A^*$*, and* $\lambda^* = M\bar A^* \in \mathcal{B}$. *(c)* $M^{-1}\lambda^* \in R(\lambda^*)$*, where* $R(\lambda) = \partial^+ \sum_i V_i(\lambda)$.

Proof. (a) $V = \sum_i V_i$ is concave and finite and $\frac12\|\lambda\|_{M^{-1}}^2$ strongly convex, so $\mathcal{D}$ is strongly concave, continuous and coercive, with a unique maximiser. For $L(x,\lambda) = \sum_i \langle x_i, \tilde c_i\rangle + \langle \lambda, \bar A(x)\rangle - \frac12\|\lambda\|_{M^{-1}}^2$, affine in x on a compact convex set and concave in λ, Sion's theorem gives $\min_x \sup_\lambda L = \sup_\lambda \min_x L$. For fixed x the supremum is at $\lambda = M\bar A(x)$ with value $\bar\Phi(x)$; for fixed λ, $\min_x L = \mathcal{D}(\lambda)$ because a linear function on a product of simplices is minimised at a vertex carrier by carrier. Pure profiles are vertices, where $\bar\Phi = \Phi$. (b) Two minimisers with different aggregates would contradict strict convexity of $\frac12\|\cdot\|_M^2$. A minimiser x^* and λ^* form a saddle point, and the unique maximiser of $L(x^*,\cdot)$ is $M\bar A(x^*)$. Bounds on $\bar A^*$ follow from nonnegativity. (c) This is $0 \in \partial^+\mathcal{D}(\lambda^*) = R(\lambda^*) - M^{-1}\lambda^*$, by the sum rule for concave functions [91]. □

Proof of Theorem 50. For any pure σ and λ, with $A = A(\sigma)$ and $e = M^{-1}\lambda - A$,

$$\begin{aligned}\textstyle\sum_i[\tilde c_i(\sigma_i) + \langle\lambda, a_i(\sigma_i)\rangle] - \tfrac12\|\lambda\|_{M^{-1}}^2 &= \textstyle\sum_i \tilde c_i(\sigma_i) + \langle M(A+e), A\rangle - \tfrac12\|A+e\|_M^2\\ &= \Phi(\sigma) - \tfrac12\|e\|_M^2. \qquad (*)\end{aligned}$$

(a) If $\sigma_i \in O_i(\lambda)$ the left side is $\mathcal{D}(\lambda) \le \bar\Phi^* \le \min\Phi$. (b) Fix i, $a = a_i(\sigma_i)$, $a' = a_i(p)$, $\Delta = a' - a$ (zero outside $\mathcal{Z}_i$). Since $MA_{-i} = \lambda - M(a+e)$, the change in carrier i's cost $c_i + \langle a_i, MA_{-i}\rangle$ equals $[\tilde c_i(p) + \langle\lambda, a'\rangle] - [\tilde c_i(\sigma_i) + \langle\lambda, a\rangle] + \frac12\|\Delta\|_M^2 - \langle\Delta, Me\rangle \ge -\frac12\sum_{k\in\mathcal{Z}_i} M_k e_k^2$, minimising coordinatewise at $\Delta_k = e_k$. (c) For any p, $c_i(p) + \langle a', Mm_i\rangle = [\tilde c_i(p) + \langle\lambda, a'\rangle] + \frac12\|a'-a\|_M^2 - \frac12\|a\|_M^2$, minimised by σ_i. Coordinates outside $\mathcal{Z}_i$ shift all plans' costs equally, so the belief can be replaced by one equal to A_{-i} there; within $\mathcal{Z}_i$ it deviates from A_{-i} by $|e_k| \le \delta w_k$, and under D1 the Dirac law at it lies in the credal set (Lemma 23). (d) If $e = 0$, $\Phi(\sigma) = \mathcal{D}(\lambda) \le \bar\Phi^* \le \min\Phi \le \Phi(\sigma)$; a global minimiser of an exact potential is a pure equilibrium, and uniqueness gives $\lambda = \lambda^*$. (e) If $\Phi(\sigma^\circ) = \bar\Phi^*$, Lemma 53(b) gives $\lambda^* = MA(\sigma^\circ)$, so $e = 0$, and $(*)$ shows that $\sum_i\{\tilde c_i(\sigma_i^\circ) + \langle\lambda^*, a_i(\sigma_i^\circ)\rangle - V_i(\lambda^*)\} = 0$, a sum of nonnegative terms. □

Proof of Theorem 51. Let $\chi_k = \sup\{\sup_{r\in R(\lambda)} \inf_{r'\in R(\lambda')} |r_k - r'_k| : \lambda, \lambda' \in \mathcal{B},\ \lambda_k = \lambda'_k\}$. *Key inequality.* Let a query $\lambda \in \mathcal{B}$ have $\lambda_k = \mu$ and response $r \in R(\lambda)$. Suppose $s_k > 0$ but $\mu - \lambda_k^* > M_k\chi_k$. Let λ' equal λ^* except $\lambda'_k = \mu$; then $\lambda' \in \mathcal{B}$. With $r^* = M^{-1}\lambda^* \in R(\lambda^*)$, antitonicity of R gives $r'_k \le \lambda_k^*/M_k$ for every $r' \in R(\lambda')$, and the definition of χ_k gives $r' \in R(\lambda')$ with $r_k \le r'_k + \chi_k + \eta$ for any $\eta > 0$. Hence $s_k = r_k - \mu/M_k \le -(\mu - \lambda_k^* - M_k\chi_k)/M_k + \eta < 0$ for small η, a contradiction. The case $s_k < 0$ is symmetric, and $s_k = 0$ combines both.

Induction. Initially $\lambda_k^* \in [0, M_k A_k^{\max}]$. If $s_k > 0$ the new bracket is $[\mu, u_k]$; if $\lambda_k^* \geq \mu$ the distance is unchanged, and otherwise it is $\mu - \lambda_k^* \leq M_k \chi_k$. The other cases are symmetric. *Rate.* Midpoint rounds halve every bracket and other rounds never enlarge it, so after T rounds with at least $\lceil T/2 \rceil$ midpoints the width is at most $M_k A_k^{\max} 2^{-\lceil T/2 \rceil}$. If plan sets are separable, V_i is separable and the projection of $R(\lambda)$ on coordinate k depends on λ_k only, so $\chi_k = 0$. □

Example 54 (The floor is genuine)**.** Let $d = 2$, $M = I$ and one carrier with plans 0 ($\tilde{c} = 0$, $a = (0, 0)$) and b ($\tilde{c} = -1.5$, $a = (1, 1)$). Then $\lambda^* = (0.75, 0.75)$, $\chi_1 = \chi_2 = 1$, and the queries $(0.7, 0.9)$, $(0.6, 1)$, $(0.5, 1)$ all return plan 0 (tie at the last resolved towards 0), so the bracket update sets $u_1 = 0.5$ and excludes λ_1^* by 0.25, within the floor but positive.

Proof of Theorem 52. Each round makes one pricing call per carrier. The polish is sequential best-response dynamics in a finite exact-potential game; every accepted change strictly decreases Φ, which takes finitely many values, so it stops at a pure Nash equilibrium. The polish does not increase Φ, and the start has $\Phi \leq \Phi(\sigma^{\text{price}}) = \mathcal{D}(\lambda^{t^\star}) + \frac{1}{2}\|e^{t^\star}\|_M^2 \leq \bar{\Phi}^* + \frac{1}{2}\min_t \|e^t\|_M^2$ by Theorem 50(a) and Lemma 53(a). An exact pure equilibrium makes each plan optimal at the centre belief, which lies in every credal box; the Γ case is Lemma 25. □

Proposition 55 (Lipschitz dependence on charges)**.** *With charges t on the contributions, let $\mu^*(t) = \lambda^*(t) + t$. Then $\|\mu^*(t) - \mu^*(t')\|_{M^{-1}} \leq \|t - t'\|_{M^{-1}}$ and $\langle \bar{A}^*(t) - \bar{A}^*(t'), t - t' \rangle \leq 0$.*

Proof. With charges, $V^t(\lambda) = V(\lambda + t)$, so the equilibrium condition reads $r = M^{-1}(\mu - t) \in R(\mu)$. For $r' = M^{-1}(\mu' - t') \in R(\mu')$, antitonicity gives $\langle r - r', \mu - \mu' \rangle \leq 0$, i.e. $\|\mu - \mu'\|_{M^{-1}}^2 \leq \langle M^{-1}(t - t'), \mu - \mu' \rangle \leq \|t - t'\|_{M^{-1}} \|\mu - \mu'\|_{M^{-1}}$. For the second claim, $\langle r - r', t - t' \rangle = \langle r - r', (\mu - \mu') - M(r - r') \rangle \leq -\|r - r'\|_M^2 \leq 0$. □

Appendix D. Proofs for dual implementation pricing

Notation as in Section 4.4; $m_z^S = b_z + 2\kappa_z$ is the social slope, and $q_{iz}^2 = d_{iz} q_{iz}$ is used throughout. The proofs are taken from the theory notes of the repository, file `theory/new_algorithm_C/theory.tex`.

Proof of Proposition 6. (a) $\tau_{iz} = r q_{iz}$. (b) $s + r t_z = u_z$ and $t_z \geq 0$ because $s \leq u_z$; zones without lots carry no plan. (c) Substitution of $s_z \equiv s$ and $t_z = \vartheta g_z X_z^{\text{NE}}$. □

Proof of Lemma 7. With $q_{iz}^2 = d_{iz} q_{iz}$, $X_z = r Q_z$ and $m_z^S - m_z = \kappa_z$,

$$\Phi_u - \text{SC} = \sum_z \left[\tfrac{m_z}{2} \textstyle\sum_i d_{iz} q_{iz} - \tfrac{\kappa_z}{2} Q_z^2 + u_z Q_z \right].$$

Insert $u_z = \kappa_z Q_z^\circ - \frac{m_z}{2} \bar{d}_z + v_z$ and complete the square: $-\frac{\kappa_z}{2} Q_z^2 + \kappa_z Q_z^\circ Q_z = -\frac{\kappa_z}{2}(Q_z - Q_z^\circ)^2 + \frac{\kappa_z}{2}(Q_z^\circ)^2$, and $\frac{m_z}{2} \sum_i d_{iz} q_{iz} - \frac{m_z}{2} \bar{d}_z Q_z = \sum_i \frac{m_z}{2}(d_{iz} - \bar{d}_z) q_{iz}$. □

Proposition 56 (Carrier-specific charges; classical)**.** *Let σ^* be a pure profile and $U_{iz} = \kappa_z Q_{-i,z}^* - \frac{b_z}{2} d_{iz}$. Under carrier-specific per-delivery charges U_i, for all i and p, p': $J_i^U(p; \sigma_{-i}^*) - J_i^U(p'; \sigma_{-i}^*) = \text{SC}(p, \sigma_{-i}^*) - \text{SC}(p', \sigma_{-i}^*)$. Hence every profile that no single carrier can improve socially is a pure equilibrium.*

Proof. $\text{SC}(p, \sigma_{-i}^*) = R_i(p) + \sum_z q_{iz}(-\alpha_z + m_z^S Q_{-i,z}^* + \frac{m_z^S}{2} d_{iz}) + \text{const}$ and $J_i^U(p) = R_i(p) + \sum_z q_{iz}(-\alpha_z + m_z d_{iz} + m_z Q_{-i,z}^* + U_{iz})$; the brackets coincide because $m_z + \kappa_z = m_z^S$ and $m_z - \frac{b_z}{2} = \frac{m_z^S}{2}$. □

Convexified game. With mixtures $x_i \in \Delta(\mathcal{P}_i)$, $\bar{q}_i = \mathbb{E}_{x_i} q_i$ and $\bar{Q} = \sum_i \bar{q}_i$, the convexified potential and social cost are $\overline{\Phi}_u(x) = \sum_i \mathbb{E}_{x_i}[R_i + \sum_z q_{iz}(-\alpha_z + u_z + \frac{m_z}{2} d_{iz})] + \sum_z \frac{m_z}{2} \bar{Q}_z^2$ and $\overline{\mathrm{SC}}(x) = \sum_i \mathbb{E}_{x_i}[R_i - \sum_z \alpha_z q_{iz}] + \sum_z \frac{m_z^S}{2} \bar{Q}_z^2$. Both are convex, agree with Φ_u and SC at pure profiles, and the identity of Lemma 7 holds for them with Q, q replaced by $\bar{Q}, \bar{q}$.

Proof of Theorem 8. $\overline{\mathrm{SC}}$ is linear in x plus $\sum_z \frac{m_z^S}{2} \bar{Q}_z^2$ with $\bar{Q}$ linear in x, so $\overline{\mathrm{SC}}(x) - \overline{\mathrm{SC}}(x^*) = \langle \nabla \overline{\mathrm{SC}}(x^*), x - x^* \rangle + \sum_z \frac{m_z^S}{2} (\bar{Q}_z - \bar{Q}_z^*)^2$, and the inner product is non-negative by optimality of x^*. Hence $\Delta := \overline{\mathrm{SC}}(x^u) - \overline{\mathrm{SC}}(x^*) \geq \sum_z \frac{m_z^S}{2} (\Delta \bar{Q}_z)^2$. By the identity with $Q^\circ = \bar{Q}^*$, $v = 0$: $\overline{\Phi}_u(x) = \overline{\mathrm{SC}}(x) - \sum_z \frac{\kappa_z}{2} (\bar{Q}_z - \bar{Q}_z^*)^2 + L(x) + \text{const}$. Since x^u minimises $\overline{\Phi}_u$, $\Delta \leq \sum_z \frac{\kappa_z}{2} (\Delta \bar{Q}_z)^2 + L(x^*) - L(x^u)$. Combining, $\sum_z \frac{b_z + \kappa_z}{2} (\Delta \bar{Q}_z)^2 \leq L(x^*) - L(x^u)$; then $\sum_z \frac{\kappa_z}{2} (\Delta \bar{Q}_z)^2 \leq \varrho [L(x^*) - L(x^u)]$ and $\Delta \leq (1 + \varrho)[L(x^*) - L(x^u)]$. Finally $|L(x^*) - L(x^u)| \leq \sum_{i,z} \frac{m_z}{2} |d_{iz} - \bar{d}_z| \, |\bar{q}_{iz}^* - \bar{q}_{iz}^u| \leq E$ because $\bar{q}_{iz} \in [0, d_{iz}]$. □

Proof of Theorem 9. With $\bar{d} = d$, L vanishes on the feasible set, so $E = 0$ in Theorem 8 and every minimiser of $\overline{\Phi}_u$ has aggregate $\bar{Q}^*$ and social cost $\overline{\mathrm{SC}}(x^*)$. Moreover $\overline{\Phi}_u = \overline{\mathrm{SC}} - \sum_z \frac{\kappa_z}{2} (\bar{Q}_z - \bar{Q}_z^*)^2 + \text{const}$; the subtracted term has zero gradient at x^*, so x^* satisfies the first-order optimality condition of the convex function $\overline{\Phi}_u$. □

Proof of Theorem 10. (a) $J_i^u(p) = J_i^0(p) + \langle u, q_i(p) \rangle$, so constraint (i, p) reads $J_i^u(p) \geq J_i^u(\sigma_i^*)$; all of carrier i's constraints hold iff $\min_p J_i^u(p) \geq J_i^u(\sigma_i^*)$, and the violation $J_i^u(\sigma_i^*) - J_i^u(p)$ is maximised by a best response. (b) If a round adds a cut (i, b_i), the current u violates it and satisfies every earlier cut (it is LP-feasible), so the cut is new; there are $\sum_i (|\mathcal{P}_i| - 1)$ non-trivial constraints. Every cut is a valid constraint of $P(\sigma^*)$, so an infeasible LP shows that $P(\sigma^*)$ is empty. □

Proof of Proposition 11. σ is an equilibrium under u iff $u \in P(\sigma)$, so both descriptions of $V^\star$ range over the same pairs (u, σ). The inequality is an inclusion of feasible sets. If the loop ends with $u \in P(\sigma^*)$, the leader attains $\mathrm{SC}(\sigma^*)$. □

Proof of Theorem 12. Best-response dynamics strictly decreases Φ_u on a finite set, so it ends at a pure equilibrium with $\Phi_u(\sigma^u) \leq \Phi_u(\sigma^*)$. Substitute the identity with $Q^\circ = Q^*$ on both sides; the constants cancel and the congestion term vanishes at σ^*. □

Proof of Corollary 13. Count the calls of the three phases; Theorem 10 bounds the rounds, and each improvement of the fallback dynamics costs at most N calls. □

Proof of Theorem 14. By Lemma 25, carrier i's Γ-maximin payoff is its precise payoff minus $\delta \langle a_i(p), Mw \rangle$ (traffic already enters the route loss at its upper expectation). With $a_i = (q_i, r q_i)$, $\delta \langle a_i(p), Mw \rangle = \langle \hat{s}(\delta), q_i(p) \rangle$, so the Γ-maximin cost under u is the precise cost under $u + \hat{s}(\delta)$, and u implements σ^* iff $u + \hat{s}(\delta) \in P(\sigma^*)$. The social cost does not contain the surcharge. The last claim is Lemma 24: an equilibrium of the precise game lies in $\mathrm{ICNE}_C(\delta)$ for $C \in \{E, M, ID\}$ and every δ. □

Proof of Proposition 15. The first claim is Theorem 9 with a uniform tangent charge. For the second, $Q_z^{\text{full}} = n_z d_z$ with equal lots, and without congestion ($m_z \approx b_z$ when $\kappa_z \ll b_z$) $u_z = -\frac{b_z}{2} d_z = -\frac{\alpha_z}{4 n_z} = -0.375 \, \text{fee}/n_z$. The last claim is Proposition 6. □

Appendix E. Proofs for aggregate-response pricing

Notation as in Section 4.4: $q_{iz}^2 = d_{iz}q_{iz}$, $\kappa_z = g_z r^2$, $m_z = b_z + \kappa_z$, $m_z^S = b_z + 2\kappa_z$, and Φ_u the exact potential of the game under the per-delivery-equivalent charge u. For a reference $\bar{d} \in \mathbb{R}^Z$, $L_{\bar{d}}(\sigma) = \sum_{i,z} \frac{m_z}{2}(d_{iz} - \bar{d}_z)q_{iz}$ and $E_{\bar{d}} = \sum_{i,z} \frac{m_z}{2}|d_{iz} - \bar{d}_z|d_{iz}$. The convexified potential $\overline{\Phi}_u$ and social cost $\overline{\mathrm{SC}}$ are those of Appendix D. The proofs are taken from `theory/new_algorithm_D/theory.tex`.

Lemma 57. *G is convex and piecewise linear on $\mathcal{Q}$; F is proper, closed and 1-strongly convex in $\|\cdot\|_m$, i.e. $F - \frac{1}{2}\|\cdot\|_m^2$ is convex. Moreover $\min_{x\in\mathcal{X}} \overline{\Phi}_u(x) = \min_Q\{F(Q) + \langle u, Q\rangle\}$ and $\min_{x\in\mathcal{X}}[\overline{\mathrm{SC}}(x) + L_{\bar{d}}(x)] = \min_Q\{F(Q) + g(Q)\}$ with $g(Q) = \sum_z[\frac{\kappa_z}{2}Q_z^2 - \frac{m_z}{2}\bar{d}_z Q_z]$, and in both cases x is a minimiser on the left if and only if $\bar{Q}(x)$ is a minimiser on the right and x attains the minimum defining G at $Q = \bar{Q}(x)$.*

Proof. G is the optimal value of a linear program in x whose right-hand side Q enters through $\bar{Q}(x) = Q$; such a value function is convex and piecewise linear on the set of right-hand sides for which the program is feasible, which is $\mathcal{Q}$. The quadratic part of F is $\frac{1}{2}\|Q\|_m^2$ and the rest is convex, so $F - \frac{1}{2}\|\cdot\|_m^2$ is convex; F is closed because G is continuous on the closed set $\mathcal{Q}$. For the identities, write $\frac{m_z}{2}d_{iz} = \frac{m_z}{2}(d_{iz} - \bar{d}_z) + \frac{m_z}{2}\bar{d}_z$ in $\overline{\Phi}_u$ and $\frac{m_z^S}{2} = \frac{m_z}{2} + \frac{\kappa_z}{2}$ in $\overline{\mathrm{SC}} + L_{\bar{d}}$; in both the objective depends on x only through $\bar{Q}(x)$ plus the quantity minimised in the definition of G. □

Proof of Theorem 17. By Lemma 57, $F + \langle u, \cdot\rangle$ is strongly convex on the compact set $\mathcal{Q}$, so it has a unique minimiser $Q(u)$, and every convexified equilibrium has $\bar{Q}(x) = Q(u)$. For closed proper strongly convex F the conjugate F^* is finite (the domain is bounded) and differentiable with $\nabla F^*(y) = \arg\max_Q\{\langle y, Q\rangle - F(Q)\}$ [91, Theorem 26.3]; with $y = -u$ this is $Q(u)$. The optimality conditions read $-u_k \in \partial F(Q_k)$; strong convexity gives $\langle(-u_1) - (-u_2), Q_1 - Q_2\rangle \geq \|Q_1 - Q_2\|_m^2$, and by Cauchy–Schwarz in the m-inner product $\langle u_2 - u_1, Q_1 - Q_2\rangle \leq \|u_2 - u_1\|_{m^{-1}}\|Q_1 - Q_2\|_m$, which gives the Lipschitz bound.

Let $D(u) = F^*(-u) + g^*(u)$ with $g^*(u) = \sum_z(u_z + \frac{m_z}{2}\bar{d}_z)^2/(2\kappa_z)$ (maximise $\langle u, Q\rangle - g(Q)$ coordinatewise). D is differentiable with $\nabla D(u) = -Q(u) + \kappa^{-1} \odot (u + \frac{m}{2}\bar{d})$, which is observable, and strongly convex because g^* has Hessian $\mathrm{diag}(1/\kappa)$. Fenchel duality [91, Corollary 31.2.1], with attainment because g is finite and continuous, gives $\min_Q\{F + g\} = -\min_u D(u)$. The minimiser u^* is unique and characterised by $\nabla D(u^*) = 0$, i.e. $u^* = \kappa \odot Q(u^*) - \frac{m}{2} \odot \bar{d} =: T(u^*)$; any fixed point of T is such a zero. At the saddle point the primal minimiser is $Q(u^*)$, which minimises $F + g$, i.e. $\overline{\mathrm{SC}} + L_{\bar{d}}$ by Lemma 57. For welfare, let x^* minimise $\overline{\mathrm{SC}}$ and x^∞ be a convexified equilibrium under u^*; then $\overline{\mathrm{SC}}(x^\infty) - \overline{\mathrm{SC}}(x^*) \leq L_{\bar{d}}(x^*) - L_{\bar{d}}(x^\infty) \leq \sum_{i,z} \frac{m_z}{2}|d_{iz} - \bar{d}_z||\bar{q}_{iz}^* - \bar{q}_{iz}^\infty| \leq E_{\bar{d}}$, because $\bar{q}_{iz} \in [0, d_{iz}]$; with symmetric lots $L_{\bar{d}} \equiv 0$. □

Proof of Theorem 18. Let $K = \mathrm{diag}(\kappa)$ and equip $\mathbb{R}^Z$ with $\langle\cdot,\cdot\rangle_{\kappa^{-1}}$. The gradient of D in this geometry is $K\nabla D(u) = u + \frac{m}{2}\bar{d} - \kappa \odot Q(u)$, so the rule (5) is $u^{t+1} = u^t - \theta K \nabla D(u^t)$, gradient descent with step θ. Since $\kappa_z \leq \varrho m_z$ and $1/m_z \leq \varrho/\kappa_z$ on active zones, the Lipschitz bound of Theorem 17 gives $\|Q(u_1) - Q(u_2)\|_\kappa \leq \sqrt{\varrho}\|Q(u_1) - Q(u_2)\|_m \leq \sqrt{\varrho}\|u_1 - u_2\|_{m^{-1}} \leq \varrho\|u_1 - u_2\|_{\kappa^{-1}}$. Hence $u \mapsto \kappa \odot Q(u)$ is ϱ-Lipschitz from $\|\cdot\|_{\kappa^{-1}}$ to itself (note $\|\kappa \odot v\|_{\kappa^{-1}} = \|v\|_\kappa$), $K\nabla D$ is $(1 + \varrho)$-Lipschitz, and D is strongly convex with modulus 1 in this geometry. (a) T is ϱ-Lipschitz and $u^* = T(u^*)$. (b) For a μ-strongly convex function with L-Lipschitz gradient,

gradient descent with step $\theta \le 2/(\mu+L)$ satisfies $\|u^{t+1}-u^*\|^2 \le (1-\frac{2\theta\mu L}{\mu+L})\|u^t-u^*\|^2$ [92, Theorem 2.1.15]; the proof uses only an inequality valid in any inner-product space. With $\mu=1$, $L=1+\varrho$ and $\theta=2/(2+\varrho)$ the factor is $1-4(1+\varrho)/(2+\varrho)^2=\varrho^2/(2+\varrho)^2$. □

Proof of Theorem 19. (a) Apply Lemma 7 with $Q^\circ = Q(\sigma)$. Since σ is an equilibrium, $\Phi_u(\sigma) \le \Phi_u(\sigma')$ [43]. The term $\sum_z \frac{\kappa_z}{2}(Q_z - Q_z^\circ)^2$ vanishes at σ and equals $\sum_z \frac{\kappa_z}{2}\Delta Q_z^2$ at σ'; rearranging the identity for σ and σ' gives the inequality. The last bound uses $|q_{iz}(p) - q_{iz}(\sigma_i)| \le d_{iz}$.

(b) $u^{t+1} = \kappa \odot Q^t - \frac{m}{2}\bar{d} = T(u^t) + \kappa \odot e^t$ and $u^* = T(u^*)$, so the first inequality follows from Theorem 18(a); iterate for the limsup. For the gap bound let $h = F + \langle u^t, \cdot\rangle$, which is 1-strongly convex in $\|\cdot\|_m$ and minimised at $Q(u^t)$, so $h(Q) \ge h(Q(u^t)) + \frac{1}{2}\|Q - Q(u^t)\|_m^2$. A pure profile is a point of $\mathcal{X}$ with $\bar{Q} = Q^t$, so $\overline{\Phi}_{u^t}(\sigma^t) \ge h(Q^t)$ while $\min \overline{\Phi}_{u^t} = h(Q(u^t))$ (Lemma 57); with $\Phi_{u^t}(\sigma^t) = \overline{\Phi}_{u^t}(\sigma^t)$ at a pure profile, $\eta_t \ge \frac{1}{2}\|e^t\|_m^2$ and $\|e^t\|_\kappa \le \sqrt{\varrho}\|e^t\|_m \le \sqrt{2\varrho\eta_t}$.

(c) By Theorem 17, $Q = Q(u)$ iff $-u \in \partial F(Q)$. F is polyhedral plus a quadratic on the polytope $\mathcal{Q}$, so $\partial F(Q) \ne \emptyset$ on $\mathcal{Q}$ and the graph of the response map is the graph of $-\partial F$ on $\mathcal{Q}$. A closed proper convex function is determined by its subdifferential up to an additive constant [91, Theorem 24.9]. The convexified optimum minimises $F+g$ with g known, and u^* is the unique zero of ∇D; both are unchanged by a constant shift of F. Example 58 shows that the pure optimum is not determined. □

Example 58. One zone, $\kappa = 0$, slope $b = m$, intercept α, reference $\bar{d} = d$. Instance A: two carriers, each with lot d and plans "stay out" ($R=0$) and "serve" ($R=r$). Instance B: one carrier with lot $2d$ and plans "stay out" ($R=0$) and "serve" ($R = 2r - md^2$), with $r > md^2/2$. In both, $\mathcal{Q} = [0, 2d]$ and G is linear with $G_A(Q) = G_B(Q) = \frac{r}{d}Q$, so $F_A = F_B$ and the response maps coincide. The pure social costs are $\mathrm{SC}_A(Q) = \frac{r}{d}Q - \alpha Q + \frac{m}{2}Q^2$ on $Q \in \{0, d, 2d\}$ and $\mathrm{SC}_B(Q) = (2r - md^2)\mathbf{1}[Q = 2d] - \alpha Q + \frac{m}{2}Q^2$ on $\{0, 2d\}$. With $\alpha = \frac{r}{d} + md$, $Q = d$ is the unique pure optimum of A with value $-\frac{m}{2}d^2$, while $Q = 2d$ is the unique pure optimum of B with value $-md^2$. The optima differ in aggregate and value although nothing observable in the convexified game differs.

Lemma 59 (Lot estimates)**.** *(a) If lots are symmetric within zones, $Q_z/n_z = \bar{d}_z$ in every served zone. (b) If between two observations n_z changes by exactly one, $|Q_z^t - Q_z^{t-1}|$ is the lot of the carrier that entered or left; if $n_z = 1$, Q_z is the lot of the serving carrier. (c) For a zone never served the observation carries no information about its lots; the bound of Theorem 17 holds for any reference vector, with $E_{\bar{d}}$ for the vector actually used.*

Proof. (a) $Q_z = n_z \bar{d}_z$. (b) The other lots in z are unchanged, so the difference of the sums is the changed term. (c) Two instances that differ only in the lots of a never-served zone are indistinguishable; the bound was shown for an arbitrary reference vector. □

Proof of Proposition 20. By Lemma 25, the Γ-maximin cost gains the surcharge

$$\delta\langle a_i(p), Mw\rangle = \langle \hat{s}(\delta), q_i(p)\rangle,$$

with $\hat{s}_z(\delta) = \delta\hat{w}_z$, $\hat{w}_z = b_z w_{z,Q} + g_z r\, w_{z,X}$, linear in deliveries; so the response to u is the precise response to $u + \hat{s}$, the map $u \mapsto T(u + \hat{s})$ is ϱ-Lipschitz and the rule contracts as before, with a fixed point at which $u + \hat{s}$ is a zero of the dual gradient of $\min_Q F + g + \langle \hat{s}, Q\rangle$.

For the loss, let Q_0 and Q_δ minimise $H = F + g$ and $H + \langle \hat{s}, \cdot \rangle$, both 1-strongly convex in $\|\cdot\|_m$. Strong convexity at Q_δ gives $H(Q_\delta) - H(Q_0) \le \langle \hat{s}, Q_0 - Q_\delta \rangle - \frac{1}{2}\|Q_0 - Q_\delta\|_m^2 \le \|\hat{s}\|_{m^{-1}}\|Q_0 - Q_\delta\|_m - \frac{1}{2}\|Q_0 - Q_\delta\|_m^2 \le \frac{1}{2}\|\hat{s}\|_{m^{-1}}^2$.

For E-admissibility, write $\Delta_z = u_z - u_z^*$ for a posted charge u and a pure equilibrium σ^* under u^* with precise beliefs. Carrier i's cost of plan p at charge u and belief (μ_Q, μ_X) about rivals is $R_i(p) + \sum_z q_{iz}(p)(-\alpha_z + m_z d_{iz} + b_z \mu_{Q,z} + g_z r \mu_{X,z} + u_z)$. Choose $\mu_{Q,z} = Q_{-i,z} - \Delta_z w_{z,Q}/\hat{w}_z$ and $\mu_{X,z} = X_{-i,z} - \Delta_z w_{z,X}/\hat{w}_z$. Then $b_z\mu_{Q,z} + g_z r \mu_{X,z} + u_z = b_z Q_{-i,z} + g_z r X_{-i,z} + u_z^*$, so every plan's cost at this belief and charge u equals its cost at the realised aggregate and charge u^*, where σ_i^* is optimal. The belief lies in the box iff $|\Delta_z| \le \delta \hat{w}_z = \hat{s}_z(\delta)$, and it is nonnegative when $\Delta_z \le 0$, or when $Q_{-i,z} \ge \Delta_z w_{z,Q}/\hat{w}_z$ and $rQ_{-i,z} \ge \Delta_z w_{z,X}/\hat{w}_z$ for $\Delta_z > 0$. By Lemma 23, σ_i^* is E-admissible. The round count follows from Theorem 18(a) and $|u_z - u_z^*| \le \sqrt{\kappa_z}\|u - u^*\|_{\kappa^{-1}}$. □

Proof of Theorem 16. (a) $J_i^T(p; \sigma_{-i}) = R_i(p) + \sum_z q_{iz}(-\alpha_z + m_z d_{iz} + m_z Q_{-i,z} + \kappa_z Q_{-i,z} - \frac{b_z}{2} d_{iz}) = R_i(p) + \sum_z q_{iz}(-\alpha_z + \frac{m_z^S}{2} d_{iz} + m_z^S Q_{-i,z})$, using $q_{iz}^2 = d_{iz} q_{iz}$, $m_z - \frac{b_z}{2} = \frac{m_z^S}{2}$ and $m_z + \kappa_z = m_z^S$. On the other side $\mathrm{SC}(p, \sigma_{-i}) - \mathrm{SC}(\emptyset_i, \sigma_{-i}) = R_i(p) + \sum_z [-\alpha_z q_{iz} + \frac{m_z^S}{2}((q_{iz} + Q_{-i,z})^2 - Q_{-i,z}^2)]$, which is the same expression. The right side differs from $\mathrm{SC}(p, \sigma_{-i})$ by a term independent of p, which is the definition of an exact potential. (b) Monderer–Shapley for the potential SC on a finite profile set. (c) Immediate from the first expression in (a). □

Appendix F. Additional tables

Table F.11: Methods on the nine city games (seed 0, price-and-branch oracle). d_{eff}: active coordinates. Oracle calls and relative potential gap to the convexified optimum, in %, for best response (BR), ZMC with eight rounds, Frank–Wolfe and column generation with eight rounds (FW, CG8) and column generation to convergence (CG); all methods end with the same best-response polish.

City	N	Z	d_{eff}	BR	ZMC	FW	CG8	CG	BR	ZMC	FW	CG8	CG
New Delhi	5	120	64	14	45	45	50	65	13.8	10.2	10.2	10.2	10.2
Cairo	4	42	32	8	36	24	16	16	$1.0 \cdot 10^{-5}$	$1.0 \cdot 10^{-5}$	$1.0 \cdot 10^{-5}$	$1.0 \cdot 10^{-5}$	$1.0 \cdot 10^{-5}$
Moscow	5	160	71	12	45	45	45	60	0.220	0.220	0.220	0.220	0.220
Buenos Aires	4	208	62	10	39	37	40	91	17.9	16.3	16.6	16.0	16.0
New York	4	234	64	8	36	36	36	76	0.451	0.451	0.437	0.437	0.437
Mumbai	5	143	55	14	48	48	48	60	4.60	1.98	1.98	1.98	1.98
London	5	348	89	15	49	49	50	70	20.0	18.0	18.0	18.0	18.0
Paris	5	198	68	13	50	50	50	108	2.74	2.78	2.74	2.74	2.74
Berlin	6	593	82	17	65	59	59	143	2.32	2.43	2.32	2.46	2.46

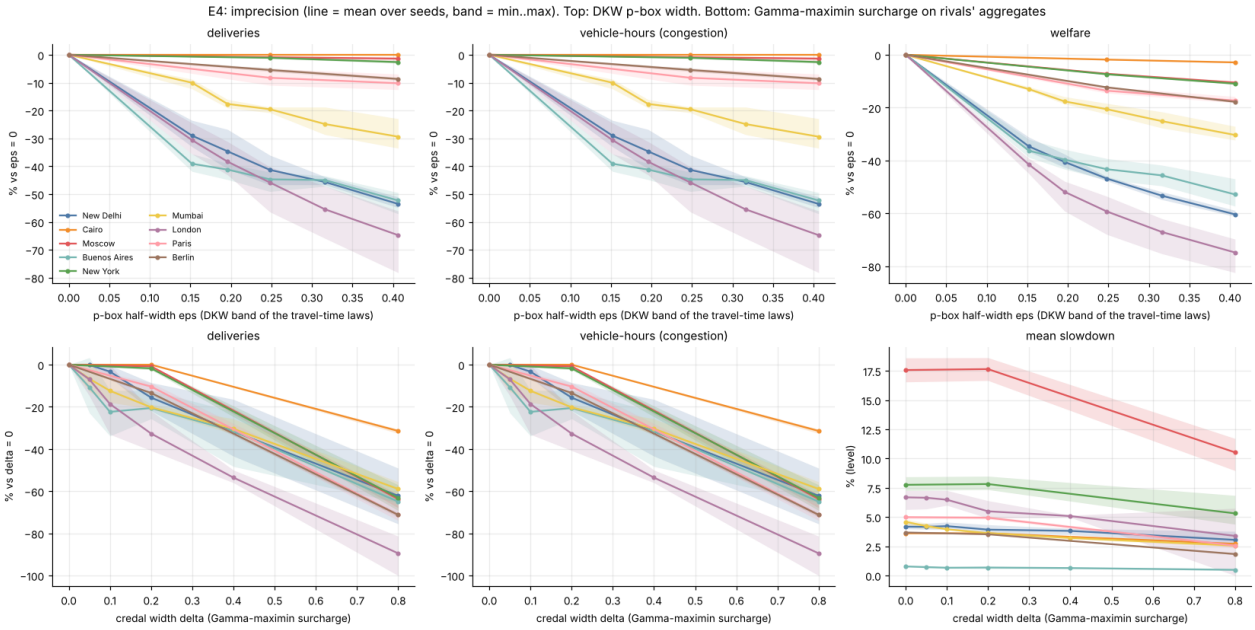


Figure F.3: Imprecision, exact oracle, mean over the seeds (line) with the minimum–maximum band. Top: probability-box half-width (route costs through BRL). Bottom: Γ-maximin credal width.

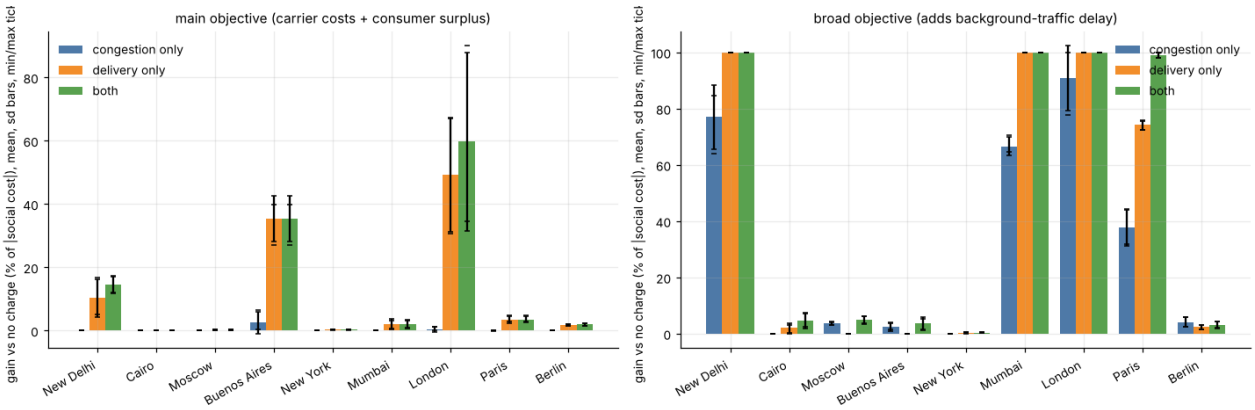


Figure F.4: Gain of the grid leader's best exact charge over no charge, by instrument: mean over the seeds with standard-deviation bars and minimum/maximum ticks. Left: main objective. Right: broad objective with the background-delay proxy (shut-down optima are an artifact). Cairo, Moscow, New York, Mumbai, Paris, Berlin use a reduced exact grid (lower bounds).

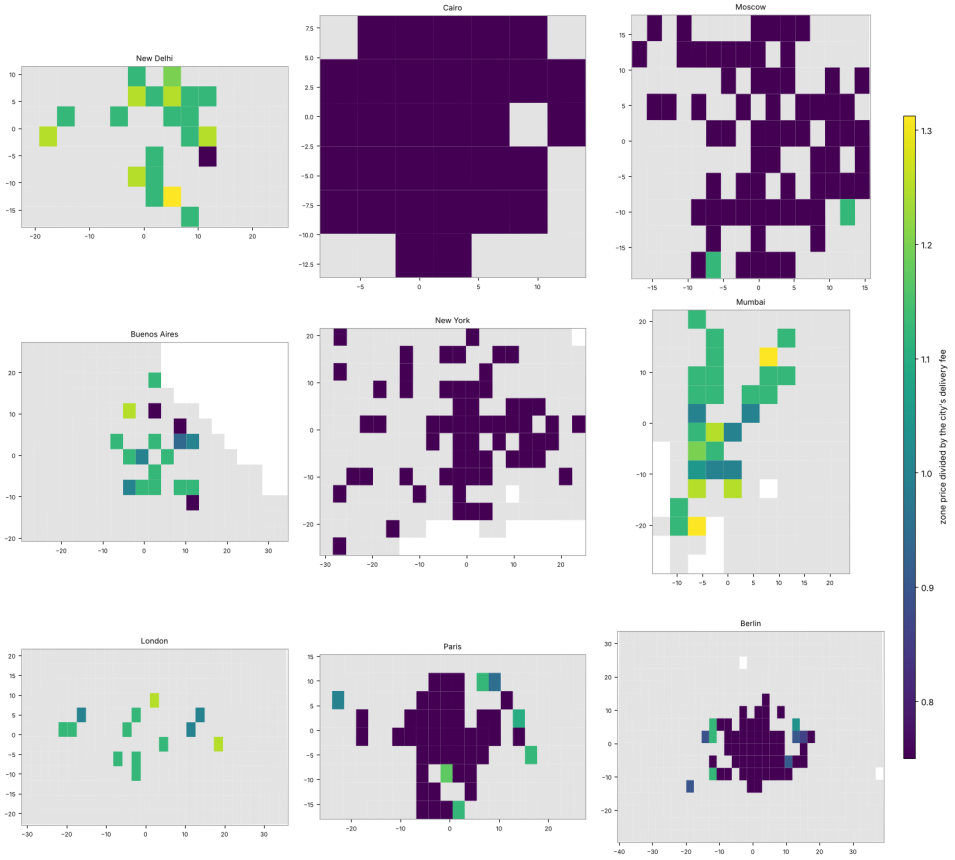


Figure F.5: Unpriced equilibrium zone prices divided by the city's delivery fee, exact oracle, seed 0 (light grey: zones without deliveries).

Table F.12: City instances. Z: zones; population from GHS-POP inside the zones; major road and lane kilometres inside the zones; nodes of the major-road graph; N: number of carriers.

City	Z	pop. (M)	major road km	lane-km	graph nodes	N
New Delhi	120	25.0	5 556	8 590	138 542	5
Cairo	42	14.9	2 649	4 498	119 027	4
Moscow	160	12.6	4 475	12 325	146 234	5
Buenos Aires	208	13.3	5 764	12 032	130 526	4
New York	234	12.1	8 798	17 770	382 014	4
Mumbai	143	22.0	4 259	7 092	109 804	5
London	348	11.3	7 110	13 628	350 931	5
Paris	198	9.02	5 990	10 854	309 175	5
Berlin	593	4.65	5 490	11 147	192 577	6

Table F.13: Coverage of the exact-oracle game experiments per city and seed: exact grid points of the leader experiment (out of 63), price-of-anarchy experiment computed (yes/–), and exact widths of the imprecision experiment (out of 12); – means not computed with the exact oracle. The base equilibria are exact for all city-seeds.

City	E2 s0	E3 s0	E4 s0	E2 s1	E3 s1	E4 s1	E2 s2	E3 s2	E4 s2
New Delhi	63/63	yes	12/12	63/63	yes	12/12	63/63	yes	12/12
Cairo	13/63	yes	6/12	17/63	yes	6/12	13/63	yes	6/12
Moscow	12/63	yes	6/12	10/63	yes	6/12	11/63	yes	6/12
Buenos Aires	63/63	yes	12/12	63/63	yes	12/12	63/63	yes	12/12
New York	7/63	yes	6/12	7/63	yes	6/12	9/63	yes	6/12
Mumbai	15/63	yes	12/12	14/63	yes	12/12	14/63	yes	12/12
London	63/63	yes	12/12	63/63	yes	12/12	63/63	yes	12/12
Paris	14/63	yes	6/12	13/63	yes	6/12	13/63	yes	6/12
Berlin	13/63	yes	6/12	13/63	yes	6/12	13/63	yes	6/12

Table F.14: Calibration per city (base values; ranges and sources in `data/parameters/SOURCES.md`). Parcels per capita are country-level; TomTom congestion is the 2025 city value (Moscow: 2020); drops per vehicle and van capacity are proxies outside the United States; fees and wages are graded C for most cities.

City	parcels/cap./day	cong. %	drops/veh.	shift h	van cap.	fee USD	wage USD/h	fuel USD/L	N	veh./1000
New Delhi	0.00964	60.2	40.0	10	40.0	0.690	1.07	1.15	5	400
Cairo	0.00561	40.5	40.0	8.4	40.0	1.18	0.610	0.440	4	220
Moscow	0.156	54.0	70.0	8.4	126	3.67	4.53	0.900	5	297
Buenos Aires	0.0150	48.1	70.0	8.4	126	0.980	1.28	1.45	4	331
New York	0.184	48.8	120	8.4	226	9.09	22.3	1.22	4	242
Mumbai	0.00964	63.2	40.0	10	40.0	0.690	1.07	1.15	5	330
London	0.166	51.6	72.0	7.3	126	4.21	18.8	2.34	5	288
Paris	0.0650	40.0	70.0	8.4	126	6.04	15.7	2.42	5	575
Berlin	0.143	44.6	126	8.4	206	4.40	18.8	2.39	6	330

Table F.15: Sensitivity of unpriced welfare (% change, low to high value of each weak parameter, seed 0). The scenarios use the exact oracle; the reference base case is the earlier price-and-branch equilibrium of the same seed, so differences of the order of the oracle effect (Section 7.1) are not meaningful.

City	fee	wage	drops	N	parcels
New Delhi	-58 to 317	-53 to 43	-90 to 294	-38 to 53	-9 to 0
Cairo	19 to 19	-5 to 2	-31 to 59	-25 to 24	-0 to 0
Moscow	-75 to 48	-26 to 6	-43 to 105	-62 to 21	-0 to 1
Buenos Aires	-59 to 452	-48 to -48	-87 to 474	-13 to 72	-1 to 6
New York	-41 to 55	-8 to 1	-57 to 9	-27 to 23	-0 to 0
Mumbai	-43 to 208	-36 to 26	-72 to 169	-19 to 29	-5 to -1
London	-95 to 363	-3 to 28	-95 to 795	-26 to 107	-2 to 4
Paris	-58 to 83	-4 to 10	-57 to 149	-16 to 27	-0 to 1
Berlin	-58 to 24	-5 to 3	-82 to 4	-3 to -3	-0 to 2

Table F.16: City leader, broad objective with the background-delay proxy (3 seeds). Gains as in Table 4 (mean [min, max]); combined gain per seed; served share at the best point with both instruments. Shut-down optima (served share 0) are an artifact of the proxy.

City	cong. only	deliv. only	both	both by seed	served
New Delhi	77.1 [64, 85]	100	100	100 / 100 / 100	0
Cairo†	0	2.06 [0, 3.3]	4.59 [2.5, 7.5]	2.47 / 7.50 / 3.81	0.67 [0.65, 0.69]
Moscow†	3.67 [3.1, 4.1]	0.0230 [0, 0.069]	4.87 [3.6, 6.3]	3.59 / 4.72 / 6.31	0.92 [0.91, 0.93]
Buenos Aires	2.41 [1.4, 4.0]	0	3.71 [1.1, 5.4]	4.62 / 5.42 / 1.09	0.35 [0.29, 0.46]
New York†	0.00636 [0, 0.019]	0.320 [0.18, 0.52]	0.344 [0.26, 0.52]	0.259 / 0.518 / 0.256	1.00 [0.99, 1.0]
Mumbai†	66.7 [65, 71]	100	100	100 / 100 / 100	0
London	90.9 [78, 100]	100	100	100 / 100 / 100	0
Paris†	37.7 [32, 44]	74.2 [72, 76]	99.1 [98, 100]	99.3 / 100 / 98.1	0.042 [0, 0.085]
Berlin†	4.07 [2.6, 5.9]	2.39 [1.5, 3.2]	3.10 [2.1, 4.5]	2.13 / 2.66 / 4.50	0.74 [0.65, 0.88]

Table F.17: ZMC on synthetic D1 games with non-separable plan sets (five seeds per row). Entries: mean oracle calls / mean potential gap to the convexified optimum; last column: instances in which ZMC has the smallest gap among FW8, CG8 and multi-start BR.

N	Z	BR	BR multi-start	FW8	CG8	ZMC	CG full	ZMC best
4	30	12 / 5.33	45 / 3.22	38 / 3.64	39 / 3.20	39 / 2.77	58 / 2.69	4/5
5	60	21 / 15.8	56 / 11.8	50 / 7.78	51 / 7.80	49 / 6.90	159 / 6.13	4/5
6	120	28 / 50.2	70 / 38.4	61 / 21.4	63 / 27.5	60 / 17.1	365 / 14.2	4/5
6	250	26 / 91.5	71 / 80.2	64 / 47.9	67 / 55.4	60 / 33.4	797 / 27.8	5/5

Table F.18: Tiny games solved by exhaustive enumeration (earlier price-and-branch oracle): share of instances in which the method's profile is a Nash equilibrium of the enumerated game and reaches the global potential minimum, largest independent-oracle regret and mean excess of the potential over the global minimum.

Method	games	calls	is NE	max regret	global min.	excess
ZMC (6 rounds)	40	17.5	1.00	0	1.00	0
BR	40	4.85	1.00	0	0.875	0.93
FW (6 rounds)	40	16.7	1.00	0	0.950	0.067
CG	40	11.1	1.00	0	0.975	0.020

Table F.19: Checks of ZMC on the city games (seed 0, earlier price-and-branch oracle): largest residual of the clearing identity (Theorem 50(a)); share of rounds in which its bound holds; inverted brackets; active coordinates; largest relative bracket width after eight rounds; in bisection mode, share of coordinates within the bound of Theorem 51 at $T = 8$, largest normalised error and the bound 2^{-4}.

City	identity resid.	bound holds	inverted	active	width_8	within_8	err_8	bound_8
New Delhi	$2.8 \cdot 10^{-14}$	1.00	0	128	0.0435	1.00	0.00385	0.0625
Cairo	$1.6 \cdot 10^{-13}$	0.875	0	64	0.0435	1.00	0.00385	0.0625
Moscow	$8.1 \cdot 10^{-13}$	1.00	0	142	0.0435	1.00	0.00385	0.0625
Buenos Aires	$5.7 \cdot 10^{-14}$	1.00	0	124	0.0435	1.00	0.00385	0.0625
New York	$3.4 \cdot 10^{-12}$	1.00	0	128	0.0435	1.00	0.00385	0.0625
Mumbai	$4.3 \cdot 10^{-14}$	1.00	0	110	0.0435	1.00	0.00385	0.0625
London	$1.1 \cdot 10^{-13}$	1.00	0	178	0.0435	1.00	0.00385	0.0625
Paris	$1.5 \cdot 10^{-12}$	1.00	0	136	0.0415	1.00	0.00385	0.0625
Berlin	$1.8 \cdot 10^{-12}$	1.00	0	164	0.0435	1.00	0.00385	0.0625

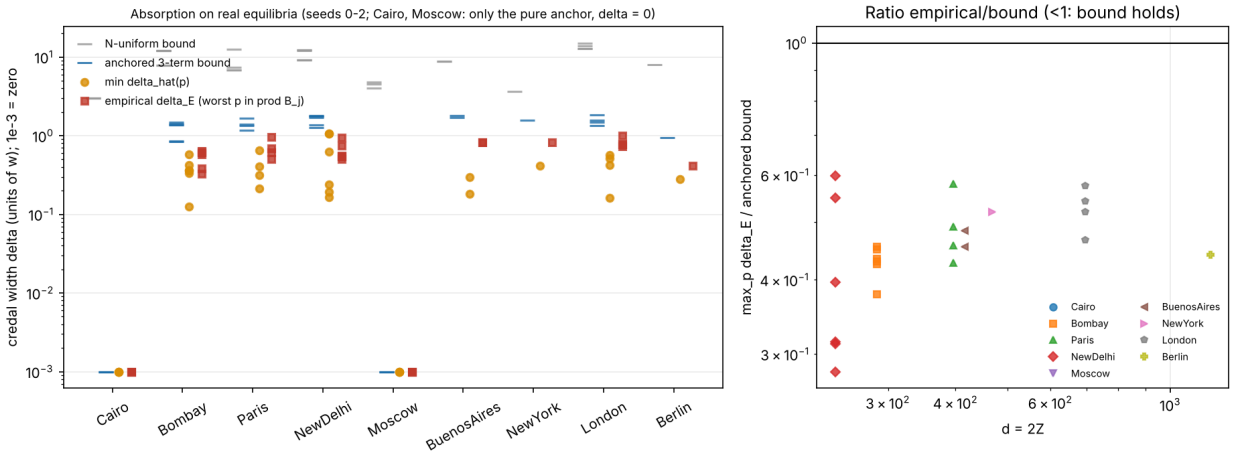


Figure F.6: Credal absorption on the city equilibria (three seeds). Left: N-uniform bound, anchored bound, smallest anchored radius and largest empirical E-threshold per anchor (Cairo and Moscow: pure anchors only, plotted at 10^{-3} for zero). Right: ratio of the largest empirical E-threshold to the anchored bound against $d = 2Z$; values below one mean the bound holds.

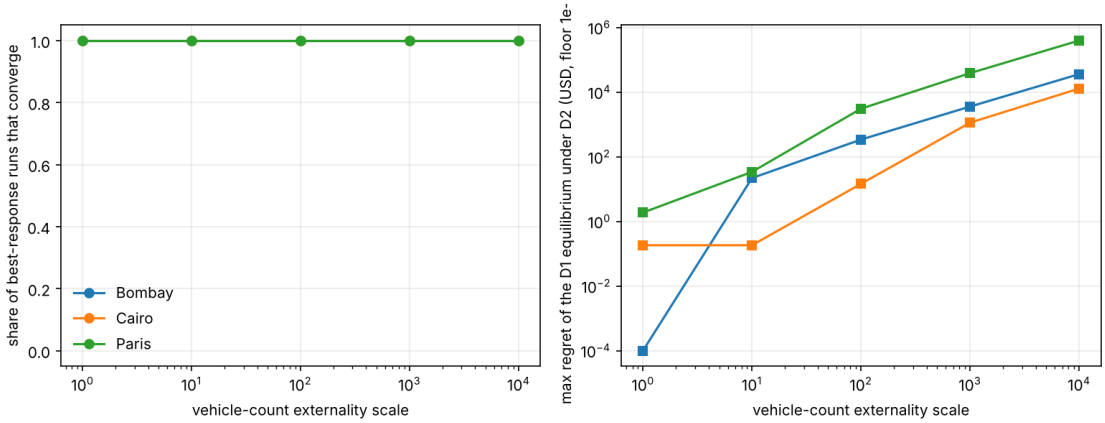


Figure F.7: City games of Cairo, Mumbai and Paris (three seeds) with a vehicle-count externality of increasing scale. Left: share of best-response runs that converge to a pure equilibrium. Right: largest regret of the D1 equilibrium when the game is of type D2 (USD).